

None of the 650 profiles can be approved as published

The verdict split below is withdrawn as a verdict and kept only as an index of which profiles show which symptom; see Corrections.

Every profile passes its tests in both implementations, and every packet digest matches its profile. The tests prove the rules run. They do not prove the rules are right, and running the engine on real scientific cases shows several rules returning the wrong answer in both directions.



STATUS OF THIS REVIEW

What this is	An independent pre-sign-off analysis and change request, produced by an AI assistant (Claude Opus 5) at the repository owner's request. It is not the named, domain-qualified scientific sign-off that PROFILE_REVIEW.md requires, and must not be recorded as one. Git history attributes all profile source to one author, so the named scientific and schema reviewers must both come from outside that authorship.
Scope	All 650 profiles. No assignment list exists in the repository, packets or prior sessions, and every profile is produced by one generator, so the generator's rules were reviewed across all profiles, with representative profiles checked against the running engine.
Repository	Biosimulant/model-compatibility-standard @ 9a4edf8 (working tree with uncommitted changes)
Evidence files	No file was written under source/reviews/. Every profile carries at least one blocking change, and review evidence may only be created after blocking changes are incorporated. None. No file under spec/v0.1/ or source/ was modified.
Digests	650 of 650 packet profile_sha256 values reproduce from the profile definitions.
Source pinning	Sources are pinned in docs/scientific-remediation/sources/sources.lock.json. Read the pinned_as field before relying on a digest: a document pin covers the reviewed text, a citation-metadata pin covers only the Crossref record for the DOI, and licensed standards (ISO 80000-1, ISO 20776-1, SNOMED CT) cannot be pinned here at all. Where a decision rests on a paywalled or licensed text, the reviewer pins their own copy and records the terms.

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Companion data. The per-profile contract-area rationales, the full item-by-item decision lists and the answers to every reviewer question in every packet are recorded in `change-request.json`, which accompanies this report. This PDF carries every decision rule, every profile's verdict and audit results, and every profile-specific finding in full.

READ THE REPORT WITH THESE CORRECTIONS

Corrections

Raised in review of this report and accepted. The report stands as a fault report; these parts do not. Each correction opens a design decision, tracked in the remediation plan.

E1 The per-profile verdict split is withdrawn

136 BLOCK / 514 CHANGES_REQUIRED implied a per-profile precision the evidence does not support, because nearly every defect originates in shared generator and comparison code. The single conclusion is that all 650 profiles are unsuitable for GA review until those foundations are fixed. The per-profile table below survives only as an index of which profiles show which symptom.

E2 S12 miscounted the security profiles

The 50 profiles requiring security.data_use and security.consent_scope are 25 clinical and 25 epidemiology, not 50 clinical.

E3 The blanket candidate-item decisions are withdrawn

Several were over-broad: uncertainty.censoring was marked required in 400 packets although its own reason limits it to time-to-event data; semantic.ontology_terms was marked required for all 650; artifact.sha256 required wherever an artifact representation is allowed. Item levels are now decided per profile or profile class (decision D10), with this table as input only.

E4 Four conclusions were stated too absolutely

Dense-to-sparse re-encoding is lossless only when implicit-entry meaning, ordering, shape and dtype are declared equal (D6). Celsius to kelvin is exact for absolute temperatures; a temperature difference converts with factor 1 and no offset (D1). Identifier mapping is not always lossy: a pinned, total, bijective mapping is lossless (D7). 'Any species' has no defined meaning yet, so the report should not have prescribed UNKNOWN (D5).

E5 No source is pinned

Every source has a title, DOI or permanent URL and version, but none has a SHA-256 of the reviewed bytes. Pinning is a Phase 1 task.

E6 The change request now lives in the repository

It is at docs/scientific-remediation/pre-review-2026-09-15/ with the tools that produce it.

E7 The change request answered questions the packets do not ask

Corrected: it now carries 5,829 answers, exactly the set the 650 packets ask.

E8 S12 overstated the data-use ordering defect

security.data_use is already a set-like path in the normalization rules, and normalised contracts compare as EXACT whatever the element order. The narrower real defect: compare_contracts does not normalise internally, so a caller skipping that stage gets INCOMPATIBLE from ordering alone, and equal on a set of DUO codes carries no subsumption.

E9 The findings that survived replay are now executable

Fifteen defects and eight guards live in scientific-checks/v0.1/cases.json and run in both implementations. Prefer those tests over this report's prose where the two disagree.

WHAT THE PASSING TESTS LEAVE OUT

The engine, run on cases a scientist would try

Each row was executed against `compare_contracts` or `validate_contract` in the Python reference implementation. The engine result is what the standard returns today; the correct outcome is what the science requires.

PROFILE	CASE	ENGINE RETURNS	CORRECT OUTCOME	FINDING
transcriptome/ single-cell-expression-matrix	Raw read counts vs log1p-normalised expression, declared on both ports	DIRECT_COMPATIBLE	Lossy transformation, approval required	S3
transcriptome/ single-cell-expression-matrix	ensembl.gene vs hgnc.symbol namespaces, declared on both ports	DIRECT_COMPATIBLE	Identifier mapping required; lossy	S3
core/ scalar-quantity	A value in mg vs a value in s, unit declared on both ports	DIRECT_COMPATIBLE	INCOMPATIBLE (different dimension)	S3, S4
neuroscience/ firing-rate	Firing rate in nM vs uM - the profile's own conversion fixture	LOSSLESS_CONVERSION_AVAILABLE	INCOMPATIBLE (a rate has no molar unit)	S7
neuroscience/ firing-rate	Contract declaring a firing rate in kg	valid, no findings	Invalid: unit dimension contradicts quantity	S4
physiology/ body-mass	g vs kg	INCOMPATIBLE	LOSSLESS_CONVERSION_ AVAILABLE (factor 1000)	S4, S8
physiology/ body-temperature	Cel vs K	INCOMPATIBLE	Lossless for absolute values (offset 273.15)	S4
physiology/ heart-rate	1/s vs /min	INCOMPATIBLE	LOSSLESS_CONVERSION_ AVAILABLE (factor 60)	S4
pharmacology/ plasma-concentration	uM vs mol/L	INCOMPATIBLE	LOSSLESS_CONVERSION_ AVAILABLE (factor 10⁶)	S4
neuroscience/ firing-rate	Species 'any' on the source vs NCBITaxon:9606 on the target	INCOMPATIBLE	UNKNOWN (absent evidence, not contradiction)	S5
metabolome/ metabolite-concentration	Subject 'biological_sample' vs 'blood_plasma'	INCOMPATIBLE	Decided by UBERON/OBI subsumption against a pinned snapshot	S2
transcriptome/ single-cell-expression-matrix	dense_vector vs sparse_vector	INCOMPATIBLE	LOSSLESS_CONVERSION_ AVAILABLE	S11
pharmacology/ plasma-concentration	The published positive fixture: a uM concentration on a probability scale	valid, no findings	Invalid: a concentration is ratio-scaled	S6

DECISION REQUESTED ON THE PROVISIONAL IDENTIFIER

What to do with the #concept identifier

Keep a Biosimulant-controlled term, re-minted in a version-independent namespace, and require a mapping to external ontology terms.

TODAY	<code>https://biosimulant.com/standards/model-compatibility/profiles/neuroscience/firing-rate/v0.1#concept</code>
PROPOSED	<code>https://biosimulant.com/standards/model-compatibility/terms/neuroscience/firing-rate</code>
MAPPING	<code>semantic.ontology_terms[]: { uri, ontology, version, relation: exact-match broad-match narrow-match }</code>

REJECT	Keep as is. The equal rule is tautological, since validation already forces both contracts to hold the same constant. And because the IRI embeds /v0.1, every v0.1 port would be INCOMPATIBLE with a v0.2 port whose meaning is unchanged.
REJECT	External identifier only. No ontology defines a Biosimulant contract for a given quantity. Forcing one would overstate equivalence and lose the contract boundary the profile exists to draw.
ADOPT	Controlled term plus mapping. The Biosimulant term keeps profile identity stable across versions; required ontology_terms, compared by subsumption against a pinned snapshot, carry the scientific meaning. With no snapshot the comparison returns UNKNOWN.

Sources: [obo-fp-003](#) [w3c-cooluris](#) [bioregistry-2022](#) [obi-2016](#)

APPLY TO THE GENERATOR, SO THEY APPLY TO EVERY PROFILE IT EMITS

Systemic findings

Each finding gives the exact field, its current definition, the proposed definition, the reason, the supporting sources and the fixtures that must change. All changes belong in `source/catalogue.review.json` or `scripts/build_standard.py`, followed by regeneration; nothing under `spec/vo.1/` should be edited by hand.

S1 BLOCKING

S1. semantic.concept is a version-bound self-reference, so it carries no scientific meaning and breaks across versions

Applies to all 650

FIELD	<code>semantic.concept</code> (fixed value and the equal comparison rule); <code>source: scripts/build_standard.py profile_concept()</code>
CURRENT	Fixed const ' <code><profile ref>/vo.1#concept</code> ', i.e. <code>https://biosimulant.com/.../profiles/<domain>/<name>/vo.1#concept</code> , compared with operator 'equal'. The term has no definition, no label, no axioms and no mapping to any external vocabulary. It is also minted inside the versioned profile document IRI.
PROPOSED	Two changes. (a) Keep a Biosimulant-controlled term but mint it in a version-independent namespace, e.g. <code>https://biosimulant.com/standards/model-compatibility/terms/<domain>/<name></code> , with a label, a textual definition, and its own term-level version; the profile document stays versioned. (b) Because the concept only restates profile identity, add <code>semantic.ontology_terms</code> as a required item for every non-core domain profile: at least one CURIE from a Bioregistry-registered prefix (OBI, UBERON, CL, CHEBI, PATO, NCIT, EFO, GO, SO, MONDO as appropriate), each with ontology, uri, version and relation (exact-match broad-match narrow-match), compared with term-equivalent / term-subsumes against a pinned ontology snapshot, missing snapshot -> UNKNOWN.
REASON	As built, the rule is tautological: validation already forces both contracts to hold the same const, so the equal rule can never fail between two valid contracts of the same profile - it duplicates <code>profile_refs</code> and checks nothing about science. Worse, because the IRI embeds <code>/vo.1</code> , a <code>vo.2</code> profile mints a different concept IRI, so every <code>vo.1</code> port becomes INCOMPATIBLE with a <code>vo.2</code> port even when the scientific meaning is unchanged. Identifier stability across versions is exactly what OBO FP-003 and the W3C hash-URI guidance are for. A compatibility contract that claims to encode scientific meaning must anchor that meaning in a maintained vocabulary, otherwise two independently built ports can never agree except by adopting the same Biosimulant profile.
FIXTURES	Add a cross-version fixture: <code>vo.1 concept</code> vs <code>vo.2 concept</code> of the same term must not be INCOMPATIBLE. Add <code>comparison-unknown-semantic-ontology-terms</code> (no pinned snapshot -> UNKNOWN) and <code>comparison-direct-ontology-term-subsumes</code> (child term -> parent term). Every positive fixture gains a real <code>ontology_terms</code> entry.
SOURCES	<code>obo-fp-003 w3c-cooluris bioregistry-2022 obi-2016 uo-2012</code>

S2 BLOCKING

S2. semantic.subject is uncontrolled free text compared by string equality

Applies to all 650

FIELD	<code>semantic.subject</code> (required item, schema {type: string, minLength: 1, maxLength: 4096}, operator 'equal')
CURRENT	Any string up to 4096 characters. Generated positive fixtures use one placeholder per domain: 'biological_sample', 'nervous_system', 'data_value', 'exposed_biological_system', and so on. Comparison is exact string equality.
PROPOSED	Replace the free string with a CURIE from a registered prefix (UBERON, CL, OBI, ENVO, NCBITaxon, CHEBI as fits the domain), pattern <code>^[A-Za-z][A-Za-zo-9._-]*:[A-Za-zo-9._-]+/\$</code> with the prefix resolvable in a pinned Bioregistry snapshot. Compare with term-subsumes (the target's subject must subsume the source's subject) against a pinned ontology snapshot; no snapshot -> UNKNOWN. Demote to conditional-required: required only where the sampled entity changes the meaning of the value (tissue, cell type, biofluid, environment), not for profiles whose value is entity-independent.

REASON	String equality on uncontrolled text produces both error directions. False INCOMPATIBLE: 'blood_plasma' vs 'plasma' vs 'UBERON:0001969' describe the same specimen and compare unequal (verified against the engine: metabolite-concentration with subject 'biological_sample' vs 'blood_plasma' returns INCOMPATIBLE). False DIRECT_COMPATIBLE: the domain placeholder 'biological_sample' matches itself across plasma, CSF, tumour biopsy and faecal samples, which are not interchangeable for any concentration or abundance measurement. Anatomical and specimen semantics are exactly what UBERON, CL and OBI maintain, including the subsumption hierarchy that makes 'is this sample type acceptable' answerable.
FIXTURES	Replace every positive fixture placeholder with a real CURIE. Add comparison-direct-semantic-subsumes (UBERON:0001969 blood plasma -> UBERON:0000178 blood must not be INCOMPATIBLE when the target is the broader term) and comparison-unknown-semantic-subject-no-snapshot.
SOURCES	uber-on-2012 cl-2016 obi-2016 bioregistry-2022

S3 BLOCKING**S3. Direct compatibility hides required transformations: optional fields present on both contracts are never compared**

Applies to all 650; demonstrated on the 526 profiles that do not require measurement.unit and the 552 that do not require identifiers.namespace

FIELD	<code>Comparison rule generation (scripts/build_standard.py, rules derived only from required_items) and python/src/.../compare.py _rules()</code>
CURRENT	When profile refs are supplied, the engine evaluates only the rules the profile declares, which are generated only from required_items. spec/v0.1/catalogue/items.json states the opposite for all 266 items: 'Checked only when the profile requires it or both contracts include it.'
PROPOSED	Generate comparison rules with missing:'ignore' for every declared-on-both-sides item that can change interpretation, at minimum: measurement.unit (unit-convertible), measurement.scale, measurement.normalization, measurement.baseline, measurement.aggregation, identifiers.namespace and identifiers.namespace_version, representation.ordering, representation.feature_space, dimensions.axes, lifecycle.time_unit, origin.type. missing:'ignore' gives exactly the documented semantics: skip when either side is silent, compare when both declare.
REASON	Verified against the engine on transcriptome/single-cell-expression-matrix: source declaring measurement {quantity: read_count, scale: count} and target declaring {quantity: log1p_normalized_expression, scale: log} returns DIRECT_COMPATIBLE with no finding. Raw counts and log-normalised expression are not interchangeable inputs to any model; feeding one where the other is expected is a silent scientific error, and library-size normalisation plus a log transform is precisely a declared, lossy, non-invertible transformation. The same check returns DIRECT_COMPATIBLE for identifiers.namespace 'ensembl.gene' vs 'hgnc.symbol', which is a many-to-many mapping with real loss. This directly violates the standard's own promise that direct compatibility never hides a required transformation, and it contradicts the published item semantics.
FIXTURES	Every profile that permits a numeric value needs comparison-lossy-normalization (counts vs log-normalised -> LOSSY_CONVERSION_REQUIRES_APPROVAL or INCOMPATIBLE) and comparison-mapping-required (ensembl.gene vs hgnc.symbol -> mapping required, not DIRECT).
SOURCES	wagner-2012-tpm robinson-2010-tmm ensembl-2024 bioregistry-2022

S4 BLOCKING**S4. unit-convertible is a four-row lookup table, not dimensional analysis, and no rule ties a unit to its quantity**

Applies to all 650; the 124 profiles that require measurement.unit are directly affected

FIELD	<code>spec/v0.1/rules/unit-conversions.json</code> and the unit-convertible operator in both implementations
CURRENT	The whole conversion table is nM->uM, uM->nM, mM->uM, uM->mM. measurement.unit is validated only as {type: string, minLength: 1, maxLength: 128}. Verified engine results: g vs kg -> INCOMPATIBLE; Cel vs K -> INCOMPATIBLE; 1/s vs /min -> INCOMPATIBLE; uM vs mol/L -> INCOMPATIBLE; a firing rate declared in kg validates with no finding.

PROPOSED	Adopt UCUM 2.2 as the unit grammar: validate measurement.unit as a parseable UCUM expression, and implement unit-convertible as UCUM dimensional analysis (equal dimension vectors -> convertible; factor from the canonical form). Handle affine units explicitly (Cel/[degF] convert to K with an offset, so only whole quantities convert, never differences or ratios). Constrain each profile's unit to the dimension of its quantity (e.g. amount-of-substance concentration N.L ⁻³ for concentration profiles) and reject dimensionally impossible units. Mass-concentration to molar-concentration (mg/L vs uM) requires an external molar mass and must resolve to a declared parameterised transformation, not DIRECT and not INCOMPATIBLE.
REASON	Every one of those verified results is scientifically wrong. g and kg are the same quantity with a factor of 1000; /min and 1/s are the same quantity; uM and mol/L are the same quantity; Cel and K differ by an additive offset. Returning INCOMPATIBLE for convertible units makes the standard block valid couplings, and accepting kg for a firing rate makes it approve impossible ones. UCUM exists to answer exactly this question and is the unit code system used by HL7/FHIR and LOINC.
FIXTURES	Add per-dimension conversion fixtures: g<->kg and s<->min and 1/s<->/min lossless; Cel<->K lossless for absolute values but flagged for differences; uM<->mol/L lossless; mg/L vs uM requires a declared molar-mass parameter; kg vs Hz INCOMPATIBLE. Remove the blanket nM->uM fixture from profiles whose quantity is not a molar concentration.
SOURCES	ucum-2.2 si-brochure-9 vim-jcgm-200-2012 iso-80000-1-2022 uc-2012

S5 BLOCKING

S5. biological_context.species is a single taxon compared by string equality, with no subsumption and an unsafe wildcard direction

Applies to 575 profiles requiring species; the 75 without it (core, chemical, simulation) are unaffected

FIELD	<code>biological_context.species</code> (required in 575 profiles), operator 'context-compatible'
CURRENT	Schema: 'any' or ^NCBITaxon:[1-9][0-9]*\$, one value. The operator is implemented as source == target or target in (None, 'any', 'unspecified'). Verified: source 'any' vs target NCBITaxon:9606 -> INCOMPATIBLE; source NCBITaxon:9606 vs target 'any' -> DIRECT_COMPATIBLE. Every generated positive fixture uses NCBITaxon:9606, including ecology, evolution, microbiology and virology profiles.
PROPOSED	(a) Compare with term-subsumes over a pinned NCBI Taxonomy snapshot so that a strain or subspecies satisfies a port that asks for the species (NCBITaxon:511145 E. coli K-12 MG1655 satisfies NCBITaxon:562 E. coli), and so a broader source does not silently satisfy a narrower target. (b) A source of 'any' against a specific target must return UNKNOWN, not INCOMPATIBLE - it is absent evidence, not a contradiction. (c) Replace the single value with a role-typed set for profiles where more than one organism is in scope: <code>biological_context.taxa[]</code> with {taxon, role: host pathogen community_member donor}, required for virology, microbiology, epidemiology and host-pathogen profiles. (d) For community-level profiles (ecology/species-abundance, microbiology/taxonomic-abundance, microbiology/community-composition and similar), a single species is the wrong cardinality: require the taxonomy namespace and release plus a feature universe, and make species conditional.
REASON	Absent evidence and contradiction are different states, and the current asymmetry gets both backwards in the unsafe direction: a port that declares nothing about species is allowed to feed a port that requires human. Equality without the taxonomic hierarchy blocks correct couplings between strain-level and species-level declarations, which is the normal case in microbiology. And a human taxon on an ecology abundance vector or a viral titre is not a valid scientific example: for virology and microbiology the host and the organism measured are different entities, so one species field cannot express either.
FIXTURES	Rebuild positive fixtures per domain with a plausible taxon (virology: pathogen NCBITaxon:2697049 with host NCBITaxon:9606; ecology: a real community with a taxonomy release). Add comparison-unknown-species-any-source, comparison-direct-species-strain-satisfies-species, and keep comparison-incompatible-species for a genuine cross-species contradiction.
SOURCES	ncbitaxon-schoch-2020 bioregistry-2022 obi-2016

S6 BLOCKING

S6. Unit, scale and axis values in the generated fixtures are chosen by substring matching on the profile name

Applies to 124 profiles requiring measurement.unit; 89 profiles requiring dimensions.axes

FIELD	<code>scripts/build_standard.py measurement_unit(), example_value() scale branch, axis_names()</code>
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CURRENT	Unit and scale are selected by testing whether the profile name contains a token. 'concentration' contains the substring 'ratio', so all 14 concentration profiles get scale 'probability'. 'integrated-embedding' contains 'rate', so it gets unit 1/s. 'proliferation-rate', 'migration-rate' and 'glomerular-filtration-rate' contain 'ratio' and get scale 'probability'. Axis names come from the same mechanism: any name containing 'volume' gets image axes [x, y], so stroke-volume and volume-of-distribution get spatial axes, while ct-volume and mri-volume - which are three-dimensional - get only two.
PROPOSED	Delete the substring heuristics. Declare quantity kind, unit dimension, measurement scale and axis structure explicitly per profile in source/catalogue.review.json, reviewed one profile at a time. Separate Stevens' measurement level (nominal ordinal interval ratio) from the value domain and transform: add measurement.transform (identity log2 log10 ln logit) and use 'proportion' for bounded [0,1] fractions, reserving 'probability' for values that are probabilities of an event. The log base must be explicit, because log2 and log10 fold changes are different numbers.
REASON	A generated example that is scientifically wrong is worse than no example, because downstream implementers copy it: the positive fixture is the profile's worked example of a valid contract. A micromolar concentration is a ratio-scale quantity with a true zero, not a probability; a fraction bounded in [0,1] is a proportion, which is not the same claim as a probability; and Celsius is an interval scale, so 'ratio' is wrong for body temperature even though the unit is right. Stevens' levels also determine which operations are meaningful, so the scale value is not cosmetic - it decides whether averaging or ratio comparison is defensible.
FIXTURES	Every positive fixture for the 124 measurement profiles must be regenerated from reviewed per-profile values; see the per-profile unit and scale corrections.
SOURCES	stevens-1946 vim-jcgm-200-2012 ucum-2.2 iso-80000-1-2022

S7 BLOCKING

S7. The single transformation fixture asserts a molar-concentration conversion for every measurement profile

Applies to 124 profiles requiring measurement.unit

FIELD	<code>scripts/build_standard.py</code> fixture generation; fixture comparison-lossless-unit-conversion in 124 profiles
CURRENT	All 124 profiles that require measurement.unit receive the same fixture: source unit nM, target unit uM, expected LOSSLESS_CONVERSION_AVAILABLE - including firing-rate, body-temperature, heart-rate, simulation-time, retention-time, body-mass and polygenic-score. Verified: neuroscience/firing-rate with nM vs uM returns LOSSLESS_CONVERSION_AVAILABLE.
PROPOSED	Generate the conversion fixture from the profile's own declared unit dimension (mass: g<->kg; time: s<->min; frequency: 1/s<->/min; molar concentration: nM<->uM; temperature: Cel<->K flagged as affine), and add a dimensional-mismatch fixture (the profile's unit vs a unit of another dimension -> INCOMPATIBLE).
REASON	The fixture certifies, as a passing conformance test in both implementations, that a neuronal firing rate can be converted from nanomolar to micromolar. It is dimensionally impossible. Because the fixtures are the cross-language conformance suite, this false claim is what the Python and TypeScript packages are being tested to agree on.
FIXTURES	Replace comparison-lossless-unit-conversion in all 124 profiles; add comparison-incompatible-unit-dimension.
SOURCES	ucum-2.2 vim-jcgm-200-2012 iso-80000-1-2022

S8 BLOCKING**S8. The unit contradiction fixture asserts that g and kg are incompatible**

Applies to 6 profiles with unit g; the pattern is wrong for any profile whose unit shares a dimension with kg

FIELD	<code>scripts/build_standard.py incompatible_value(); fixture comparison-incompatible-measurement-unit</code>
CURRENT	The contradiction unit is hard-coded as 'kg' (or 's' when the profile's unit is already kg). For the six profiles whose unit is g - physiology/body-mass, ecology/biomass, microbiology/biofilm-biomass, chemical/exact-mass, metabolome/mass-spectral-peak, metabolome/mass-spectrum - the fixture therefore asserts that kg vs g must be INCOMPATIBLE. Verified: the engine returns INCOMPATIBLE.
PROPOSED	Choose the contradiction unit as one of a different dimension from the profile's own unit (for a mass quantity use s or Hz, never another mass unit), and add the matching g<->kg lossless conversion fixture.
REASON	Kilograms and grams are the same quantity; a contract expressed in one converts exactly to the other. Certifying that pairing as a contradiction means the standard blocks a correct coupling and, being a fixture, freezes the error into the conformance suite.
FIXTURES	Fix comparison-incompatible-measurement-unit for the 6 mass profiles; add comparison-lossless g<->kg.
SOURCES	ucum-2.2 si-brochure-9

S9 BLOCKING**S9. No profile has a lossy or inference fixture, although the transformation policy declares both paths**

Applies to all 650

FIELD	<code>transformation_policy {lossless: allow, lossy: approval, inference: approval}; fixture_groups.transformations</code>
CURRENT	526 profiles have an empty transformations group and the remaining 124 have exactly one lossless fixture. No profile anywhere exercises LOSSY_CONVERSION_REQUIRES_APPROVAL or the inference path, so the approval branches of the policy are untested in both implementations.
PROPOSED	Require, for every profile whose transformation_policy admits them, at least one lossy fixture (e.g. continuous value binned to an ordinal category; log-normalised from counts) and one inference fixture (a value that can only be produced by a model, e.g. imputed or deconvolved), each with the expected approval status.
REASON	The review instruction that lossy conversions and inference must be clearly identified cannot be verified for any profile in the release. An untested approval path is where silent scientific substitution happens, because the failure mode is a status that quietly reads as compatible.
FIXTURES	Add lossy and inference fixtures to all 650 profiles.
SOURCES	wagner-2012-tpm robinson-2010-tmm svensson-2020-zeros

S10 NON-BLOCKING**S10. UNKNOWN is only tested for a missing source; a missing target is never exercised**

Applies to all 650

FIELD	<code>fixture comparison-unknown-<field></code>
CURRENT	Each unknown fixture removes the field from the source contract only. The engine treats either side missing as UNKNOWN, but no fixture pins that behaviour for the target side.
PROPOSED	Add the mirrored fixture (field absent from the target) for each required field.
REASON	Missing evidence must return UNKNOWN regardless of which port is silent; a one-sided test lets an asymmetric regression through. The behaviour is currently correct in both implementations, so this is a coverage gap rather than a defect.
FIXTURES	Add comparison-unknown-target-<field> for every required field.
SOURCES	vim-jcgm-200-2012

S11 NON-BLOCKING**S11. representation.kind uses exact equality, so lossless re-encodings are reported as contradictions**

Applies to all 650; the allowed-set contradiction affects the 395 profiles carrying the full 8-kind set

FIELD	representation.kind (required in all 650), operator 'equal'
CURRENT	Verified: dense_vector vs sparse_vector on transcriptome/single-cell-expression-matrix returns INCOMPATIBLE. Allowed kind sets are derived from the coarse applies_to families, so core/scalar-quantity permits matrix, tensor and table.
PROPOSED	Declare lossless representation transformations (dense_vector <-> sparse_vector, matrix <-> table with a declared schema) and return LOSSLESS_CONVERSION_AVAILABLE; narrow each profile's allowed kinds to those its scientific meaning admits (core/scalar-quantity should allow scalar only).
REASON	Sparse and dense encodings of the same vector carry identical information, so INCOMPATIBLE is a false negative. It is the safe direction of error, which is why this is not blocking on its own - but for scalar-quantity the allowed set contradicts the profile's own name and intended use.
FIXTURES	Add comparison-lossless-representation-kind; fix the allowed-kind enum for core/scalar-quantity and the other scalar profiles.
SOURCES	edam-2013

S12 BLOCKING**S12. Identifier, consent and data-use fixtures carry placeholder values, and no namespace is constrained to a real registry**

Applies to 98 profiles with identifiers; 50 clinical profiles with security fields

FIELD	identifiers.namespace, identifiers.namespace_version (98 profiles); security.data_use, security.consent_scope (50 clinical profiles)
CURRENT	Positive fixtures use 'example-namespace', 'example-namespace-version', 'example-consent-scope' and ['example-data_use']. The schemas accept any string; data_use is compared with 'equal' on an array, so element order changes the result.
PROPOSED	Constrain identifiers.namespace to a Bioregistry-registered prefix resolved against a pinned registry snapshot, and namespace_version to the namespace's own release identifier (e.g. ensembl.gene with release 114). Replace security.data_use with DUO CURIEs (DUO:0000042 general research use and similar), normalise as a set, and compare by DUO subsumption: the target's permitted uses must subsume the source's declared use. Give consent_scope the same treatment.
REASON	A positive fixture is the normative worked example; a placeholder namespace teaches implementers that any string is acceptable, which defeats the purpose of requiring a versioned identifier namespace. For the clinical profiles the consequence is governance, not just tidiness: order-sensitive equality on a data-use array can wrongly block a lawful data flow or, when the arrays happen to match, pass one that consent does not cover. DUO exists as the GA4GH standard for exactly this comparison.
FIXTURES	Replace all placeholder identifier and security values with real registered values; add set-order and DUO-subsumption comparison fixtures.
SOURCES	bioregistry-2022 ensembl-2024 duo-2021

S13 NON-BLOCKING**S13. Uncertainty, detection limits and censoring are never required, though they change whether values are comparable**

Applies to conditional across the 124 measurement profiles, matrix profiles and survival profiles

FIELD	<code>uncertainty.*</code> and <code>measurement.detection_limits</code> , currently candidate-recommended only
CURRENT	No profile requires any uncertainty item. missingness appears as a candidate in 475 packets, censoring in 400, <code>detection_limits</code> in 450.
PROPOSED	Make them conditional-required where the science demands it: <code>uncertainty.censoring</code> for phenotype/time-to-event and clinical/survival-outcome; <code>measurement.detection_limits</code> for every concentration, titre and MIC profile; <code>uncertainty.missingness</code> for every matrix profile, distinguishing a structural zero from an unobserved value.
REASON	A survival time without its censoring indicator is not interpretable and will be silently treated as an observed event. A concentration below the lower limit of quantification is not a measurement of zero, which is why bioanalytical guidance requires the limits to travel with the data. In single-cell matrices, zeros are mostly genuine low counts rather than missing values, so a port that treats them as missing and one that treats them as zero are not interchangeable.
FIXTURES	Add missing-evidence fixtures (censoring absent -> UNKNOWN) for the affected profiles.
SOURCES	clark-2003-survival fda-bmv-2018 svensson-2020-zeros

S14 NON-BLOCKING**S14. Time-series and event profiles do not require a time unit, time origin or sampling rule**

Applies to 25 profiles with a time axis; 76 event/state profiles

FIELD	<code>lifecycle.time_unit</code> , <code>lifecycle.time_origin</code> , <code>lifecycle.sampling</code> , <code>lifecycle.interpolation</code> (candidates in 350 packets); <code>dimensions.axes ['time']</code> in 25 profiles
CURRENT	A profile can require a time axis while leaving the time unit, the origin of the clock and the sampling regime undeclared. <code>core/regular-time-series</code> and <code>core/irregular-time-series</code> differ in exactly this respect yet impose the same requirements.
PROPOSED	For every profile whose axes include time, require <code>lifecycle.time_unit</code> (UCUM), <code>lifecycle.time_origin</code> , and <code>lifecycle.sampling</code> ; require <code>lifecycle.interpolation</code> for irregular series; and record the domain's own conventions where they exist (BIDS/NWB for electrophysiology).
REASON	Two time series that agree on every declared field can still be a 1 kHz recording and a daily series. Sampling rate and epoch are part of what makes time-indexed values comparable, and the electrophysiology community already standardises them.
FIXTURES	Add comparison-unknown-lifecycle-time-unit and a sampling-mismatch fixture for the time-axis profiles.
SOURCES	bids-ieeg-2019 nwb-2022 ucum-2.2

FINDINGS THAT BELONG TO INDIVIDUAL PROFILES

Profile-specific findings

136 profiles carry at least one finding of their own, and every one of them is BLOCK: P-MEAS (unit or scale, 124), P-CONV (conversion fixture, 124), P-KG (g vs kg fixture, 6) and P-AXES (axes or representation, 22). A non-blocking P-MEAS means the declared unit and scale are acceptable but under-specified; the profile is still blocked by its conversion fixture. The remaining profiles have no finding of their own and are CHANGES_REQUIRED because of the systemic blockers.

P-CONV is shown as current and proposed only, because its reason is the same everywhere: the fixture certifies a nanomolar-to-micromolar conversion regardless of the quantity the profile measures, and both implementations are tested to agree on it (S7). For the 14 molar-concentration profiles the conversion is dimensionally right, but the fixture pair inherits the wrong probability scale from the positive example. Fixture to change: comparison-lossless-unit-conversion. Sources: ucum-2.2, vim-jcgm-200-2012.

cardiopulmonary-renal 3 profiles

Ejection Fraction cardiopulmonary-renal/ejection-fraction@0.1 BLOCK

P-MEAS	Declared scale wrong for this quantity
FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale probability
PROPOSED	unit 1 or %; scale proportion
REASON	Ejection fraction is a bounded proportion of end-diastolic volume, not a probability; the imaging modality must be declared because EF is modality-dependent.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV	Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity
CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 or %)

Pressure Waveform cardiopulmonary-renal/pressure-waveform@0.1 BLOCK

P-MEAS	Unit and scale acceptable but under-specified
FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit mm[Hg], scale ratio
PROPOSED	unit mm[Hg]; scale ratio
REASON	Unit correct; the waveform needs its sampling rate and time unit, and the measurement site.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV	Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity
CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: mm[Hg])

Ventricular Volume cardiopulmonary-renal/ventricular-volume@0.1 **BLOCK****P-MEAS** Unit and scale acceptable but under-specified

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit mL, scale ratio
PROPOSED	unit mL; scale ratio
REASON	Unit correct, but this is a scalar or a time series of scalars, not an image with [x, y] axes.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: mL)

P-AXES Declared axes or representation contradict the data

FIELD	dimensions.axes
CURRENT	[x, y]
PROPOSED	no spatial axes; scalar per cardiac phase or a time series
REASON	Ventricular volume is a scalar per phase.
FIXTURES	positive, comparison-direct
SOURCES	ucum-2.2

cell 4 profiles**Growth Rate** cell/growth-rate@0.1 **BLOCK****P-MEAS** Unit and scale acceptable but under-specified

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit 1/h; scale ratio
REASON	Correct dimension; the profile should state that it is the specific (per-capita) growth rate rather than an absolute increase.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1/h)

Invasion Score cell/invasion-score@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ordinal
PROPOSED	unit 1; scale interval
REASON	An assay-derived invasion score is continuous and assay-specific; ordinal understates it and the assay must be declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Migration Rate cell/migration-rate@0.1 **BLOCK****P-MEAS Declared unit and scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale probability
PROPOSED	unit um/min for speed, or 1/h for a transition rate; scale ratio
REASON	Cell migration is normally a speed (length per time); as declared, the unit fits only a transition rate and the scale is wrong.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: um/min for speed, or 1/h for a transition rate)

Proliferation Rate cell/proliferation-rate@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale probability
PROPOSED	unit 1/h or 1/d; scale ratio
REASON	A proliferation rate is per-capita per time on a ratio scale; 'probability' comes from the substring bug and would license treating a rate as a bounded [0,1] value.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946 ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1/h or 1/d)

chemical 2 profiles

Exact Mass chemical/exact-mass@0.1 **BLOCK**

P-MEAS Unit and scale acceptable but under-specified

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit g, scale ratio
PROPOSED	unit Da (u); scale ratio
REASON	Exact monoisotopic mass is reported in daltons; grams is dimensionally valid but off by 24 orders of magnitude from any value a model will carry.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	goldbook-mr ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: Da (u))

P-KG Contradiction fixture treats g and kg as incompatible

FIELD	fixture comparison-incompatible-measurement-unit
CURRENT	source kg vs target g, expected INCOMPATIBLE
PROPOSED	a contradiction unit of another dimension (e.g. s), plus a g<->kg lossless fixture
REASON	Grams and kilograms are the same quantity; certifying them as contradictory blocks a correct coupling.
FIXTURES	comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 si-brochure-9

Molecular Weight chemical/molecular-weight@0.1 **BLOCK**

P-MEAS Unit and scale acceptable but under-specified

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit g/mol for molar mass, or 1 for relative molecular mass; scale ratio
REASON	Relative molecular mass is dimensionless and molar mass is g/mol; they are numerically equal but different quantities, so the profile must declare which it means.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	goldbook-mr iso-80000-1-2022

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: g/mol for molar mass, or 1 for relative molecular mass)

core 6 profiles**Bounded Score** core/bounded-score@0.1 **BLOCK****P-MEAS** Declared scale wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ordinal
PROPOSED	unit 1; scale interval
REASON	A bounded score is a dimensionless value on a declared interval; its level must be declared per instrument rather than assumed ordinal, and the bounds belong in the contract.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Categorical Value core/categorical-value@0.1 **BLOCK****P-MEAS** Declared unit and scale wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ordinal
PROPOSED	unit (no unit; remove measurement requirement); scale nominal
REASON	A categorical value has no unit and no order; 'ordinal' asserts a ranking the profile does not define, and unit '1' asserts a dimensionless magnitude.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946 vim-jcgm-200-2012

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: (no unit; remove measurement requirement))

Irregular Time Series core/irregular-time-series@0.1 **BLOCK****P-MEAS** Declared unit wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit s, scale ratio
PROPOSED	unit (value unit, profile-declared); scale ratio
REASON	As for the regular series: 's' is the axis unit. An irregular series must additionally declare its interpolation rule.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 vim-jcgm-200-2012

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: (value unit, profile-declared))

Regular Time Series `core/regular-time-series@0.1` **BLOCK****P-MEAS** **Declared unit wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit s, scale ratio
PROPOSED	unit (value unit, profile-declared); scale ratio
REASON	Unit 's' describes the time axis, not the values. The series values carry their own quantity and unit; the sampling interval belongs in lifecycle.time_unit.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 vim-jcgm-200-2012

P-CONV **Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity**

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: (value unit, profile-declared))

Scalar Quantity `core/scalar-quantity@0.1` **BLOCK****P-AXES** **Allowed representations contradict a scalar quantity**

FIELD	representation.kind (allowed enum)
CURRENT	scalar, dense_vector, sparse_vector, matrix, tensor, array, record, table
PROPOSED	representation.kind restricted to scalar
REASON	The allowed set admits matrix, tensor and table, contradicting the profile's own name and intended use.
FIXTURES	positive, comparison-direct
SOURCES	vim-jcgm-200-2012

Tensor `core/tensor@0.1` **BLOCK****P-AXES** **Declared axes or representation contradict the data**

FIELD	dimensions.axes
CURRENT	[feature]
PROPOSED	n named axes with per-axis meaning, size and order
REASON	A tensor by definition has an arbitrary number of axes; one 'feature' axis cannot describe it.
FIXTURES	positive, comparison-direct
SOURCES	edam-2013

developmental 1 profiles**Morphogen Concentration** `developmental/morphogen-concentration@0.1` **BLOCK****P-MEAS** **Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit uM, scale probability
PROPOSED	unit uM or nM; scale ratio
REASON	Unit acceptable, scale wrong; a morphogen gradient may also need a spatial basis.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion

CURRENT source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE

PROPOSED keep nM<->uM, but only after the positive fixture's scale is corrected to ratio

ecology 6 profiles**Biomass** ecology/biomass@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD measurement.unit, measurement.scale (profile requirement and positive fixture)

CURRENT unit g, scale ratio

PROPOSED unit g, or g/m2 for standing stock; scale ratio

REASON Ecological biomass is normally per unit area or volume; a bare mass loses the basis. Dry versus wet mass must also be declared.

FIXTURES positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit

SOURCES [ucum-2.2](#)**P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity**

CURRENT source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE

PROPOSED a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: g, or g/m2 for standing stock)

P-KG Contradiction fixture treats g and kg as incompatible

FIELD fixture comparison-incompatible-measurement-unit

CURRENT source kg vs target g, expected INCOMPATIBLE

PROPOSED a contradiction unit of another dimension (e.g. s), plus a g<->kg lossless fixture

REASON Grams and kilograms are the same quantity; certifying them as contradictory blocks a correct coupling.

FIXTURES comparison-incompatible-measurement-unit

SOURCES [ucum-2.2](#) [si-brochure-9](#)**Birth Rate** ecology/birth-rate@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD measurement.unit, measurement.scale (profile requirement and positive fixture)

CURRENT unit 1/s, scale ratio

PROPOSED unit 1/d or 1/a, per capita; scale ratio

REASON Per-capita versus absolute birth rate must be declared; the dimension is right.

FIXTURES positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit

SOURCES [ucum-2.2](#)**P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity**

CURRENT source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE

PROPOSED a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1/d or 1/a, per capita)

Climate Time Series ecology/climate-time-series@0.1 **BLOCK****P-MEAS Declared unit wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit s, scale ratio
PROPOSED	unit (value unit, profile-declared); scale ratio
REASON	's' is the time axis unit, not the unit of temperature, precipitation or radiation values.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: (value unit, profile-declared))

Death Rate ecology/death-rate@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit 1/d or 1/a, per capita; scale ratio
REASON	As for birth rate.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1/d or 1/a, per capita)

Predator Prey Rate ecology/predator-prey-rate@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit 1/d per predator, or a declared functional-response form; scale ratio
REASON	Attack and consumption rates depend on the functional response; the rate basis must be declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1/d per predator, or a declared functional-response form)

Species Abundance ecology/species-abundance@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1 (counts) or a declared density unit; scale count
REASON	Abundance is a count or a density per area or volume; the profile must declare which, and a community vector cannot be described by one NCBITaxon species.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ncbitaxon-schoch-2020

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 (counts) or a declared density unit)

epidemiology 7 profiles**Attack Rate** epidemiology/attack-rate@0.1 **BLOCK****P-MEAS Declared unit and scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit 1; scale proportion
REASON	An attack rate is a proportion of a population at risk over an outbreak period - a risk, not a rate per unit time. 1/s is the wrong dimension.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	cdc-epi-3ed

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Contact Matrix epidemiology/contact-matrix@0.1 **BLOCK****P-AXES Declared axes or representation contradict the data**

FIELD	dimensions.axes
CURRENT	[sample, feature]
PROPOSED	[age_group, age_group] with the contact definition and setting
REASON	A contact matrix is square over population strata.
FIXTURES	positive, comparison-direct
SOURCES	cdc-epi-3ed

Hospitalization Rate epidemiology/hospitalization-rate@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit 1/{person-year}, or 1 as a proportion of cases; scale ratio or proportion
REASON	The term covers both a rate per person-time and a risk (proportion of cases); the profile must declare which.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	cdc-epi-3ed

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1/{person-year}, or 1 as a proportion of cases)

Incidence Rate epidemiology/incidence-rate@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit 1/{person-year} or 1/{person-day}; scale ratio
REASON	An incidence rate is events per person-time; the person-time denominator is part of the quantity and 1/s omits it.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	cdc-epi-3ed

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1/{person-year} or 1/{person-day})

Mobility Matrix epidemiology/mobility-matrix@0.1 **BLOCK****P-AXES Declared axes or representation contradict the data**

FIELD	dimensions.axes
CURRENT	[sample, feature]
PROPOSED	[origin_location, destination_location] with the geography and its version
REASON	A mobility matrix is origin x destination over places.
FIXTURES	positive, comparison-direct
SOURCES	cdc-epi-3ed

Mortality Rate epidemiology/mortality-rate@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit 1/{person-year}; scale ratio
REASON	As for incidence rate; crude and age-standardised rates must be distinguished.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	cdc-epi-3ed

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1/{person-year})

Recovery Rate epidemiology/recovery-rate@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit 1/d; scale ratio
REASON	Dimension correct; for a compartmental model this is the reciprocal of the infectious period and that interpretation should be declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	cdc-epi-3ed

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1/d)

epigenome 6 profiles**Chromatin Contact Matrix** epigenome/chromatin-contact-matrix@0.1 **BLOCK****P-AXES Declared axes or representation contradict the data**

FIELD	dimensions.axes
CURRENT	[sample, feature]
PROPOSED	[genomic_bin, genomic_bin] with bin size, assembly and coordinate convention
REASON	A contact matrix is square over genomic bins.
FIXTURES	positive, comparison-direct
SOURCES	lieberman-aiden-2009

DNA Methylation Fraction epigenome/dna-methylation-fraction@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale probability
PROPOSED	unit 1; scale proportion
REASON	A methylation fraction is a bounded proportion of methylated molecules, not the probability of an event.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	du-2010-mvalue stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Footprint Score epigenome/footprint-score@0.1 **BLOCK****P-MEAS** Declared scale wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ordinal
PROPOSED	unit 1; scale interval
REASON	A footprint score is a continuous algorithm output; its level and its producing method must be declared rather than assumed ordinal.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Hi-C Contact Map epigenome/hic-contact-map@0.1 **BLOCK****P-AXES** Declared axes or representation contradict the data

FIELD	dimensions.axes
CURRENT	[sample, feature]
PROPOSED	[genomic_bin, genomic_bin] with bin size, assembly and normalisation method
REASON	As for the contact matrix; the balancing/normalisation method must be declared.
FIXTURES	positive, comparison-direct
SOURCES	lieberman-aiden-2009

Methylation Beta Value epigenome/methylation-beta-value@0.1 **BLOCK****P-MEAS** Declared scale wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1; scale proportion
REASON	Beta is bounded in [0,1] and heteroscedastic; declaring it ratio-scaled invites averaging and ratio comparison that the beta scale does not support.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	du-2010-mvalue

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Methylation M Value epigenome/methylation-m-value@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1; scale interval with transform log2
REASON	The M-value is $\log_2(\beta/(1-\beta))$: unbounded, can be negative, so it is not ratio-scaled, and the log base must be declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	du-2010-mvalue

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

evolution 6 profiles**Adaptation Score** evolution/adaptation-score@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ordinal
PROPOSED	unit 1; scale interval
REASON	A continuous statistic; the test and null model must be declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Divergence Time evolution/divergence-time@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit s, scale ratio
PROPOSED	unit a (years), Ma; scale ratio
REASON	Seconds is dimensionally correct but never used; the calibration and the molecular clock model must be declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: a (years), Ma)

Evolutionary Distance evolution/evolutionary-distance@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1 (substitutions per site); scale ratio
REASON	Dimensionless is right, but the substitution model that produced the distance must be declared, because distances from different models are not comparable.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	lynch-2010-mutrate

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 (substitutions per site))

Fitness Value evolution/fitness-value@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1; scale ratio
REASON	Relative fitness is dimensionless; the reference genotype must be declared, since fitness is only defined relative to one.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Mutation Rate evolution/mutation-rate@0.1 **BLOCK****P-MEAS Declared unit wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit 1 per site per generation, or 1/a; scale ratio
REASON	Mutation rates are per site per generation (or per genome per generation); 1/s has no site or generation basis.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	lynch-2010-mutrate ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 per site per generation, or 1/a)

Recombination Rate evolution/recombination-rate@0.1 **BLOCK****P-MEAS Declared unit wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit cM/Mb, or 1 per bp per generation; scale ratio
REASON	Recombination rate is per physical distance per generation; 1/s is not that quantity.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: cM/Mb, or 1 per bp per generation)

genome 1 profiles**Polygenic Score** genome/polygenic-score@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ordinal
PROPOSED	unit 1; scale interval
REASON	A polygenic score is a continuous, usually standardised weighted sum; it is not ordinal, and its value is only interpretable with the scoring file and allele-weight provenance.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	pgs-catalog-2021 stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

imaging 4 profiles**CT Volume** imaging/ct-volume@0.1 **BLOCK****P-MEAS Declared unit and scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit mL, scale ratio
PROPOSED	unit [hnsf'U] (Hounsfield units); scale interval
REASON	A CT volume's voxel values are Hounsfield units on an interval scale (water = 0, air = -1000), not millilitres; mL is the unit of a measured volume, a different quantity. Three spatial axes plus pixel spacing and slice thickness are required.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	dicom-ps3.3

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: [hnsf'U] (Hounsfield units))

P-AXES Declared axes or representation contradict the data

FIELD	dimensions.axes
CURRENT	[x, y]
PROPOSED	[x, y, z] with pixel spacing, slice thickness and patient orientation
REASON	A CT volume is three-dimensional; two axes cannot index a voxel grid.
FIXTURES	positive, comparison-direct
SOURCES	dicom-ps3.3

Image Time Series [imaging/image-time-series@0.1](#) **BLOCK****P-MEAS Declared unit wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit s, scale ratio
PROPOSED	unit (voxel value unit, profile-declared); scale ratio
REASON	's' is the frame interval, not the voxel unit.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	dicom-ps3.3

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: (voxel value unit, profile-declared))

MRI Volume [imaging/mri-volume@0.1](#) **BLOCK****P-MEAS Declared unit and scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit mL, scale ratio
PROPOSED	unit 1 (arbitrary intensity) unless a quantitative map is declared; scale interval
REASON	MRI voxel intensities are arbitrary unless the sequence yields a quantitative map (T1 in ms, ADC in mm ² /s); mL is wrong, and three spatial axes are required.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	dicom-ps3.3

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 (arbitrary intensity) unless a quantitative map is declared)

P-AXES Declared axes or representation contradict the data

FIELD	dimensions.axes
CURRENT	[x, y]
PROPOSED	[x, y, z] (plus echo/coil dimensions where present)
REASON	An MRI volume is three-dimensional.
FIXTURES	positive, comparison-direct
SOURCES	dicom-ps3.3

PET Volume `imaging/pet-volume@0.1` **BLOCK****P-MEAS** **Declared unit wrong for this quantity**

FIELD	<code>measurement.unit, measurement.scale</code> (profile requirement and positive fixture)
CURRENT	<code>unit mL, scale ratio</code>
PROPOSED	<code>unit Bq/mL, or 1 for SUV; scale ratio</code>
REASON	PET voxels are activity concentration or a standardised uptake value; SUV additionally requires the body-weight basis and the uptake time.
FIXTURES	<code>positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit</code>
SOURCES	boellaard-2015-suv dicom-ps3.3

P-CONV **Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity**

CURRENT	<code>source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE</code>
PROPOSED	<code>a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: Bq/mL, or 1 for SUV)</code>

P-AXES **Declared axes or representation contradict the data**

FIELD	<code>dimensions.axes</code>
CURRENT	<code>[x, y]</code>
PROPOSED	<code>[x, y, z] plus frame time for dynamic studies</code>
REASON	A PET volume is three-dimensional and often four-dimensional.
FIXTURES	<code>positive, comparison-direct</code>
SOURCES	dicom-ps3.3 boellaard-2015-suv

immunology 7 profiles**Antibody Titer** `immunology/antibody-titer@0.1` **BLOCK****P-MEAS** **Unit and scale acceptable but under-specified**

FIELD	<code>measurement.unit, measurement.scale</code> (profile requirement and positive fixture)
CURRENT	<code>unit 1, scale ratio</code>
PROPOSED	<code>unit 1 (reciprocal dilution) or IU/mL, BAU/mL; scale ordinal with transform log2 for dilution titres; ratio for standardised units</code>
REASON	A dilution titre is a reciprocal dilution on a two-fold series, so it is log-spaced and not ratio-scaled; only WHO-standardised units are ratio-scaled and comparable across assays.
FIXTURES	<code>positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit</code>
SOURCES	who-is-antibody-2021 stevens-1946

P-CONV **Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity**

CURRENT	<code>source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE</code>
PROPOSED	<code>a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 (reciprocal dilution) or IU/mL, BAU/mL)</code>

Chemokine Concentration immunology/chemokine-concentration@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit uM, scale probability
PROPOSED	unit pg/mL or uM; scale ratio
REASON	As for cytokine concentration.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	keep nM<->uM, but only after the positive fixture's scale is corrected to ratio

Cytokine Concentration immunology/cytokine-concentration@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit uM, scale probability
PROPOSED	unit pg/mL or uM; scale ratio
REASON	Cytokines are usually reported as mass concentration; scale is ratio, not probability.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	keep nM<->uM, but only after the positive fixture's scale is corrected to ratio

Exhaustion Score immunology/exhaustion-score@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ordinal
PROPOSED	unit 1; scale interval
REASON	A signature score is a continuous method output; the gene set and method must be declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Immune Cell Fraction immunology/immune-cell-fraction@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale probability
PROPOSED	unit 1; scale proportion
REASON	A cell fraction is a bounded proportion of a declared parent population; the denominator (the gating parent) must be required, because the same number means different things under different gates.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Inflammatory Score immunology/inflammatory-score@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ordinal
PROPOSED	unit 1; scale interval
REASON	As for the exhaustion score.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Neutralization Titer immunology/neutralization-titer@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1 (reciprocal dilution, e.g. ID50/NT50) or IU/mL; scale ordinal with transform log2; ratio for standardised units
REASON	Neutralisation titres are assay-specific and only comparable after calibration to an international standard.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	who-is-antibody-2021 reed-muench-1938

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 (reciprocal dilution, e.g. ID50/NT50) or IU/mL)

lipidome-glycome 4 profiles

Glycan Abundance lipidome-glycome/glycan-abundance@0.1 **BLOCK**

P-MEAS Unit and scale acceptable but under-specified

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1 (arbitrary intensity) or a declared molar unit; scale ratio with declared normalization
REASON	Relative glycan abundance is normally a within-sample proportion of the glycan pool; that normalisation must be declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	mzml-2011

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 (arbitrary intensity) or a declared molar unit)

Glycopeptide Abundance lipidome-glycome/glycopeptide-abundance@0.1 **BLOCK**

P-MEAS Unit and scale acceptable but under-specified

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1 (arbitrary intensity), or site occupancy as a proportion; scale ratio with declared normalization
REASON	Glycopeptide abundance and site occupancy are different quantities with different domains.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	mzml-2011

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 (arbitrary intensity), or site occupancy as a proportion)

Lipid Abundance lipidome-glycome/lipid-abundance@0.1 **BLOCK**

P-MEAS Unit and scale acceptable but under-specified

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1 (arbitrary intensity) or a declared molar unit; scale ratio with declared normalization
REASON	Lipidomics abundances are intensities unless a standard-calibrated molar quantity is declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	mzml-2011

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 (arbitrary intensity) or a declared molar unit)

Lipid Concentration lipidome-glycome/lipid-concentration@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit uM, scale probability
PROPOSED	unit uM; scale ratio
REASON	Unit right, scale wrong: a concentration is ratio-scaled with a true zero.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	keep nM<->uM, but only after the positive fixture's scale is corrected to ratio

metabolome 10 profiles**Isotope Tracer Fraction** metabolome/isotope-tracer-fraction@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale probability
PROPOSED	unit 1; scale proportion
REASON	Mole percent enrichment is a bounded proportion of labelled molecules, not a probability.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Mass Spectral Peak metabolome/mass-spectral-peak@0.1 **BLOCK****P-MEAS Declared unit wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit g, scale ratio
PROPOSED	unit m/z (dimensionless ratio) plus a separate intensity; scale ratio
REASON	A peak is a mass-to-charge ratio with an intensity, not a mass in grams; m/z is the PSI-MS quantity.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	mzml-2011 goldbook-mr

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: m/z (dimensionless ratio) plus a separate intensity)

P-KG Contradiction fixture treats g and kg as incompatible

FIELD	fixture comparison-incompatible-measurement-unit
CURRENT	source kg vs target g, expected INCOMPATIBLE
PROPOSED	a contradiction unit of another dimension (e.g. s), plus a g<->kg lossless fixture
REASON	Grams and kilograms are the same quantity; certifying them as contradictory blocks a correct coupling.
FIXTURES	comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 si-brochure-9

Mass Spectrum [metabolome/mass-spectrum@0.1](#) **BLOCK****P-MEAS Declared unit wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit g, scale ratio
PROPOSED	unit m/z axis with an intensity value; scale ratio
REASON	A spectrum is an array over m/z; grams is neither the axis unit nor the value unit.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	mzml-2011

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: m/z axis with an intensity value)

P-KG Contradiction fixture treats g and kg as incompatible

FIELD	fixture comparison-incompatible-measurement-unit
CURRENT	source kg vs target g, expected INCOMPATIBLE
PROPOSED	a contradiction unit of another dimension (e.g. s), plus a g<->kg lossless fixture
REASON	Grams and kilograms are the same quantity; certifying them as contradictory blocks a correct coupling.
FIXTURES	comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 si-brochure-9

Metabolite Abundance [metabolome/metabolite-abundance@0.1](#) **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1 (arbitrary intensity) or a declared molar unit; scale ratio with declared normalization
REASON	Untargeted metabolomics abundances are arbitrary intensities; targeted values are molar. The profile must distinguish them.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	mzml-2011

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 (arbitrary intensity) or a declared molar unit)

Metabolite Concentration [metabolome/metabolite-concentration@0.1](#) **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit uM, scale probability
PROPOSED	unit uM; scale ratio
REASON	Unit is right; scale is not. A concentration has a true zero and is ratio-scaled - 'probability' comes from the substring bug in S6.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	keep nM<->uM, but only after the positive fixture's scale is corrected to ratio

Metabolite Consumption Rate [metabolome/metabolite-consumption-rate@0.1](#) **BLOCK****P-MEAS Declared unit wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit mmol/(gDW.h) or mol/(L.s); scale ratio
REASON	As for production rate: the flux basis is part of the quantity.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	orth-2010-fba ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: mmol/(gDW.h) or mol/(L.s))

Metabolite Production Rate [metabolome/metabolite-production-rate@0.1](#) **BLOCK****P-MEAS Declared unit wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit mmol/(gDW.h) or mol/(L.s); scale ratio
REASON	A metabolic flux needs an amount basis (biomass or volume); 1/s carries none, so two fluxes cannot be compared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	orth-2010-fba ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: mmol/(gDW.h) or mol/(L.s))

Metabolite Time Series metabolome/metabolite-time-series@0.1 **BLOCK****P-MEAS** **Declared unit wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit s, scale ratio
PROPOSED	unit (value unit, profile-declared); scale ratio
REASON	's' is the time axis unit, not the unit of the metabolite values.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV **Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity**

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: (value unit, profile-declared))

Pathway Metabolite Score metabolome/pathway-metabolite-score@0.1 **BLOCK****P-MEAS** **Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ordinal
PROPOSED	unit 1; scale interval
REASON	A pathway score is a continuous method output; its method and gene/metabolite set must be declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV **Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity**

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Retention Time metabolome/retention-time@0.1 **BLOCK****P-MEAS** **Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit s, scale ratio
PROPOSED	unit s or min; scale ratio
REASON	Unit is correct; retention time is only comparable within a declared chromatographic method, which must be required.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 mzml-2011

P-CONV **Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity**

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: s or min)

microbiology 6 profiles

Biofilm Biomass microbiology/biofilm-biomass@0.1 **BLOCK**

P-MEAS Unit and scale acceptable but under-specified

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit g, scale ratio
PROPOSED	unit g or g/cm ² ; scale ratio
REASON	Biofilm biomass is often per unit surface area; a bare mass hides the area basis.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: g or g/cm ²)

P-KG Contradiction fixture treats g and kg as incompatible

FIELD	fixture comparison-incompatible-measurement-unit
CURRENT	source kg vs target g, expected INCOMPATIBLE
PROPOSED	a contradiction unit of another dimension (e.g. s), plus a g<->kg lossless fixture
REASON	Grams and kilograms are the same quantity; certifying them as contradictory blocks a correct coupling.
FIXTURES	comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 si-brochure-9

Community Time Series microbiology/community-time-series@0.1 **BLOCK**

P-MEAS Declared unit wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit s, scale ratio
PROPOSED	unit (value unit, profile-declared); scale ratio
REASON	's' is the time axis unit, not the value unit.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: (value unit, profile-declared))

Functional Abundance microbiology/functional-abundance@0.1 **BLOCK**

P-MEAS Declared scale wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1 (counts) or a proportion; scale count or proportion
REASON	Gene-family abundance needs its reference database and version, and its normalisation, declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	bioregistry-2022

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 (counts) or a proportion)

Microbial Growth Rate microbiology/microbial-growth-rate@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit 1/h; scale ratio
REASON	Correct dimension; state that it is the specific growth rate mu.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1/h)

Minimum Inhibitory Concentration microbiology/minimum-inhibitory-concentration@0.1 **BLOCK****P-MEAS Declared unit and scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit uM, scale probability
PROPOSED	unit mg/L; scale ordinal with transform log2
REASON	The reference method reports MIC in mg/L on a two-fold dilution series, so values are interval-censored and log-spaced, not probability-scaled. Molar conversion needs the molar mass and the organism and method must be declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	iso-20776-1-2019 stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	keep nM<->uM, but only after the positive fixture's scale is corrected to ratio

Taxonomic Abundance microbiology/taxonomic-abundance@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1 (counts) or a proportion of the community; scale count or proportion
REASON	Relative and absolute abundance are different quantities, and a community vector needs a taxonomy namespace and release plus a feature universe, not a single species.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ncbitaxon-schoch-2020

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 (counts) or a proportion of the community)

multiomics 6 profiles

Gene Metabolite Map multiomics/gene-metabolite-map@0.1 **BLOCK**

P-AXES Declared axes or representation contradict the data

FIELD	dimensions.axes
CURRENT	[x, y]
PROPOSED	[gene, metabolite] with both namespaces and releases
REASON	An identifier mapping, not an image.
FIXTURES	positive, comparison-direct
SOURCES	bioregistry-2022

Gene Protein Map multiomics/gene-protein-map@0.1 **BLOCK**

P-AXES Declared axes or representation contradict the data

FIELD	dimensions.axes
CURRENT	[x, y]
PROPOSED	[gene, protein] with both namespaces, releases and the mapping's cardinality and coverage
REASON	This is an identifier mapping, not a spatial image.
FIXTURES	positive, comparison-direct
SOURCES	bioregistry-2022 ensembl-2024

Gene Set Score multiomics/gene-set-score@0.1 **BLOCK**

P-MEAS Declared scale wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ordinal
PROPOSED	unit 1; scale interval
REASON	As for the pathway score.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Integrated Embedding multiomics/integrated-embedding@0.1 **BLOCK**

P-MEAS Declared unit and scale wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit 1; scale interval
REASON	An embedding coordinate is dimensionless and has no natural zero, so it is not ratio-scaled. The 1/s unit is the substring bug matching 'rate' inside 'integrated'. Embeddings are only comparable within one model and version, which must be declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Multi-omics Sample Map `multiomics/multiomics-sample-map@0.1` **BLOCK****P-AXES Declared axes or representation contradict the data**

FIELD	dimensions.axes
CURRENT	[x, y]
PROPOSED	[sample, assay] with the linkage key and its scope
REASON	A sample linkage table, not an image.
FIXTURES	positive, comparison-direct
SOURCES	obi-2016

Pathway Score `multiomics/pathway-score@0.1` **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ordinal
PROPOSED	unit 1; scale interval
REASON	A pathway enrichment score is continuous and method-specific; the method and gene set must be declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

neuroscience 6 profiles**Cognitive Score** `neuroscience/cognitive-score@0.1` **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ordinal
PROPOSED	unit 1; scale ordinal or interval by instrument
REASON	Ordinal is defensible for most instruments, but the instrument and its normative sample must be declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Connectivity Matrix neuroscience/connectivity-matrix@0.1 **BLOCK****P-AXES** Declared axes or representation contradict the data

FIELD	dimensions.axes
CURRENT	[sample, feature]
PROPOSED	[region, region] (or [neuron, neuron]) over one declared parcellation or cell set
REASON	A connectivity matrix is square over one entity set; sample x feature misdescribes it.
FIXTURES	positive, comparison-direct
SOURCES	nwb-2022

Firing Rate neuroscience/firing-rate@0.1 **BLOCK****P-MEAS** Unit and scale acceptable but under-specified

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit Hz (1/s); scale ratio
REASON	Unit is correct. The blocking problem is the transformation fixture asserting a nanomolar-to-micromolar conversion for this profile (S7), and the absence of the counting window, which is part of the quantity.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 nwb-2022

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: Hz (1/s))

Local Field Potential neuroscience/local-field-potential@0.1 **BLOCK****P-AXES** Declared axes or representation contradict the data

FIELD	dimensions.axes
CURRENT	[x, y]
PROPOSED	[time, channel] with sampling rate, reference scheme and electrode coordinates
REASON	An LFP recording is time x channel; [x, y] came from the substring 'field'.
FIXTURES	positive, comparison-direct
SOURCES	bids-ieeg-2019 nwb-2022

Neurotransmitter Concentration neuroscience/neurotransmitter-concentration@0.1 **BLOCK****P-MEAS** Declared scale wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit uM, scale probability
PROPOSED	unit uM or nM; scale ratio
REASON	Unit acceptable, scale wrong; the compartment (synaptic cleft, extracellular, tissue) must be declared because it changes the quantity by orders of magnitude.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	keep nM<->uM, but only after the positive fixture's scale is corrected to ratio

Synaptic Weight neuroscience/synaptic-weight@0.1 **BLOCK****P-MEAS** Unit and scale acceptable but under-specified

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1, nS or mV (model-dependent); scale ratio or interval by convention
REASON	Synaptic weight is model-specific: a dimensionless coupling, a conductance and a postsynaptic potential amplitude are different quantities that must not compare as equal.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	nwb-2022

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1, nS or mV (model-dependent))

pharmacology 12 profiles**Area Under Curve** pharmacology/area-under-curve@0.1 **BLOCK****P-AXES** Declared axes or representation contradict the data

FIELD	dimensions.axes
CURRENT	[time]
PROPOSED	no axes; scalar with the integration interval and the integrated quantity declared
REASON	AUC is a scalar summary; its units are concentration x time.
FIXTURES	positive, comparison-direct
SOURCES	iuphar-neubig-2003

Association Rate pharmacology/association-rate@0.1 **BLOCK****P-MEAS** Declared unit wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit /(mol/L)/s i.e. M-1.s-1; scale ratio
REASON	kon is second-order: per concentration per time. 1/s is the unit of koff, so the two rate constants are currently declared identically.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	iuphar-neubig-2003 ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: /(mol/L)/s i.e. M-1.s-1)

Binding Affinity pharmacology/binding-affinity@0.1 **BLOCK****P-MEAS Declared unit wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit mol/L for Kd/Ki, or 1 for pKd/pKi; scale ratio for Kd, interval for pKd
REASON	An affinity constant is a molar concentration; its negative logarithm is a different quantity on a different scale. Unit '1' conflates them, and Kd versus IC50 versus EC50 are not interchangeable.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	iuphar-neubig-2003

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: mol/L for Kd/Ki, or 1 for pKd/pKi)

Compound Dose pharmacology/compound-dose@0.1 **BLOCK****P-MEAS Declared unit wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit mg, mg/kg or mol; scale ratio
REASON	A dose without a unit and, for body-weight dosing, a per-mass basis is not a dose; unit '1' makes two incomparable dosing conventions look identical.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 fda-bmv-2018

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: mg, mg/kg or mol)

Concentration Response Curve pharmacology/concentration-response-curve@0.1 **BLOCK****P-MEAS Declared unit and scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit uM, scale probability
PROPOSED	unit response unit over a concentration axis in uM; scale ratio
REASON	Same axis error as the dose-response curve, plus the probability-scale bug; the concentration axis unit and the response unit are two different declarations.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	iuphar-neubig-2003 stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	keep nM<->uM, but only after the positive fixture's scale is corrected to ratio

P-AXES Declared axes or representation contradict the data

FIELD	dimensions.axes
CURRENT	[time]
PROPOSED	[concentration] with the concentration unit on the axis
REASON	The independent variable is concentration, not time.
FIXTURES	positive, comparison-direct
SOURCES	iuphar-neubig-2003

Dissociation Rate pharmacology/dissociation-rate@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit 1/s; scale ratio
REASON	Unit is correct for koff; the profile must state that it is the kinetic off-rate and not the equilibrium dissociation constant Kd.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	iuphar-neubig-2003

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1/s)

Dose Response Curve pharmacology/dose-response-curve@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit response unit over a dose axis; scale ratio
REASON	The value is a response and the axis is dose; a single dimensionless unit describes neither, and the axis must be dose, not time.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	iuphar-neubig-2003

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: response unit over a dose axis)

P-AXES Declared axes or representation contradict the data

FIELD	dimensions.axes
CURRENT	[time]
PROPOSED	[dose] with the dose unit on the axis
REASON	The independent variable of a dose-response curve is dose, not time.
FIXTURES	positive, comparison-direct
SOURCES	iuphar-neubig-2003

Dose Schedule pharmacology/dose-schedule@0.1 **BLOCK****P-MEAS Declared unit wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit dose unit plus an interval (e.g. mg every 12 h); scale ratio
REASON	A schedule is a dose quantity plus a timing rule; a dimensionless '1' captures neither.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: dose unit plus an interval (e.g. mg every 12 h))

Exposure Concentration pharmacology/exposure-concentration@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit uM, scale probability
PROPOSED	unit uM or ng/mL; scale ratio
REASON	Unit acceptable, scale wrong; free versus total concentration must also be declared because they are different quantities.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	iuphar-neubig-2003 stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	keep nM<->uM, but only after the positive fixture's scale is corrected to ratio

Plasma Concentration pharmacology/plasma-concentration@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit uM, scale probability
PROPOSED	unit uM or ng/mL; scale ratio
REASON	A plasma concentration is ratio-scaled. Mass and molar units both occur in practice and interconvert only with the molar mass, which must be a declared transformation parameter.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	fda-bmv-2018 ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	keep nM<->uM, but only after the positive fixture's scale is corrected to ratio

Tissue Concentration pharmacology/tissue-concentration@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit uM, scale probability
PROPOSED	unit uM or ng/g; scale ratio
REASON	Tissue concentration is per tissue mass or volume; the basis must be declared, and the scale is ratio.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	keep nM->uM, but only after the positive fixture's scale is corrected to ratio

Volume of Distribution pharmacology/volume-of-distribution@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit mL, scale ratio
PROPOSED	unit L or L/kg; scale ratio
REASON	Vd is reported in litres, or litres per kilogram when weight-normalised; mL as a dense vector with image axes is wrong in unit, shape and axes.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	iuphar-neubig-2003 ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: L or L/kg)

P-AXES Declared axes or representation contradict the data

FIELD	dimensions.axes
CURRENT	[x, y]
PROPOSED	no axes; scalar (optionally per compartment)
REASON	Vd is a scalar PK parameter, not an image.
FIXTURES	positive, comparison-direct
SOURCES	iuphar-neubig-2003

phenotype 8 profiles**Biomarker Value** phenotype/biomarker-value@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit analyte-specific (must be declared); scale ratio
REASON	Biomarker values are analyte- and assay-specific; a bare '1' lets two different analytes compare as compatible.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	fda-bmv-2018

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: analyte-specific (must be declared))

Functional Score phenotype/functional-score@0.1 **BLOCK**

P-MEAS Declared scale wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ordinal
PROPOSED	unit 1; scale interval
REASON	Instrument-specific; the instrument and its version must be declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Risk Score phenotype/risk-score@0.1 **BLOCK**

P-MEAS Declared scale wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ordinal
PROPOSED	unit 1; scale interval
REASON	A risk score is continuous; if it is a predicted probability it must be declared as such with its time horizon.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Survival Fraction phenotype/survival-fraction@0.1 **BLOCK**

P-MEAS Declared scale wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale probability
PROPOSED	unit 1; scale proportion
REASON	A surviving fraction is a proportion with a declared reference population and time point.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	clark-2003-survival

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Symptom Score phenotype/symptom-score@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ordinal
PROPOSED	unit 1; scale ordinal
REASON	Ordinal is right for most symptom instruments, but the instrument identity is what makes scores comparable and must be required.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Time to Event phenotype/time-to-event@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit s, scale ratio
PROPOSED	unit s, d or mo; scale ratio
REASON	Unit is right, but a time to event without uncertainty.censoring is not interpretable and must not be exchangeable.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	clark-2003-survival

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: s, d or mo)

Trait Value phenotype/trait-value@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit trait-specific (must be declared); scale trait-specific
REASON	A single dimensionless unit cannot serve all traits; the trait ontology term and its unit must be declared together.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	uo-2012

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: trait-specific (must be declared))

Viability at Dose phenotype/viability-at-dose@0.1 **BLOCK****P-MEAS Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1; scale proportion
REASON	Viability is a bounded proportion relative to a declared control, which must be required.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

physiology 14 profiles**Airflow Rate** physiology/airflow-rate@0.1 **BLOCK****P-MEAS Declared unit wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit L/s; scale ratio
REASON	Airflow is volumetric flow: volume per time. 1/s has no volume, so two flows cannot be compared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ats-ers-spirometry-2019 ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: L/s)

Blood Pressure physiology/blood-pressure@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit mm[Hg], scale ratio
PROPOSED	unit mm[Hg]; scale ratio
REASON	Unit correct; systolic, diastolic and mean are distinct quantities and the profile must declare which, or carry them as named components.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: mm[Hg])

Body Mass [physiology/body-mass@0.1](#) **BLOCK****P-MEAS** **Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit g, scale ratio
PROPOSED	unit kg; scale ratio
REASON	Unit is fine in grams or kilograms - but the two must convert losslessly, and this profile's contradiction fixture currently asserts they are incompatible (S8).
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	si-brochure-9 ucum-2.2

P-CONV **Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity**

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: kg)

P-KG **Contradiction fixture treats g and kg as incompatible**

FIELD	fixture comparison-incompatible-measurement-unit
CURRENT	source kg vs target g, expected INCOMPATIBLE
PROPOSED	a contradiction unit of another dimension (e.g. s), plus a g<->kg lossless fixture
REASON	Grams and kilograms are the same quantity; certifying them as contradictory blocks a correct coupling.
FIXTURES	comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 si-brochure-9

Body Temperature [physiology/body-temperature@0.1](#) **BLOCK****P-MEAS** **Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit Cel, scale ratio
PROPOSED	unit Cel or K; scale interval
REASON	Celsius is an interval scale: ratios of Celsius values are meaningless, and conversion to kelvin is affine, not multiplicative. 'ratio' licenses invalid arithmetic. The measurement site must also be declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	si-brochure-9 stevens-1946 ucum-2.2

P-CONV **Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity**

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: Cel or K)

Electrolyte Concentration [physiology/electrolyte-concentration@0.1](#) **BLOCK****P-MEAS** **Declared scale wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit uM, scale probability
PROPOSED	unit mmol/L; scale ratio
REASON	Clinical electrolytes are millimolar, not micromolar, and the scale is ratio.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 stevens-1946

P-CONV

Conversion fixture asserts a molar-concentration conversion

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	keep nM<->uM, but only after the positive fixture's scale is corrected to ratio

Glomerular Filtration Rate

physiology/glomerular-filtration-rate@0.1

BLOCK

P-MEAS

Declared unit and scale wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale probability
PROPOSED	unit mL/min/{1.73_m2}; scale ratio
REASON	GFR is a volumetric clearance normalised to body surface area; 1/s has neither, and the probability scale is the substring bug. Measured and estimated GFR must be distinguished, with the estimating equation declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	kdigo-2024-ckd ucum-2.2

P-CONV

Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: mL/min/{1.73_m2})

Heart Rate

physiology/heart-rate@0.1

BLOCK

P-MEAS

Unit and scale acceptable but under-specified

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit /min; scale ratio
REASON	Clinically reported per minute; 1/s is the same quantity and must convert losslessly, which the current table does not support.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV

Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: /min)

Hormone Concentration

physiology/hormone-concentration@0.1

BLOCK

P-MEAS

Declared scale wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit uM, scale probability
PROPOSED	unit pmol/L, ng/mL or uM; scale ratio
REASON	Hormones are reported in both molar and mass concentrations; conversion requires the molar mass. Scale is ratio.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 stevens-1946

P-CONV

Conversion fixture asserts a molar-concentration conversion

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	keep nM<->uM, but only after the positive fixture's scale is corrected to ratio

Lung Volume physiology/lung-volume@0.1 **BLOCK****P-MEAS** **Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit mL, scale ratio
PROPOSED	unit L or mL; scale ratio
REASON	Unit acceptable; which volume (tidal, FRC, TLC) must be declared, and the value is scalar, not an image.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ats-ers-spirometry-2019

P-CONV **Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity**

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: L or mL)

P-AXES **Declared axes or representation contradict the data**

FIELD	dimensions.axes
CURRENT	[x, y]
PROPOSED	no spatial axes; scalar, optionally a time series
REASON	A lung volume measurement is a scalar.
FIXTURES	positive, comparison-direct
SOURCES	ats-ers-spirometry-2019

Mean Arterial Pressure physiology/mean-arterial-pressure@0.1 **BLOCK****P-MEAS** **Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit mm[Hg], scale ratio
PROPOSED	unit mm[Hg]; scale ratio
REASON	Unit correct; MAP is derived, so the derivation (measured versus estimated from systolic and diastolic) belongs in origin.method.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV **Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity**

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: mm[Hg])

Metabolic Rate physiology/metabolic-rate@0.1 **BLOCK****P-MEAS** **Declared unit wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit W, or kcal/d, or mL/min for VO2; scale ratio
REASON	Metabolic rate is energy per time (or oxygen consumption per time); 1/s carries neither energy nor amount.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	si-brochure-9 ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: W, or kcal/d, or mL/min for VO2)

Physiological Time Series `physiology/physiological-time-series@0.1` **BLOCK****P-MEAS Declared unit wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit s, scale ratio
PROPOSED	unit (value unit, profile-declared); scale ratio
REASON	's' is the time axis unit, not the value unit.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: (value unit, profile-declared))

Respiratory Rate `physiology/respiratory-rate@0.1` **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit /min; scale ratio
REASON	As for heart rate.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: /min)

Stroke Volume `physiology/stroke-volume@0.1` **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit mL, scale ratio
PROPOSED	unit mL; scale ratio
REASON	Unit correct, but stroke volume is a scalar per beat: the dense-vector representation with image axes [x, y] is wrong.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: mL)

P-AXES Declared axes or representation contradict the data

FIELD	dimensions.axes
CURRENT	[x, y]
PROPOSED	no spatial axes; scalar per beat, optionally a time series
REASON	Stroke volume is a scalar; image axes came from the substring 'volume'.
FIXTURES	positive, comparison-direct
SOURCES	ucum-2.2

proteome 4 profiles**Peptide Abundance** [proteome/peptide-abundance@0.1](#) **BLOCK****P-MEAS** Unit and scale acceptable but under-specified

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1 (arbitrary intensity) or a declared molar/mass unit; scale ratio with declared normalization
REASON	As for protein abundance: intensity units are instrument- and pipeline-specific.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	mzml-2011

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 (arbitrary intensity) or a declared molar/mass unit)

Phosphoprotein Abundance [proteome/phosphoprotein-abundance@0.1](#) **BLOCK****P-MEAS** Unit and scale acceptable but under-specified

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1 (arbitrary intensity), or occupancy as a proportion; scale ratio with declared normalization
REASON	Phosphoprotein abundance and phosphosite occupancy are different quantities; the profile must declare which, as occupancy is a bounded proportion.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	mzml-2011

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 (arbitrary intensity), or occupancy as a proportion)

Protein Abundance `proteome/protein-abundance@0.1` **BLOCK****P-MEAS** **Unit and scale acceptable but under-specified**

FIELD	<code>measurement.unit, measurement.scale</code> (profile requirement and positive fixture)
CURRENT	<code>unit 1, scale ratio</code>
PROPOSED	<code>unit 1 (arbitrary intensity) or a declared molar/mass unit; scale ratio with declared normalization</code>
REASON	Protein abundance is usually an arbitrary-unit intensity that is only comparable within one normalisation; unit '1' with no normalization field hides that.
FIXTURES	<code>positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit</code>
SOURCES	robinson-2010-tmm

P-CONV **Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity**

CURRENT	<code>source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE</code>
PROPOSED	<code>a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 (arbitrary intensity) or a declared molar/mass unit)</code>

Secretion Rate `proteome/secretion-rate@0.1` **BLOCK****P-MEAS** **Declared unit wrong for this quantity**

FIELD	<code>measurement.unit, measurement.scale</code> (profile requirement and positive fixture)
CURRENT	<code>unit 1/s, scale ratio</code>
PROPOSED	<code>unit pg/(cell.h) or fmol/(cell.h); scale ratio</code>
REASON	A secretion rate is amount per cell per time; 1/s has no amount or cell basis and so cannot be compared across experiments.
FIXTURES	<code>positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit</code>
SOURCES	ucum-2.2 iso-80000-1-2022

P-CONV **Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity**

CURRENT	<code>source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE</code>
PROPOSED	<code>a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: pg/(cell.h) or fmol/(cell.h))</code>

simulation 1 profiles**Simulation Time** `simulation/simulation-time@0.1` **BLOCK****P-MEAS** **Unit and scale acceptable but under-specified**

FIELD	<code>measurement.unit, measurement.scale</code> (profile requirement and positive fixture)
CURRENT	<code>unit s, scale ratio</code>
PROPOSED	<code>unit s, or declared simulation time units; scale ratio</code>
REASON	Unit is right; the time origin and whether time is wall-clock or model time must be declared.
FIXTURES	<code>positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit</code>
SOURCES	ucum-2.2

P-CONV **Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity**

CURRENT	<code>source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE</code>
PROPOSED	<code>a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: s, or declared simulation time units)</code>

spatial 3 profiles

Concentration Field spatial/concentration-field@0.1 **BLOCK**

P-MEAS Declared scale wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit uM, scale probability
PROPOSED	unit uM; scale ratio
REASON	Unit acceptable, scale wrong; a field also needs its spatial axes, spacing and possibly a third dimension.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2 stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	keep nM<->uM, but only after the positive fixture's scale is corrected to ratio

Distance Matrix spatial/distance-matrix@0.1 **BLOCK**

P-MEAS Declared unit wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit um or mm; scale ratio
REASON	A spatial distance has a length unit; '1' would also admit a unitless dissimilarity, which is a different quantity. Axes must be entity x entity, not sample x feature.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: um or mm)

P-AXES Declared axes or representation contradict the data

FIELD	dimensions.axes
CURRENT	[sample, feature]
PROPOSED	[entity, entity] with a length unit on the values
REASON	A distance matrix is square over one entity set.
FIXTURES	positive, comparison-direct
SOURCES	ucum-2.2

Spatial Abundance Map spatial/spatial-abundance-map@0.1 **BLOCK**

P-MEAS Unit and scale acceptable but under-specified

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1 (counts) or a declared normalised unit; scale count or ratio
REASON	Spot- and cell-level abundances need their normalisation and spatial resolution declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	wagner-2012-tpm

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 (counts) or a declared normalised unit)

transcriptome 1 profiles

Small RNA Abundance transcriptome/small-rna-abundance@0.1 **BLOCK**

P-MEAS Unit and scale acceptable but under-specified

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1 (counts) or a declared normalised unit; scale count or ratio with transform
REASON	Abundance is either integer counts or a normalised quantity (CPM/TPM); the profile must say which, because they are not interchangeable.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	wagner-2012-tpm robinson-2010-tmm

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 (counts) or a declared normalised unit)

virology 8 profiles

Immune Escape Score virology/immune-escape-score@0.1 **BLOCK**

P-MEAS Declared scale wrong for this quantity

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ordinal
PROPOSED	unit 1; scale interval
REASON	An escape score is a continuous model or assay output; the method and the reference serum or antibody panel must be declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	stevens-1946

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1)

Infection Rate virology/infection-rate@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit 1/h, or per-contact per-time; scale ratio
REASON	The basis (per susceptible, per contact, per cell) is part of the quantity and must be declared via measurement.rate_basis.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1/h, or per-contact per-time)

Infectious Titer virology/infectious-titer@0.1 **BLOCK****P-MEAS Declared unit wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit [PFU]/mL or TCID50/mL; scale ratio with transform log10
REASON	Infectious titre is reported per volume in plaque-forming units or TCID50 and is conventionally handled on a log10 scale; '1' declares no volume basis and no assay.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	reed-muench-1938

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: [PFU]/mL or TCID50/mL)

Replication Rate virology/replication-rate@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit 1/h; scale ratio
REASON	Correct dimension; distinguish the within-host replication rate from the basic reproduction number, which is dimensionless.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1/h)

Shedding Time Series virology/shedding-time-series@0.1 **BLOCK****P-MEAS Declared unit wrong for this quantity**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit s, scale ratio
PROPOSED	unit (value unit, profile-declared); scale ratio
REASON	's' is the time axis unit.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	ucum-2.2

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: (value unit, profile-declared))

Transmission Rate virology/transmission-rate@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1/s, scale ratio
PROPOSED	unit 1/d, per-contact per-time; scale ratio
REASON	Transmission rate beta is model-dependent; without the rate basis two models' betas are not comparable.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	cdc-epi-3ed

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1/d, per-contact per-time)

Viral Protein Abundance virology/viral-protein-abundance@0.1 **BLOCK****P-MEAS Unit and scale acceptable but under-specified**

FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1 (arbitrary intensity); scale ratio with declared normalization
REASON	Intensity units are pipeline-specific.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	mzml-2011

P-CONV Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity

CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 (arbitrary intensity))

Viral Transcript Abundance

virology/viral-transcript-abundance@0.1

BLOCK

P-MEAS	Unit and scale acceptable but under-specified
FIELD	measurement.unit, measurement.scale (profile requirement and positive fixture)
CURRENT	unit 1, scale ratio
PROPOSED	unit 1 (counts) or a declared normalised unit; scale count or ratio
REASON	As for host transcript abundance; the viral reference genome and version must be declared.
FIXTURES	positive, comparison-direct, comparison-unknown-measurement-unit, comparison-incompatible-measurement-unit
SOURCES	wagner-2012-tpm

P-CONV	Conversion fixture asserts a molar-concentration conversion for a non-concentration quantity
CURRENT	source nM, target uM, expected LOSSLESS_CONVERSION_AVAILABLE
PROPOSED	a conversion within this profile's own dimension, plus a cross-dimension contradiction fixture (proposed unit: 1 (counts) or a declared normalised unit)

EVERY PROFILE'S VERDICT, AUDIT AND ITEM DECISIONS AT A GLANCE

Verdict register: all 650 profiles

Contract areas read left to right in this order: 1 structure; 2 scientific meaning; 3 representation; 4 dimensions and feature order; 5 identifiers and versions; 6 quantity, unit, scale, normalisation, endpoint; 7 biological and experimental context; 8 time and lifecycle meaning; 9 origin and provenance; 10 uncertainty and missingness; 11 artifact format; 12 cross-field constraints; 13 security, consent, licensing, data use. Each letter is one area in that order: I = included, C = conditional, X = excluded.

Fixture audit reads left to right: 1 positive example is scientifically valid; 2 missing evidence returns unknown; 3 known contradiction returns incompatible; 4 direct compatibility hides no transformation; 5 unit and representation conversions are correct; 6 lossy conversion and inference are identified. Each character is one check in that order: P = pass, ~ = partial, W = schema-valid but scientifically weak, F = fail, - = not applicable.

Items counts the candidate items listed in the packet, as required / conditional / recommended / optional.

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
cardiopulmonary-renal 25 profiles					
Arrhythmia Event cardiopulmonary-renal/arrhythmia-event@0.1	CHANGES_REQD	-	IIIXXXIICCXXC	W~PF-F	15/35/10/5
Arterial Stiffness cardiopulmonary-renal/arterial-stiffness@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	15/35/10/5
Blood Cell Count cardiopulmonary-renal/blood-cell-count@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	15/35/10/5
Cardiac Biomarker cardiopulmonary-renal/cardiac-biomarker@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	15/35/10/5
Cardiac Geometry cardiopulmonary-renal/cardiac-geometry@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	15/35/10/5
Coagulation Marker cardiopulmonary-renal/coagulation-marker@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	15/35/10/5
Coronary Flow cardiopulmonary-renal/coronary-flow@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	15/35/10/5
ECG Interval cardiopulmonary-renal/ecg-interval@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	15/35/10/5
Ejection Fraction cardiopulmonary-renal/ejection-fraction@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCXXC	F~PFFF	12/35/10/5
Electrocardiogram cardiopulmonary-renal/electrocardiogram@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	15/35/10/5
Flow Waveform cardiopulmonary-renal/flow-waveform@0.1	CHANGES_REQD	-	IIICCIICCIXC	W~PF-F	18/40/11/5
Heart Rhythm cardiopulmonary-renal/heart-rhythm@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	15/35/10/5
Hemodynamic State cardiopulmonary-renal/hemodynamic-state@0.1	CHANGES_REQD	-	IIIXXXIICCIXC	W~PF-F	15/35/10/5
Myocardial Strain cardiopulmonary-renal/myocardial-strain@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	15/35/10/5
Oxygen Consumption cardiopulmonary-renal/oxygen-consumption@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	15/35/10/5
Perfusion Map cardiopulmonary-renal/perfusion-map@0.1	CHANGES_REQD	-	IIICCIICCIXC	W~PF-F	18/40/11/5

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Pressure Waveform cardiopulmonary-renal/pressure-waveform@0.1	BLOCK	P-MEAS P-CONV	IIIIICIIICCIIXC	F~PFFF	15/40/11/5
Pulse Wave Velocity cardiopulmonary-renal/pulse-wave-velocity@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	15/35/10/5
Renal Clearance Marker cardiopulmonary-renal/renal-clearance-marker@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	15/35/10/5
Respiratory Waveform cardiopulmonary-renal/respiratory-waveform@0.1	CHANGES_REQD	–	IIIIICIIICCIIXC	W~PF-F	18/40/11/5
Spirometry Curve cardiopulmonary-renal/spirometry-curve@0.1	CHANGES_REQD	–	IIIIICIIICCIIXC	W~PF-F	18/40/11/5
Thrombus State cardiopulmonary-renal/thrombus-state@0.1	CHANGES_REQD	–	IIIXXXICCCIXC	W~PF-F	15/35/10/5
Vascular Geometry cardiopulmonary-renal/vascular-geometry@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	15/35/10/5
Ventricular Volume cardiopulmonary-renal/ventricular-volume@0.1	BLOCK	P-MEAS P-CONV P-AXES	IIIIICIIICCIIXC	F~PFFF	15/40/11/5
Wall Shear Stress cardiopulmonary-renal/wall-shear-stress@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	15/35/10/5
cell 25 profiles					
Apoptosis State cell/apoptosis-state@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	10/35/12/6
ATP Level cell/atp-level@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	10/35/12/6
Cell Count cell/cell-count@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	10/35/12/6
Cell Cycle State cell/cell-cycle-state@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	10/35/12/6
Cell Density cell/cell-density@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	10/35/12/6
Cell Identity cell/cell-identity@0.1	CHANGES_REQD	–	IIIXIXICCCIXX	W~PF-F	10/35/12/6
Cell Line Identity cell/cell-line-identity@0.1	CHANGES_REQD	–	IIIXIXICCCIXX	W~PF-F	10/35/12/6
Cell Size cell/cell-size@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	10/35/12/6
Cell State cell/cell-state@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	10/35/12/6
Cell Type cell/cell-type@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	10/35/12/6
Colony Formation cell/colony-formation@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	10/35/12/6
Differentiation State cell/differentiation-state@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	10/35/12/6
DNA Damage State cell/dna-damage-state@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	10/35/12/6
Growth Rate cell/growth-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	7/35/12/6
Intracellular pH cell/intracellular-ph@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	10/35/12/6

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Invasion Score cell/invasion-score@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	7/35/12/6
Lineage Trajectory cell/lineage-trajectory@0.1	CHANGES_REQD	–	IIIIICCIICCIXX	W~PF-F	13/40/13/6
Membrane Potential cell/membrane-potential@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	10/35/12/6
Migration Rate cell/migration-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	7/35/12/6
Morphology Features cell/morphology-features@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	10/35/12/6
Organelle State cell/organelle-state@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	10/35/12/6
Proliferation Rate cell/proliferation-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	7/35/12/6
Reactive Oxygen Species cell/reactive-oxygen-species@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	10/35/12/6
Senescence State cell/senescence-state@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	10/35/12/6
Viability cell/viability@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	10/35/12/6
chemical 25 profiles					
Canonical SMILES chemical/canonical-smiles@0.1	CHANGES_REQD	–	IIIXCXXXXCIXX	W~PF-F	18/31/8/4
Chemical Descriptor Vector chemical/chemical-descriptor-vector@0.1	CHANGES_REQD	–	IIIIICCCCCIXX	W~PF-F	18/36/9/4
Chemical Identity chemical/chemical-identity@0.1	CHANGES_REQD	–	IIIXIXCCCCIXX	W~PF-F	16/31/8/4
Chemical Similarity chemical/chemical-similarity@0.1	CHANGES_REQD	–	IIICICCCCCXXX	W~PF-F	16/31/8/4
Chemical Structure chemical/chemical-structure@0.1	CHANGES_REQD	–	IIICICCCCCXXX	W~PF-F	16/31/8/4
Circular Fingerprint chemical/circular-fingerprint@0.1	CHANGES_REQD	–	IIICCCCCCCCXXX	W~PF-F	18/31/8/4
Conformer Ensemble chemical/conformer-ensemble@0.1	CHANGES_REQD	–	IIICCCCCCCCXXX	W~PF-F	18/31/8/4
Exact Mass chemical/exact-mass@0.1	BLOCK	P-MEAS P-CONV P-KG	IIICCIICCCXXX	F~FFFF	15/31/8/4
Formal Charge chemical/formal-charge@0.1	CHANGES_REQD	–	IIICCCCCCCCXXX	W~PF-F	18/31/8/4
InChIKey chemical/inchi-key@0.1	CHANGES_REQD	–	IIIXCXXXXCIXX	W~PF-F	18/31/8/4
InChI chemical/inchi@0.1	CHANGES_REQD	–	IIIXCXXXXCIXX	W~PF-F	18/31/8/4
Isomeric SMILES chemical/isomeric-smiles@0.1	CHANGES_REQD	–	IIIXCXXXXCIXX	W~PF-F	18/31/8/4
Lipophilicity chemical/lipophilicity@0.1	CHANGES_REQD	–	IIICCCCCCCCXXX	W~PF-F	18/31/8/4
Molecular Formula chemical/molecular-formula@0.1	CHANGES_REQD	–	IIICCCCCCCCXXX	W~PF-F	18/31/8/4

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Molecular Graph chemical/molecular-graph@0.1	CHANGES_REQD	-	IIIXCXXXXCIXX	W~PF-F	23/36/8/4
Molecular Weight chemical/molecular-weight@0.1	BLOCK	P-MEAS P-CONV	IIICCCXXXXCXX	F~PFFF	15/31/8/4
Path Fingerprint chemical/path-fingerprint@0.1	CHANGES_REQD	-	IIICCCXXXXCXX	W~PF-F	18/31/8/4
Pharmacophore Fingerprint chemical/pharmacophore-fingerprint@0.1	CHANGES_REQD	-	IIICCCXXXXCXX	W~PF-F	18/31/8/4
pKa chemical/pka@0.1	CHANGES_REQD	-	IIICCCXXXXCXX	W~PF-F	18/31/8/4
Protonation State chemical/protonation-state@0.1	CHANGES_REQD	-	IIIXCXXXXCIXX	W~PF-F	18/31/8/4
Reaction SMILES chemical/reaction-smiles@0.1	CHANGES_REQD	-	IIIXCXXXXCIXX	W~PF-F	18/31/8/4
Solubility chemical/solubility@0.1	CHANGES_REQD	-	IIICCCXXXXCXX	W~PF-F	18/31/8/4
Stereochemistry chemical/stereochemistry@0.1	CHANGES_REQD	-	IIICCCXXXXCXX	W~PF-F	18/31/8/4
Synthesis Route chemical/synthesis-route@0.1	CHANGES_REQD	-	IIIXCXXXXCIXX	W~PF-F	18/31/8/4
Tautomer State chemical/tautomer-state@0.1	CHANGES_REQD	-	IIIXCXXXXCIXX	W~PF-F	18/31/8/4
clinical 25 profiles					
Admission Event clinical/admission-event@0.1	CHANGES_REQD	-	IIIXCXIICCXXI	W~PF-F	18/38/7/3
Adverse Event clinical/adverse-event@0.1	CHANGES_REQD	-	IIIXCXIICCXXI	W~PF-F	18/38/7/3
Audit Event clinical/audit-event@0.1	CHANGES_REQD	-	IIIXCXIICCXXI	W~PF-F	18/38/7/3
Clinical Endpoint clinical/clinical-endpoint@0.1	CHANGES_REQD	-	IIICCCICCCXXI	W~PF-F	18/38/7/3
Clinical Observation clinical/clinical-observation@0.1	CHANGES_REQD	-	IIICCCICCCXXI	W~PF-F	18/38/7/3
Cohort Definition clinical/cohort-definition@0.1	CHANGES_REQD	-	IIICCCICCCXXI	W~PF-F	18/38/7/3
Consent Scope clinical/consent-scope@0.1	CHANGES_REQD	-	IIICCCICCCXXI	W~PF-F	18/38/7/3
Data Use Restriction clinical/data-use-restriction@0.1	CHANGES_REQD	-	IIICCCICCCXXI	W~PF-F	18/38/7/3
Deidentification Level clinical/deidentification-level@0.1	CHANGES_REQD	-	IIICCCICCCXXI	W~PF-F	18/38/7/3
Diagnosis Code clinical/diagnosis-code@0.1	CHANGES_REQD	-	IIIXIXICCCIXI	W~PF-F	16/38/7/3
Discharge Event clinical/discharge-event@0.1	CHANGES_REQD	-	IIIXCXIICCXXI	W~PF-F	18/38/7/3
Dosing Record clinical/dosing-record@0.1	CHANGES_REQD	-	IIICCCICCCXXI	W~PF-F	18/38/7/3
Encounter Event clinical/encounter-event@0.1	CHANGES_REQD	-	IIIXCXIICCXXI	W~PF-F	18/38/7/3

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Exclusion Criteria clinical/exclusion-criteria@0.1	CHANGES_REQD	–	IIIXCXICCCIXI	W~PF-F	18/38/7/3
Inclusion Criteria clinical/inclusion-criteria@0.1	CHANGES_REQD	–	IIIXCXICCCIXI	W~PF-F	18/38/7/3
Laboratory Result clinical/laboratory-result@0.1	CHANGES_REQD	–	IIICCCICCCXXI	W~PF-F	18/38/7/3
Medication Code clinical/medication-code@0.1	CHANGES_REQD	–	IIIXIXICCCIXI	W~PF-F	16/38/7/3
Outcome Measure clinical/outcome-measure@0.1	CHANGES_REQD	–	IIICCCICCCXXI	W~PF-F	18/38/7/3
Patient Pseudonym clinical/patient-pseudonym@0.1	CHANGES_REQD	–	IIICCCICCCXXI	W~PF-F	18/38/7/3
Patient Reported Outcome clinical/patient-reported-outcome@0.1	CHANGES_REQD	–	IIICCCICCCXXI	W~PF-F	18/38/7/3
Procedure Code clinical/procedure-code@0.1	CHANGES_REQD	–	IIIXIXICCCIXI	W~PF-F	16/38/7/3
Provenance Record clinical/provenance-record@0.1	CHANGES_REQD	–	IIICCCICCCXXI	W~PF-F	18/38/7/3
Survival Outcome clinical/survival-outcome@0.1	CHANGES_REQD	–	IIICCCICCCXXI	W~PF-F	18/38/7/3
Treatment Regimen clinical/treatment-regimen@0.1	CHANGES_REQD	–	IIIXCXICCCIXI	W~PF-F	18/38/7/3
Vital Sign clinical/vital-sign@0.1	CHANGES_REQD	–	IIICCCICCCXXI	W~PF-F	18/38/7/3
core 25 profiles					
Binary Flag core/binary-flag@0.1	CHANGES_REQD	–	IIICCCXCCXCX	W~PF-F	3/14/2/2
Bounded Score core/bounded-score@0.1	BLOCK	P-MEAS P-CONV	IIICCXCCXCX	F~PFFF	3/14/2/2
Categorical Value core/categorical-value@0.1	BLOCK	P-MEAS P-CONV	IIICCXCCXCX	F~PFFF	3/14/2/2
Checkpoint Artifact core/checkpoint-artifact@0.1	CHANGES_REQD	–	IIICCCXCCXCX	W~PF-F	3/14/2/2
Dataset Reference core/dataset-reference@0.1	CHANGES_REQD	–	IIICCCXCCXCX	W~PF-F	3/14/2/2
Event Stream core/event-stream@0.1	CHANGES_REQD	–	IIIXXXICCCX	W~PF-F	3/14/2/2
Event core/event@0.1	CHANGES_REQD	–	IIIXXXICCCX	W~PF-F	3/14/2/2
Graph core/graph@0.1	CHANGES_REQD	–	IIIXXXCCICX	W~PF-F	8/19/2/2
Irregular Time Series core/irregular-time-series@0.1	BLOCK	P-MEAS P-CONV	IIICIXICCCX	F~PFFF	2/14/2/2
Labeled Vector core/labelled-vector@0.1	CHANGES_REQD	–	IIICCXCCICX	W~PF-F	5/19/3/2
Matrix core/matrix@0.1	CHANGES_REQD	–	IIICCXCCICX	W~PF-F	5/19/3/2
Model State core/model-state@0.1	CHANGES_REQD	–	IIIXXXCCICX	W~PF-F	3/14/2/2

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Ordered Vector core/ordered-vector@0.1	CHANGES_REQD	–	IIICCCXCCCICX	W~PF-F	5/19/3/2
Parameter Set core/parameter-set@0.1	CHANGES_REQD	–	IIICCCXCCCXCX	W~PF-F	3/14/2/2
Probability Distribution core/probability-distribution@0.1	CHANGES_REQD	–	IIICCCXCCCXCX	W~PF-F	3/14/2/2
Probability core/probability@0.1	CHANGES_REQD	–	IIICCCXCCCXCX	W~PF-F	3/14/2/2
Record core/record@0.1	CHANGES_REQD	–	IIICCCXCCCXCX	W~PF-F	3/14/2/2
Regular Time Series core/regular-time-series@0.1	BLOCK	P-MEAS P-CONV	IIICIXICCCXCX	F~PFFF	2/14/2/2
Scalar Quantity core/scalar-quantity@0.1	BLOCK	P-AXES	IIICCCXCCCXCX	F~PF-F	3/14/2/2
Simulation Clock core/simulation-clock@0.1	CHANGES_REQD	–	IIICCCXICCCXCX	W~PF-F	3/14/2/2
Sparse Matrix core/sparse-matrix@0.1	CHANGES_REQD	–	IIICCCXCCCICX	W~PF-F	5/19/3/2
Sparse Vector core/sparse-vector@0.1	CHANGES_REQD	–	IIICCCXCCCICX	W~PF-F	5/19/3/2
Spatial Field core/spatial-field@0.1	CHANGES_REQD	–	IIICCCXCCCICX	W~PF-F	5/19/3/2
Table Artifact core/table-artifact@0.1	CHANGES_REQD	–	IIICCCXCCCXCX	W~PF-F	3/14/2/2
Tensor core/tensor@0.1	BLOCK	P-AXES	IIICCCXCCCXCX	F~PF-F	5/19/3/2
developmental 25 profiles					
Age Equivalent developmental/age-equivalent@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/34/7/5
Birth Outcome developmental/birth-outcome@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/34/7/5
Branching Morphogenesis developmental/branching-morphogenesis@0.1	CHANGES_REQD	–	IIICICICCCXXX	W~PF-F	19/34/7/5
Cell Fate developmental/cell-fate@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/34/7/5
Cell Lineage Tree developmental/cell-lineage-tree@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	19/34/7/5
Developmental Anomaly developmental/developmental-anomaly@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/34/7/5
Developmental Clock developmental/developmental-clock@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/34/7/5
Developmental Gene Signature developmental/developmental-gene-signature@0.1	CHANGES_REQD	–	IIICICICCCXXX	W~PF-F	19/34/7/5
Developmental Stage developmental/developmental-stage@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	19/34/7/5
Developmental Timing developmental/developmental-timing@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/34/7/5
Differentiation Potential developmental/differentiation-potential@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/34/7/5

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Embryo Stage developmental/embryo-stage@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	19/34/7/5
Fate Transition developmental/fate-transition@0.1	CHANGES_REQD	–	IIIXXXICCCXXX	W~PF-F	19/34/7/5
Gastrulation State developmental/gastrulation-state@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	19/34/7/5
Gestational Age developmental/gestational-age@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/34/7/5
Growth Trajectory developmental/growth-trajectory@0.1	CHANGES_REQD	–	IIIIICCIICCIXX	W~PF-F	22/39/8/5
Lineage Identity developmental/lineage-identity@0.1	CHANGES_REQD	–	IIIXIXICCCIXX	W~PF-F	19/34/7/5
Lineage Probability developmental/lineage-probability@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/34/7/5
Maturation State developmental/maturation-state@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	19/34/7/5
Morphogen Concentration developmental/morphogen-concentration@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	19/34/7/5
Morphogen Gradient developmental/morphogen-gradient@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/34/7/5
Organoid State developmental/organoid-state@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	19/34/7/5
Segmentation State developmental/segmentation-state@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	19/34/7/5
Spatial Lineage Map developmental/spatial-lineage-map@0.1	CHANGES_REQD	–	IIIIICCIICCIXX	W~PF-F	22/39/8/5
Tissue Pattern developmental/tissue-pattern@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/34/7/5
ecology 25 profiles					
Biodiversity Index ecology/biodiversity-index@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	21/35/4/5
Biomass ecology/biomass@0.1	BLOCK	P-MEAS P-CONV P-KG	IIICCIICCCXXX	F~FFFF	18/35/4/5
Birth Rate ecology/birth-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	18/35/4/5
Carrying Capacity ecology/carrying-capacity@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	21/35/4/5
Climate Time Series ecology/climate-time-series@0.1	BLOCK	P-MEAS P-CONV	IIIIICIIICCCXXX	F~PFFF	18/35/4/5
Community Composition ecology/community-composition@0.1	CHANGES_REQD	–	IIICCCICCCIXX	W~PF-F	21/35/4/5
Death Rate ecology/death-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	18/35/4/5
Dispersal Kernel ecology/dispersal-kernel@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	21/35/4/5
Ecological Network ecology/ecological-network@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	21/35/4/5
Ecosystem Flux ecology/ecosystem-flux@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	21/35/4/5

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Environmental Covariate ecology/environmental-covariate@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	21/35/4/5
Evenness ecology/evenness@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	21/35/4/5
Food Web ecology/food-web@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	21/35/4/5
Geographic Range ecology/geographic-range@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	21/35/4/5
Habitat Suitability ecology/habitat-suitability@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	21/35/4/5
Migration Event ecology/migration-event@0.1	CHANGES_REQD	–	IIIXXXIICCXXX	W~PF-F	21/35/4/5
Nutrient Cycle ecology/nutrient-cycle@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	21/35/4/5
Occupancy ecology/occupancy@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	21/35/4/5
Organism Taxon ecology/organism-taxon@0.1	CHANGES_REQD	–	IIICICICCCXXX	W~PF-F	19/35/4/5
Population Density ecology/population-density@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	21/35/4/5
Population Size ecology/population-size@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	21/35/4/5
Predator Prey Rate ecology/predator-prey-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	18/35/4/5
Richness ecology/richness@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	21/35/4/5
Species Abundance ecology/species-abundance@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCIXX	F~PFFF	18/35/4/5
Trophic Interaction ecology/trophic-interaction@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	21/35/4/5
epidemiology 25 profiles					
Attack Rate epidemiology/attack-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXI	F~PFFF	13/32/6/5
Case Count epidemiology/case-count@0.1	CHANGES_REQD	–	IIICCCICCCXXI	W~PF-F	16/32/6/5
Case Fatality Ratio epidemiology/case-fatality-ratio@0.1	CHANGES_REQD	–	IIICCCICCCXXI	W~PF-F	16/32/6/5
Compartment State epidemiology/compartment-state@0.1	CHANGES_REQD	–	IIIXXXICCCIXI	W~PF-F	16/32/6/5
Contact Matrix epidemiology/contact-matrix@0.1	BLOCK	P-AXES	IIIIICCCICCXI	F~PF-F	19/37/7/5
Diagnostic Test Result epidemiology/diagnostic-test-result@0.1	CHANGES_REQD	–	IIICCCICCCXXI	W~PF-F	16/32/6/5
Epidemiological Forecast epidemiology/epidemiological-forecast@0.1	CHANGES_REQD	–	IIICCCICCCXXI	W~PF-F	16/32/6/5
Exposure Event epidemiology/exposure-event@0.1	CHANGES_REQD	–	IIIXXXIICCXXI	W~PF-F	16/32/6/5
Generation Interval epidemiology/generation-interval@0.1	CHANGES_REQD	–	IIICICICCCXXI	W~PF-F	16/32/6/5

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Hospitalization Rate epidemiology/hospitalization-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXI	F~PFFF	13/32/6/5
Immunity Level epidemiology/immunity-level@0.1	CHANGES_REQD	–	IIICCIICCCXXI	W~PF-F	16/32/6/5
Incidence Rate epidemiology/incidence-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXI	F~PFFF	13/32/6/5
Infection Fatality Ratio epidemiology/infection-fatality-ratio@0.1	CHANGES_REQD	–	IIICCIICCCXXI	W~PF-F	16/32/6/5
Intervention Schedule epidemiology/intervention-schedule@0.1	CHANGES_REQD	–	IIIXXXIICCIXI	W~PF-F	16/32/6/5
Mobility Matrix epidemiology/mobility-matrix@0.1	BLOCK	P-AXES	IIICCIICCCIXI	F~PF-F	19/37/7/5
Mortality Rate epidemiology/mortality-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXI	F~PFFF	13/32/6/5
Outbreak Trajectory epidemiology/outbreak-trajectory@0.1	CHANGES_REQD	–	IIICCIICCCIXI	W~PF-F	19/37/7/5
Population Stratum epidemiology/population-stratum@0.1	CHANGES_REQD	–	IIICCIICCCXXI	W~PF-F	16/32/6/5
Prevalence epidemiology/prevalence@0.1	CHANGES_REQD	–	IIICCIICCCXXI	W~PF-F	16/32/6/5
Recovery Rate epidemiology/recovery-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXI	F~PFFF	13/32/6/5
Reproduction Number epidemiology/reproduction-number@0.1	CHANGES_REQD	–	IIICCIICCCXXI	W~PF-F	16/32/6/5
Serial Interval epidemiology/serial-interval@0.1	CHANGES_REQD	–	IIICCIICCCXXI	W~PF-F	16/32/6/5
Surveillance Signal epidemiology/surveillance-signal@0.1	CHANGES_REQD	–	IIICCIICCCIXI	W~PF-F	16/32/6/5
Test Positivity epidemiology/test-positivity@0.1	CHANGES_REQD	–	IIICCIICCCXXI	W~PF-F	16/32/6/5
Vaccination Coverage epidemiology/vaccination-coverage@0.1	CHANGES_REQD	–	IIICCIICCCXXI	W~PF-F	16/32/6/5
epigenome 25 profiles					
Allele Specific Methylation epigenome/allele-specific-methylation@0.1	CHANGES_REQD	–	IIICCIICCCXXC	W~PF-F	20/29/11/2
ATAC Peak Counts epigenome/atac-peak-counts@0.1	CHANGES_REQD	–	IIICCIICCCIXC	W~PF-F	20/29/11/2
ChIP-seq Peaks epigenome/chip-seq-peaks@0.1	CHANGES_REQD	–	IIICCIICCCXXC	W~PF-F	20/29/11/2
Chromatin Accessibility epigenome/chromatin-accessibility@0.1	CHANGES_REQD	–	IIICCIICCCXXC	W~PF-F	20/29/11/2
Chromatin Contact Matrix epigenome/chromatin-contact-matrix@0.1	BLOCK	P-AXES	IIICCIICCCIXC	F~PF-F	22/34/12/2
Chromatin State epigenome/chromatin-state@0.1	CHANGES_REQD	–	IIIXXXIICCIXC	W~PF-F	20/29/11/2
CpG Methylation Profile epigenome/cpg-methylation-profile@0.1	CHANGES_REQD	–	IIICCIICCCIXC	W~PF-F	20/29/11/2
DNA Methylation Fraction epigenome/dna-methylation-fraction@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	17/29/11/2

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Enhancer Activity epigenome/enhancer-activity@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	20/29/11/2
Epigenetic Clock epigenome/epigenetic-clock@0.1	CHANGES_REQD	–	IIICICIIICCCXXC	W~PF-F	18/29/11/2
Epigenetic Trajectory epigenome/epigenetic-trajectory@0.1	CHANGES_REQD	–	IIIIICIIICCIIXC	W~PF-F	20/34/12/2
Footprint Score epigenome/footprint-score@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	17/29/11/2
Hi-C Contact Map epigenome/hic-contact-map@0.1	BLOCK	P-AXES	IIICCIICCCIXC	F~PF-F	22/34/12/2
Histone Modification Signal epigenome/histone-modification-signal@0.1	CHANGES_REQD	–	IIICCCICCCIXC	W~PF-F	20/29/11/2
Methylation Age epigenome/methylation-age@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	20/29/11/2
Methylation Beta Value epigenome/methylation-beta-value@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	17/29/11/2
Methylation M Value epigenome/methylation-m-value@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	17/29/11/2
Motif Accessibility epigenome/motif-accessibility@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	20/29/11/2
Nucleosome Positioning epigenome/nucleosome-positioning@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	20/29/11/2
Promoter Activity epigenome/promoter-activity@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	20/29/11/2
Regulatory Element Annotation epigenome/regulatory-element-annotation@0.1	CHANGES_REQD	–	IIIXXXIICCCIXC	W~PF-F	20/29/11/2
Single-cell Epigenome epigenome/single-cell-epigenome@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	20/29/11/2
Spatial Epigenome epigenome/spatial-epigenome@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	20/29/11/2
Topological Domain epigenome/topological-domain@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	20/29/11/2
Transcription Factor Binding epigenome/transcription-factor-binding@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	20/29/11/2
evolution 25 profiles					
Adaptation Score evolution/adaptation-score@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	23/32/5/4
Allele Frequency Trajectory evolution/allele-frequency-trajectory@0.1	CHANGES_REQD	–	IIIIICCIICCIXX	W~PF-F	26/37/6/4
Ancestral Sequence evolution/ancestral-sequence@0.1	CHANGES_REQD	–	IIIXIXIICCCIXX	W~PF-F	21/32/5/4
Ancestral State evolution/ancestral-state@0.1	CHANGES_REQD	–	IIIXXXIICCCIXX	W~PF-F	23/32/5/4
Coalescent Tree evolution/coalescent-tree@0.1	CHANGES_REQD	–	IIIXXXIICCCIXX	W~PF-F	23/32/5/4
Divergence Time evolution/divergence-time@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	23/32/5/4
Effective Population Size evolution/effective-population-size@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	23/32/5/4

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Evolutionary Distance evolution/evolutionary-distance@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	23/32/5/4
Evolutionary Simulation State evolution/evolutionary-simulation-state@0.1	CHANGES_REQD	–	IIIXXXIICCIXX	W~PF-F	23/32/5/4
Extinction Event evolution/extinction-event@0.1	CHANGES_REQD	–	IIIXXXIICCXXX	W~PF-F	23/32/5/4
Fitness Landscape evolution/fitness-landscape@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	23/32/5/4
Fitness Value evolution/fitness-value@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	23/32/5/4
Gene Family evolution/gene-family@0.1	CHANGES_REQD	–	IIICICICCCXXX	W~PF-F	21/32/5/4
Horizontal Transfer Event evolution/horizontal-transfer-event@0.1	CHANGES_REQD	–	IIIXXXIICCXXX	W~PF-F	23/32/5/4
Linkage Disequilibrium evolution/linkage-disequilibrium@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	23/32/5/4
Mutation Rate evolution/mutation-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	23/32/5/4
Orthology Relation evolution/orthology-relation@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	23/32/5/4
Paralogy Relation evolution/paralogy-relation@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	23/32/5/4
Phylogenetic Network evolution/phylogenetic-network@0.1	CHANGES_REQD	–	IIIXIXIICCIXX	W~PF-F	21/32/5/4
Phylogenetic Tree evolution/phylogenetic-tree@0.1	CHANGES_REQD	–	IIIXIXIICCIXX	W~PF-F	21/32/5/4
Population Genetic Statistic evolution/population-genetic-statistic@0.1	CHANGES_REQD	–	IIICICICCCXXX	W~PF-F	21/32/5/4
Recombination Rate evolution/recombination-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	23/32/5/4
Selection Coefficient evolution/selection-coefficient@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	23/32/5/4
Speciation Event evolution/speciation-event@0.1	CHANGES_REQD	–	IIIXXXIICCXXX	W~PF-F	23/32/5/4
Substitution Model evolution/substitution-model@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	23/32/5/4
genome 25 profiles					
Allele Frequency genome/allele-frequency@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	26/29/11/3
Copy Number State genome/copy-number-state@0.1	CHANGES_REQD	–	IIIXCXIICCIXC	W~PF-F	26/29/11/3
Copy Number Variant genome/copy-number-variant@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	24/29/11/3
CRISPR Edit genome/crispr-edit@0.1	CHANGES_REQD	–	IIIXCXIICCXXC	W~PF-F	26/29/11/3
Gene Mutation Vector genome/gene-mutation-vector@0.1	CHANGES_REQD	–	IIIIIICCCIXC	W~PF-F	26/34/12/3
Gene Presence/Absence genome/gene-presence-absence@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	24/29/11/3

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Genome Annotation genome/genome-annotation@0.1	CHANGES_REQD	–	IIIXCXICCCIXC	W~PF-F	26/29/11/3
Genomic Region genome/genomic-region@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	26/29/11/3
Genomic Variant genome/genomic-variant@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	24/29/11/3
Genotype genome/genotype@0.1	CHANGES_REQD	–	IIIXCXICCCIXC	W~PF-F	26/29/11/3
Guide RNA genome/guide-rna@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	26/29/11/3
Haplotype genome/haplotype@0.1	CHANGES_REQD	–	IIIXCXICCCIXC	W~PF-F	26/29/11/3
Indel genome/indel@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	26/29/11/3
Paired-end Reads genome/paired-end-reads@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	26/29/11/3
Phased Genotype genome/phased-genotype@0.1	CHANGES_REQD	–	IIIXCXICCCIXC	W~PF-F	26/29/11/3
Ploidy State genome/ploidy-state@0.1	CHANGES_REQD	–	IIIXCXICCCIXC	W~PF-F	26/29/11/3
Polygenic Score genome/polygenic-score@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	26/29/11/3
Raw Sequence genome/raw-sequence@0.1	CHANGES_REQD	–	IIIXIXICCCIXC	W~PF-F	24/29/11/3
Read Pileup genome/read-pileup@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	26/29/11/3
Reference Sequence genome/reference-sequence@0.1	CHANGES_REQD	–	IIIXIXICCCIXC	W~PF-F	24/29/11/3
Sequence Alignment genome/sequence-alignment@0.1	CHANGES_REQD	–	IIIXIXICCCIXC	W~PF-F	24/29/11/3
Sequencing Read genome/sequencing-read@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	26/29/11/3
Single Nucleotide Variant genome/single-nucleotide-variant@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	24/29/11/3
Structural Variant genome/structural-variant@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	24/29/11/3
Variant Consequence genome/variant-consequence@0.1	CHANGES_REQD	–	IIIXIXICCCIXC	W~PF-F	24/29/11/3
imaging 25 profiles					
Bounding Box imaging/bounding-box@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	17/34/11/3
Brightfield Image imaging/brightfield-image@0.1	CHANGES_REQD	–	IIICCCICCCIXC	W~PF-F	20/39/12/3
Cell Tracking imaging/cell-tracking@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	17/34/11/3
Channel Definition imaging/channel-definition@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	17/34/11/3
CT Volume imaging/ct-volume@0.1	BLOCK	P-MEAS P-CONV P-AXES	IIICCIICCCIXC	F~PFFF	20/39/12/3

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Fluorescence Image imaging/fluorescence-image@0.1	CHANGES_REQD	-	IIIIICCICCCIXC	W~PF-F	20/39/12/3
Histology Slide imaging/histology-slide@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	17/34/11/3
Image Artifact imaging/image-artifact@0.1	CHANGES_REQD	-	IIIIICCICCCIXC	W~PF-F	20/39/12/3
Image Features imaging/image-features@0.1	CHANGES_REQD	-	IIIIICCICCCIXC	W~PF-F	20/39/12/3
Image Time Series imaging/image-time-series@0.1	BLOCK	P-MEAS P-CONV	IIIIICIICCIXC	F~PFFF	20/39/12/3
Imaging Provenance imaging/imaging-provenance@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	17/34/11/3
Instance Mask imaging/instance-mask@0.1	CHANGES_REQD	-	IIICCCICCCIXC	W~PF-F	17/34/11/3
Label Map imaging/label-map@0.1	CHANGES_REQD	-	IIIIICCICCCIXC	W~PF-F	20/39/12/3
Landmark Set imaging/landmark-set@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	17/34/11/3
Microscopy Image imaging/microscopy-image@0.1	CHANGES_REQD	-	IIIIICCICCCIXC	W~PF-F	20/39/12/3
Morphological Features imaging/morphological-features@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	17/34/11/3
MRI Volume imaging/mri-volume@0.1	BLOCK	P-MEAS P-CONV P-AXES	IIIIICIICCIXC	F~PFFF	20/39/12/3
PET Volume imaging/pet-volume@0.1	BLOCK	P-MEAS P-CONV P-AXES	IIIIICIICCIXC	F~PFFF	20/39/12/3
Probability Map imaging/probability-map@0.1	CHANGES_REQD	-	IIIIICCICCCIXC	W~PF-F	20/39/12/3
Radiology Image imaging/radiology-image@0.1	CHANGES_REQD	-	IIIIICCICCCIXC	W~PF-F	20/39/12/3
Radiomics Vector imaging/radiomics-vector@0.1	CHANGES_REQD	-	IIIIICCICCCIXC	W~PF-F	20/39/12/3
Region of Interest imaging/region-of-interest@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	17/34/11/3
Segmentation Mask imaging/segmentation-mask@0.1	CHANGES_REQD	-	IIICCCICCCIXC	W~PF-F	17/34/11/3
Spatial Resolution imaging/spatial-resolution@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	17/34/11/3
Ultrasound Image imaging/ultrasound-image@0.1	CHANGES_REQD	-	IIIIICCICCCIXC	W~PF-F	20/39/12/3
immunology 25 profiles					
Activation Marker immunology/activation-marker@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	10/36/12/3
Antibody Identity immunology/antibody-identity@0.1	CHANGES_REQD	-	IIIXIXICCCIXC	W~PF-F	8/36/12/3
Antibody Titer immunology/antibody-titer@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCXXC	F~PFFF	7/36/12/3
Antigen Identity immunology/antigen-identity@0.1	CHANGES_REQD	-	IIIXIXICCCIXC	W~PF-F	8/36/12/3

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
B Cell Receptor Sequence immunology/b-cell-receptor-sequence@0.1	CHANGES_REQD	–	IIIXIXICCCIXC	W~PF-F	8/36/12/3
Chemokine Concentration immunology/chemokine-concentration@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	7/36/12/3
Clonotype immunology/clonotype@0.1	CHANGES_REQD	–	IIIXXXICCCIXC	W~PF-F	10/36/12/3
Cytokine Concentration immunology/cytokine-concentration@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	7/36/12/3
Cytokine Panel immunology/cytokine-panel@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	10/36/12/3
Cytotoxicity immunology/cytotoxicity@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	10/36/12/3
Epitope immunology/epitope@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	10/36/12/3
Exhaustion Score immunology/exhaustion-score@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	7/36/12/3
Immune Cell Count immunology/immune-cell-count@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	10/36/12/3
Immune Cell Fraction immunology/immune-cell-fraction@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	7/36/12/3
Immune Cell State immunology/immune-cell-state@0.1	CHANGES_REQD	–	IIIXXXICCCIXC	W~PF-F	10/36/12/3
Immune Cell Type immunology/immune-cell-type@0.1	CHANGES_REQD	–	IIIXXXICCCIXC	W~PF-F	10/36/12/3
Immune Infiltration immunology/immune-infiltration@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	10/36/12/3
Immune Response Trajectory immunology/immune-response-trajectory@0.1	CHANGES_REQD	–	IIICCIICCCIXC	W~PF-F	13/41/13/3
Inflammatory Score immunology/inflammatory-score@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	7/36/12/3
MHC Allele immunology/mhc-allele@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	10/36/12/3
Neutralization Titer immunology/neutralization-titer@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	7/36/12/3
Peptide MHC Binding immunology/peptide-mhc-binding@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	10/36/12/3
Repertoire Frequency immunology/repertoire-frequency@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	10/36/12/3
T Cell Receptor Sequence immunology/t-cell-receptor-sequence@0.1	CHANGES_REQD	–	IIIXIXICCCIXC	W~PF-F	8/36/12/3
Vaccine Response immunology/vaccine-response@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	10/36/12/3
lipidome-glycome 25 profiles					
Eicosanoid Profile lipidome-glycome/eicosanoid-profile@0.1	CHANGES_REQD	–	IIICCCICCCIXC	W~PF-F	16/34/11/2
Fatty Acid Composition lipidome-glycome/fatty-acid-composition@0.1	CHANGES_REQD	–	IIICCCICCCIXC	W~PF-F	16/34/11/2
Glycan Abundance lipidome-glycome/glycan-abundance@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCIXC	F~PFFF	13/34/11/2

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Glycan Identity lipidome-glycome/glycan-identity@0.1	CHANGES_REQD	-	IIIXIXICCCIXC	W~PF-F	14/34/11/2
Glycan Motif lipidome-glycome/glycan-motif@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	16/34/11/2
Glycan Structure lipidome-glycome/glycan-structure@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	16/34/11/2
Glycomics Matrix lipidome-glycome/glycomics-matrix@0.1	CHANGES_REQD	-	IIIIICCCICXC	W~PF-F	18/39/12/2
Glycopeptide Abundance lipidome-glycome/glycopeptide-abundance@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCIXC	F~PFFF	13/34/11/2
Glycoprotein Site lipidome-glycome/glycoprotein-site@0.1	CHANGES_REQD	-	IIICICICCCXXC	W~PF-F	14/34/11/2
Glycosylation State lipidome-glycome/glycosylation-state@0.1	CHANGES_REQD	-	IIIXXXICCCIXC	W~PF-F	16/34/11/2
Lectin Binding lipidome-glycome/lectin-binding@0.1	CHANGES_REQD	-	IIICCCICCCXXC	W~PF-F	16/34/11/2
Lipid Abundance lipidome-glycome/lipid-abundance@0.1	BLOCK	P-MEAS P-CONV	IIICIIICCCIXC	F~PFFF	11/34/11/2
Lipid Class Profile lipidome-glycome/lipid-class-profile@0.1	CHANGES_REQD	-	IIICICICCCIXC	W~PF-F	14/34/11/2
Lipid Concentration lipidome-glycome/lipid-concentration@0.1	BLOCK	P-MEAS P-CONV	IIICIIICCCXXC	F~PFFF	11/34/11/2
Lipid Droplet Content lipidome-glycome/lipid-droplet-content@0.1	CHANGES_REQD	-	IIICICICCCXXC	W~PF-F	14/34/11/2
Lipid Identity lipidome-glycome/lipid-identity@0.1	CHANGES_REQD	-	IIIXIXICCCIXC	W~PF-F	14/34/11/2
Lipid Metabolism Flux lipidome-glycome/lipid-metabolism-flux@0.1	CHANGES_REQD	-	IIICICICCCXXC	W~PF-F	14/34/11/2
Lipid Species Vector lipidome-glycome/lipid-species-vector@0.1	CHANGES_REQD	-	IIIIICCCICXC	W~PF-F	16/39/12/2
Lipidomics Matrix lipidome-glycome/lipidomics-matrix@0.1	CHANGES_REQD	-	IIIIICCCICXC	W~PF-F	16/39/12/2
Lipoprotein Particle lipidome-glycome/lipoprotein-particle@0.1	CHANGES_REQD	-	IIICICICCCXXC	W~PF-F	14/34/11/2
Membrane Lipid Composition lipidome-glycome/membrane-lipid-composition@0.1	CHANGES_REQD	-	IIICICICCCIXC	W~PF-F	14/34/11/2
Monosaccharide Composition lipidome-glycome/monosaccharide-composition@0.1	CHANGES_REQD	-	IIICCCICCCIXC	W~PF-F	16/34/11/2
Phospholipid Profile lipidome-glycome/phospholipid-profile@0.1	CHANGES_REQD	-	IIICICICCCIXC	W~PF-F	14/34/11/2
Sphingolipid Profile lipidome-glycome/sphingolipid-profile@0.1	CHANGES_REQD	-	IIICICICCCIXC	W~PF-F	14/34/11/2
Sterol Profile lipidome-glycome/sterol-profile@0.1	CHANGES_REQD	-	IIICCCICCCIXC	W~PF-F	16/34/11/2
metabolome 25 profiles					
Chromatographic Peak metabolome/chromatographic-peak@0.1	CHANGES_REQD	-	IIIXXXICCCIXC	W~PF-F	19/39/11/5
Compartment Metabolite metabolome/compartment-metabolite@0.1	CHANGES_REQD	-	IIICICICCCXXC	W~PF-F	12/34/11/5

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Energy Charge metabolome/energy-charge@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	14/34/11/5
Exchange Flux metabolome/exchange-flux@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	14/34/11/5
Extracellular Metabolite metabolome/extracellular-metabolite@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	12/34/11/5
Flux Vector metabolome/flux-vector@0.1	CHANGES_REQD	–	IIIIICCCICXC	W~PF-F	17/39/12/5
Intracellular Metabolite metabolome/intracellular-metabolite@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	12/34/11/5
Isotope Tracer Fraction metabolome/isotope-tracer-fraction@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	11/34/11/5
Mass Spectral Peak metabolome/mass-spectral-peak@0.1	BLOCK	P-MEAS P-CONV P-KG	IIICCIICCCXXC	F~FFFF	11/34/11/5
Mass Spectrum metabolome/mass-spectrum@0.1	BLOCK	P-MEAS P-CONV P-KG	IIICCIICCCIXC	F~FFFF	11/34/11/5
Metabolic Flux metabolome/metabolic-flux@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	14/34/11/5
Metabolite Abundance metabolome/metabolite-abundance@0.1	BLOCK	P-MEAS P-CONV	IIICIIICCCIXC	F~PFFF	9/34/11/5
Metabolite Clearance metabolome/metabolite-clearance@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	12/34/11/5
Metabolite Concentration metabolome/metabolite-concentration@0.1	BLOCK	P-MEAS P-CONV	IIICIIICCCXXC	F~PFFF	9/34/11/5
Metabolite Consumption Rate metabolome/metabolite-consumption-rate@0.1	BLOCK	P-MEAS P-CONV	IIICIIICCCXXC	F~PFFF	9/34/11/5
Metabolite Identity metabolome/metabolite-identity@0.1	CHANGES_REQD	–	IIIXIXICCCIXC	W~PF-F	12/34/11/5
Metabolite Production Rate metabolome/metabolite-production-rate@0.1	BLOCK	P-MEAS P-CONV	IIICIIICCCXXC	F~PFFF	9/34/11/5
Metabolite Ratio metabolome/metabolite-ratio@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	12/34/11/5
Metabolite Time Series metabolome/metabolite-time-series@0.1	BLOCK	P-MEAS P-CONV	IIIIIIICCCXXC	F~PFFF	9/34/11/5
Metabolomics Feature metabolome/metabolomics-feature@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	14/34/11/5
Pathway Metabolite Score metabolome/pathway-metabolite-score@0.1	BLOCK	P-MEAS P-CONV	IIICIIICCCXXC	F~PFFF	9/34/11/5
Redox Ratio metabolome/redox-ratio@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	14/34/11/5
Retention Time metabolome/retention-time@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	11/34/11/5
Targeted Metabolomics Panel metabolome/targeted-metabolomics-panel@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	14/34/11/5
Untargeted Metabolomics Matrix metabolome/untargeted-metabolomics-matrix@0.1	CHANGES_REQD	–	IIIIICCCICXC	W~PF-F	17/39/12/5
microbiology 25 profiles					
Alpha Diversity microbiology/alpha-diversity@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	17/34/10/2

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Amplicon Sequence Variant microbiology/amplicon-sequence-variant@0.1	CHANGES_REQD	-	IIIXIXICCCIXX	W~PF-F	15/34/10/2
Antimicrobial Susceptibility microbiology/antimicrobial-susceptibility@0.1	CHANGES_REQD	-	IIICCCICCCXXX	W~PF-F	17/34/10/2
Beta Diversity microbiology/beta-diversity@0.1	CHANGES_REQD	-	IIICCCICCCXXX	W~PF-F	17/34/10/2
Biofilm Biomass microbiology/biofilm-biomass@0.1	BLOCK	P-MEAS P-CONV P-KG	IIICCIICCCXXX	F~FFFF	14/34/10/2
Colony Forming Units microbiology/colony-forming-units@0.1	CHANGES_REQD	-	IIICCCICCCXXX	W~PF-F	17/34/10/2
Community Time Series microbiology/community-time-series@0.1	BLOCK	P-MEAS P-CONV	IIICIIICCCXXX	F~PFFF	13/34/10/2
Culture Condition microbiology/culture-condition@0.1	CHANGES_REQD	-	IIICCCICCCXXX	W~PF-F	17/34/10/2
Fermentation Profile microbiology/fermentation-profile@0.1	CHANGES_REQD	-	IIICCCICCCIXX	W~PF-F	17/34/10/2
Functional Abundance microbiology/functional-abundance@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCIXX	F~PFFF	14/34/10/2
Host Microbe Interaction microbiology/host-microbe-interaction@0.1	CHANGES_REQD	-	IIICCCICCCXXX	W~PF-F	17/34/10/2
Metagenomic Counts microbiology/metagenomic-counts@0.1	CHANGES_REQD	-	IIICCCICCCIXX	W~PF-F	17/34/10/2
Metatranscriptomic Counts microbiology/metatranscriptomic-counts@0.1	CHANGES_REQD	-	IIICCCICCCIXX	W~PF-F	17/34/10/2
Microbial Community Profile microbiology/microbial-community-profile@0.1	CHANGES_REQD	-	IIICCCICCCIXX	W~PF-F	17/34/10/2
Microbial Gene Content microbiology/microbial-gene-content@0.1	CHANGES_REQD	-	IIICICICCCXXX	W~PF-F	15/34/10/2
Microbial Genome microbiology/microbial-genome@0.1	CHANGES_REQD	-	IIICCCICCCXXX	W~PF-F	17/34/10/2
Microbial Growth Rate microbiology/microbial-growth-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	14/34/10/2
Microbial Metabolite Exchange microbiology/microbial-metabolite-exchange@0.1	CHANGES_REQD	-	IIICICICCCXXX	W~PF-F	15/34/10/2
Microbial Taxon microbiology/microbial-taxon@0.1	CHANGES_REQD	-	IIICICICCCXXX	W~PF-F	15/34/10/2
Minimum Inhibitory Concentration microbiology/minimum-inhibitory-concentration@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	14/34/10/2
Operational Taxonomic Unit microbiology/operational-taxonomic-unit@0.1	CHANGES_REQD	-	IIICICICCCXXX	W~PF-F	15/34/10/2
Resistance Gene Profile microbiology/resistance-gene-profile@0.1	CHANGES_REQD	-	IIICICICCCIXX	W~PF-F	15/34/10/2
Strain Identity microbiology/strain-identity@0.1	CHANGES_REQD	-	IIIXIXICCCIXX	W~PF-F	15/34/10/2
Taxonomic Abundance microbiology/taxonomic-abundance@0.1	BLOCK	P-MEAS P-CONV	IIICIIICCCIXX	F~PFFF	12/34/10/2
Virulence Factor Profile microbiology/virulence-factor-profile@0.1	CHANGES_REQD	-	IIICCCICCCIXX	W~PF-F	17/34/10/2
multiomics 25 profiles					

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Assay Identity multiomics/assay-identity@0.1	CHANGES_REQD	–	IIIXIXICCCIXC	W~PF-F	20/40/7/2
Batch Corrected Matrix multiomics/batch-corrected-matrix@0.1	CHANGES_REQD	–	IIIIICCCICCCIXC	W~PF-F	24/45/8/2
Batch Identity multiomics/batch-identity@0.1	CHANGES_REQD	–	IIIXIXICCCIXC	W~PF-F	20/40/7/2
Biological Network multiomics/biological-network@0.1	CHANGES_REQD	–	IIIXCXICCCIXC	W~PF-F	22/40/7/2
Causal Graph multiomics/causal-graph@0.1	CHANGES_REQD	–	IIIXCXICCCIXC	W~PF-F	22/40/7/2
Cross-modal Alignment multiomics/cross-modal-alignment@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	22/40/7/2
Embedding Vector multiomics/embedding-vector@0.1	CHANGES_REQD	–	IIIIICCCICCCIXC	W~PF-F	24/45/8/2
Evidence Graph multiomics/evidence-graph@0.1	CHANGES_REQD	–	IIIXCXICCCIXC	W~PF-F	22/40/7/2
Feature Crosswalk multiomics/feature-crosswalk@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	22/40/7/2
Gene Metabolite Map multiomics/gene-metabolite-map@0.1	BLOCK	P-AXES	IIIIICCCICCCIXC	F~PF-F	22/45/8/2
Gene Protein Map multiomics/gene-protein-map@0.1	BLOCK	P-AXES	IIIIICCCICCCIXC	F~PF-F	22/45/8/2
Gene Set Score multiomics/gene-set-score@0.1	BLOCK	P-MEAS P-CONV	IIICIIICCCXXC	F~PFFF	20/40/7/2
Imputed Modality multiomics/imputed-modality@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	22/40/7/2
Integrated Embedding multiomics/integrated-embedding@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	22/40/7/2
Knowledge Graph multiomics/knowledge-graph@0.1	CHANGES_REQD	–	IIIXCXICCCIXC	W~PF-F	22/40/7/2
Latent Factor multiomics/latent-factor@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	22/40/7/2
Missing Modality Mask multiomics/missing-modality-mask@0.1	CHANGES_REQD	–	IIICCCICCCIXC	W~PF-F	22/40/7/2
Multimodal Matrix multiomics/multimodal-matrix@0.1	CHANGES_REQD	–	IIIIICCCICCCIXC	W~PF-F	24/45/8/2
Multi-omics Sample Map multiomics/multiomics-sample-map@0.1	BLOCK	P-AXES	IIIIICCCICCCIXC	F~PF-F	24/45/8/2
Normalized Feature Matrix multiomics/normalized-feature-matrix@0.1	CHANGES_REQD	–	IIIIICCCICCCIXC	W~PF-F	24/45/8/2
Pathway Network multiomics/pathway-network@0.1	CHANGES_REQD	–	IIIXCXICCCIXC	W~PF-F	22/40/7/2
Pathway Score multiomics/pathway-score@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	22/40/7/2
Regulatory Network multiomics/regulatory-network@0.1	CHANGES_REQD	–	IIIXCXICCCIXC	W~PF-F	22/40/7/2
Sample Identity multiomics/sample-identity@0.1	CHANGES_REQD	–	IIIXIXICCCIXC	W~PF-F	20/40/7/2

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Subject Identity multimics/subject-identity@0.1	CHANGES_REQD	–	IIIXIXICCCIXC	W~PF-F	20/40/7/2
neuroscience 25 profiles					
Action Potential Event neuroscience/action-potential-event@0.1	CHANGES_REQD	–	IIIXXXIICCXXC	W~PF-F	22/34/6/5
Behavioral Trial neuroscience/behavioral-trial@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	22/34/6/5
Brain Region neuroscience/brain-region@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	22/34/6/5
Calcium Imaging Signal neuroscience/calcium-imaging-signal@0.1	CHANGES_REQD	–	IIICCCICCCIXC	W~PF-F	22/34/6/5
Cognitive Score neuroscience/cognitive-score@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	19/34/6/5
Connectivity Matrix neuroscience/connectivity-matrix@0.1	BLOCK	P-AXES	IIIIICCCICCCIXC	F~PF-F	25/39/7/5
Connectome Graph neuroscience/connectome-graph@0.1	CHANGES_REQD	–	IIIXXXICCCIXC	W~PF-F	22/34/6/5
Electroencephalogram neuroscience/electroencephalogram@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	22/34/6/5
Firing Rate neuroscience/firing-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	19/34/6/5
Local Field Potential neuroscience/local-field-potential@0.1	BLOCK	P-AXES	IIIIICCCICCCIXC	F~PF-F	25/39/7/5
Magnetoencephalogram neuroscience/magnetoencephalogram@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	22/34/6/5
Membrane Voltage neuroscience/membrane-voltage@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	22/34/6/5
Neural Activity neuroscience/neural-activity@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	22/34/6/5
Neural Decoding Output neuroscience/neural-decoding-output@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	22/34/6/5
Neural Population State neuroscience/neural-population-state@0.1	CHANGES_REQD	–	IIIXXXICCCIXC	W~PF-F	22/34/6/5
Neural Trajectory neuroscience/neural-trajectory@0.1	CHANGES_REQD	–	IIIIICCIICCCIXC	W~PF-F	25/39/7/5
Neuron Identity neuroscience/neuron-identity@0.1	CHANGES_REQD	–	IIIXIXICCCIXC	W~PF-F	22/34/6/5
Neurotransmitter Concentration neuroscience/neurotransmitter-concentration@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	19/34/6/5
Oscillation Power neuroscience/oscillation-power@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	22/34/6/5
Receptive Field neuroscience/receptive-field@0.1	CHANGES_REQD	–	IIIIICCCICCCIXC	W~PF-F	25/39/7/5
Sensory Stimulus neuroscience/sensory-stimulus@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	22/34/6/5
Sleep Stage neuroscience/sleep-stage@0.1	CHANGES_REQD	–	IIIXXXICCCIXC	W~PF-F	22/34/6/5
Spike Train neuroscience/spike-train@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	22/34/6/5

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Synaptic Current neuroscience/synaptic-current@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	22/34/6/5
Synaptic Weight neuroscience/synaptic-weight@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	19/34/6/5
pharmacology 25 profiles					
Area Under Curve pharmacology/area-under-curve@0.1	BLOCK	P-AXES	IIIIICCIICCIXC	F~PF-F	16/45/11/5
Association Rate pharmacology/association-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	10/40/10/5
Binding Affinity pharmacology/binding-affinity@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	10/40/10/5
Clearance pharmacology/clearance@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	13/40/10/5
Compound Dose pharmacology/compound-dose@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	10/40/10/5
Concentration Response Curve pharmacology/concentration-response-curve@0.1	BLOCK	P-MEAS P-CONV P-AXES	IIIIICIIICCIXC	F~PFFF	13/45/11/5
Dissociation Constant pharmacology/dissociation-constant@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	13/40/10/5
Dissociation Rate pharmacology/dissociation-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	10/40/10/5
Dose Response Curve pharmacology/dose-response-curve@0.1	BLOCK	P-MEAS P-CONV P-AXES	IIIIICIIICCIXC	F~PFFF	13/45/11/5
Dose Schedule pharmacology/dose-schedule@0.1	BLOCK	P-MEAS P-CONV	IIIXXIICCCIXC	F~PFFF	10/40/10/5
EC50 pharmacology/ec50@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	13/40/10/5
Efficacy pharmacology/efficacy@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	13/40/10/5
Emax pharmacology/emax@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	13/40/10/5
Exposure Concentration pharmacology/exposure-concentration@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	10/40/10/5
Hill Coefficient pharmacology/hill-coefficient@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	13/40/10/5
IC50 pharmacology/ic50@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	13/40/10/5
Inhibition Constant pharmacology/inhibition-constant@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	13/40/10/5
Pharmacodynamic Parameters pharmacology/pharmacodynamic-parameters@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	13/40/10/5
Pharmacokinetic Parameters pharmacology/pharmacokinetic-parameters@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	13/40/10/5
Plasma Concentration pharmacology/plasma-concentration@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	10/40/10/5
Potency pharmacology/potency@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	13/40/10/5
Receptor Occupancy pharmacology/receptor-occupancy@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	13/40/10/5

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Target Engagement pharmacology/target-engagement@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	13/40/10/5
Tissue Concentration pharmacology/tissue-concentration@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	10/40/10/5
Volume of Distribution pharmacology/volume-of-distribution@0.1	BLOCK	P-MEAS P-CONV P-AXES	IIIIICIIICCCIXC	F~PFFF	13/45/11/5
phenotype 25 profiles					
Adverse Event phenotype/adverse-event@0.1	CHANGES_REQD	–	IIIXXXIICCCXXC	W~PF-F	16/35/7/5
Behavioral Phenotype phenotype/behavioral-phenotype@0.1	CHANGES_REQD	–	IIIXXXIICCCIXC	W~PF-F	16/35/7/5
Biomarker Panel phenotype/biomarker-panel@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/35/7/5
Biomarker Value phenotype/biomarker-value@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	13/35/7/5
Categorical Phenotype phenotype/categorical-phenotype@0.1	CHANGES_REQD	–	IIIXXXIICCCIXC	W~PF-F	16/35/7/5
Disease Severity phenotype/disease-severity@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/35/7/5
Disease Status phenotype/disease-status@0.1	CHANGES_REQD	–	IIIXXXIICCCIXC	W~PF-F	16/35/7/5
Drug Resistance phenotype/drug-resistance@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/35/7/5
Drug Response AUC phenotype/drug-response-auc@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/35/7/5
Drug Sensitivity phenotype/drug-sensitivity@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/35/7/5
Functional Score phenotype/functional-score@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	13/35/7/5
Hazard Ratio phenotype/hazard-ratio@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/35/7/5
Overall Survival phenotype/overall-survival@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/35/7/5
Phenotype Term phenotype/phenotype-term@0.1	CHANGES_REQD	–	IIIXXXIICCCIXC	W~PF-F	16/35/7/5
Phenotype Vector phenotype/phenotype-vector@0.1	CHANGES_REQD	–	IIIIICCIICCCIXC	W~PF-F	19/40/8/5
Progression Free Survival phenotype/progression-free-survival@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/35/7/5
Quantitative Trait phenotype/quantitative-trait@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/35/7/5
Response Class phenotype/response-class@0.1	CHANGES_REQD	–	IIIXXXIICCCIXC	W~PF-F	16/35/7/5
Risk Score phenotype/risk-score@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	13/35/7/5
Survival Fraction phenotype/survival-fraction@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	13/35/7/5
Symptom Score phenotype/symptom-score@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	13/35/7/5

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Time to Event phenotype/time-to-event@0.1	BLOCK	P-MEAS P-CONV	IIIXXIICCXXC	F~PFFF	13/35/7/5
Toxicity Endpoint phenotype/toxicity-endpoint@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/35/7/5
Trait Value phenotype/trait-value@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCXXC	F~PFFF	13/35/7/5
Viability at Dose phenotype/viability-at-dose@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCXXC	F~PFFF	13/35/7/5
physiology 25 profiles					
Airflow Rate physiology/airflow-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCXXC	F~PFFF	9/37/10/5
Blood Gas physiology/blood-gas@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	12/37/10/5
Blood Glucose physiology/blood-glucose@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	12/37/10/5
Blood Pressure physiology/blood-pressure@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCXXC	F~PFFF	9/37/10/5
Body Mass physiology/body-mass@0.1	BLOCK	P-MEAS P-CONV P-KG	IIICCIICCXXC	F~FFFF	9/37/10/5
Body Temperature physiology/body-temperature@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCXXC	F~PFFF	9/37/10/5
Cardiac Output physiology/cardiac-output@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	12/37/10/5
Electrolyte Concentration physiology/electrolyte-concentration@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCXXC	F~PFFF	9/37/10/5
Fluid Balance physiology/fluid-balance@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	12/37/10/5
Glomerular Filtration Rate physiology/glomerular-filtration-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCXXC	F~PFFF	9/37/10/5
Heart Rate physiology/heart-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCXXC	F~PFFF	9/37/10/5
Hormone Concentration physiology/hormone-concentration@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCXXC	F~PFFF	9/37/10/5
Liver Function Marker physiology/liver-function-marker@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	12/37/10/5
Lung Volume physiology/lung-volume@0.1	BLOCK	P-MEAS P-CONV P-AXES	IIICIICCIXC	F~PFFF	12/42/11/5
Mean Arterial Pressure physiology/mean-arterial-pressure@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCXXC	F~PFFF	9/37/10/5
Metabolic Rate physiology/metabolic-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCXXC	F~PFFF	9/37/10/5
Muscle Force physiology/muscle-force@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	12/37/10/5
Osmolarity physiology/osmolarity@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	12/37/10/5
Oxygen Saturation physiology/oxygen-saturation@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	12/37/10/5
pH physiology/ph@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	12/37/10/5

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Physiological Time Series physiology/physiological-time-series@0.1	BLOCK	P-MEAS P-CONV	IIIIICIIICCCXXC	F~PFFF	9/37/10/5
Respiratory Rate physiology/respiratory-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	9/37/10/5
Stroke Volume physiology/stroke-volume@0.1	BLOCK	P-MEAS P-CONV P-AXES	IIIIICIIICCCIXC	F~PFFF	12/42/11/5
Urine Flow physiology/urine-flow@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	12/37/10/5
Vascular Resistance physiology/vascular-resistance@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	12/37/10/5
proteome 25 profiles					
Active Site proteome/active-site@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	11/39/11/2
Antibody Binding proteome/antibody-binding@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	11/39/11/2
Binding Site proteome/binding-site@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	11/39/11/2
Enzyme Activity proteome/enzyme-activity@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	11/39/11/2
Peptide Abundance proteome/peptide-abundance@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCIXC	F~PFFF	8/39/11/2
Phosphoprotein Abundance proteome/phosphoprotein-abundance@0.1	BLOCK	P-MEAS P-CONV	IIICIIICCCIXC	F~PFFF	6/39/11/2
Phosphorylation Site State proteome/phosphorylation-site-state@0.1	CHANGES_REQD	–	IIIXCXICCCIXC	W~PF-F	11/39/11/2
Post-translational Modification proteome/post-translational-modification@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	11/39/11/2
Protein Abundance proteome/protein-abundance@0.1	BLOCK	P-MEAS P-CONV	IIICIIICCCIXC	F~PFFF	6/39/11/2
Protein Complex proteome/protein-complex@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	9/39/11/2
Protein Domain proteome/protein-domain@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	9/39/11/2
Protein Expression Matrix proteome/protein-expression-matrix@0.1	CHANGES_REQD	–	IIIIICICCCIXC	W~PF-F	12/44/12/2
Protein Fold Change proteome/protein-fold-change@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	9/39/11/2
Protein Identity proteome/protein-identity@0.1	CHANGES_REQD	–	IIIXIXICCCIXC	W~PF-F	9/39/11/2
Protein Interaction Edge proteome/protein-interaction-edge@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	9/39/11/2
Protein Interaction Network proteome/protein-interaction-network@0.1	CHANGES_REQD	–	IIIXIXICCCIXC	W~PF-F	14/44/11/2
Protein Localization proteome/protein-localization@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	9/39/11/2
Protein Sequence proteome/protein-sequence@0.1	CHANGES_REQD	–	IIIXIXICCCIXC	W~PF-F	9/39/11/2
Protein Stability proteome/protein-stability@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	9/39/11/2

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Protein Structure proteome/protein-structure@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	9/39/11/2
Protein Turnover proteome/protein-turnover@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	9/39/11/2
Proteoform proteome/proteoform@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	11/39/11/2
Residue Coordinate proteome/residue-coordinate@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	11/39/11/2
Secondary Structure proteome/secondary-structure@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	11/39/11/2
Secretion Rate proteome/secretion-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXC	F~PFFF	8/39/11/2
simulation 25 profiles					
Agent Population simulation/agent-population@0.1	CHANGES_REQD	–	IIICCCXCCCXXX	W~PF-F	15/35/6/6
Agent State simulation/agent-state@0.1	CHANGES_REQD	–	IIIXXXCCCIXX	W~PF-F	15/35/6/6
Boundary Condition simulation/boundary-condition@0.1	CHANGES_REQD	–	IIICCCXCCCXXX	W~PF-F	15/35/6/6
Calibration Result simulation/calibration-result@0.1	CHANGES_REQD	–	IIICCCXCCCXXX	W~PF-F	15/35/6/6
Compartment State simulation/compartment-state@0.1	CHANGES_REQD	–	IIIXXXCCCIXX	W~PF-F	15/35/6/6
Discrete Event simulation/discrete-event@0.1	CHANGES_REQD	–	IIIXXXICCCXX	W~PF-F	15/35/6/6
Event Trigger simulation/event-trigger@0.1	CHANGES_REQD	–	IIIXXXICCCXX	W~PF-F	15/35/6/6
Flux Balance Solution simulation/flux-balance-solution@0.1	CHANGES_REQD	–	IIICCCXCCCXXX	W~PF-F	15/35/6/6
Forcing Function simulation/forcing-function@0.1	CHANGES_REQD	–	IIICCCXCCCXXX	W~PF-F	15/35/6/6
Initial Condition simulation/initial-condition@0.1	CHANGES_REQD	–	IIICCCXCCCXXX	W~PF-F	15/35/6/6
Ordinary Differential Equation State simulation/ordinary-differential-equation-state@0.1	CHANGES_REQD	–	IIIXXXCCCIXX	W~PF-F	15/35/6/6
Parameter Distribution simulation/parameter-distribution@0.1	CHANGES_REQD	–	IIICCCXCCCXXX	W~PF-F	15/35/6/6
Parameter Vector simulation/parameter-vector@0.1	CHANGES_REQD	–	IIICCCXCCCIXX	W~PF-F	13/35/6/6
Partial Differential Equation Field simulation/partial-differential-equation-field@0.1	CHANGES_REQD	–	IIICCCXCCCIXX	W~PF-F	13/35/6/6
Sensitivity Index simulation/sensitivity-index@0.1	CHANGES_REQD	–	IIICCCXCCCXXX	W~PF-F	15/35/6/6
Simulation Checkpoint simulation/simulation-checkpoint@0.1	CHANGES_REQD	–	IIICCCXCCCXXX	W~PF-F	15/35/6/6
Simulation Parameter simulation/simulation-parameter@0.1	CHANGES_REQD	–	IIICCCXCCCXXX	W~PF-F	15/35/6/6
Simulation Step simulation/simulation-step@0.1	CHANGES_REQD	–	IIICCCXCCCXXX	W~PF-F	15/35/6/6

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Simulation Time simulation/simulation-time@0.1	BLOCK	P-MEAS P-CONV	IIICCXCCCXX	F~PFFF	15/35/6/6
Solver Configuration simulation/solver-configuration@0.1	CHANGES_REQD	–	IIICCXCCCXX	W~PF-F	15/35/6/6
Solver Tolerance simulation/solver-tolerance@0.1	CHANGES_REQD	–	IIICCXCCCXX	W~PF-F	15/35/6/6
State Trajectory simulation/state-trajectory@0.1	CHANGES_REQD	–	IIICCXICCIX	W~PF-F	13/35/6/6
State Variable simulation/state-variable@0.1	CHANGES_REQD	–	IIIXXXCCCIX	W~PF-F	15/35/6/6
State Vector simulation/state-vector@0.1	CHANGES_REQD	–	IIICCXCCCIX	W~PF-F	13/35/6/6
Stochastic Process State simulation/stochastic-process-state@0.1	CHANGES_REQD	–	IIIXXXCCCIX	W~PF-F	15/35/6/6
spatial 25 profiles					
Adjacency Graph spatial/adjacency-graph@0.1	CHANGES_REQD	–	IIIXXXICCIX	W~PF-F	23/31/6/5
Anatomical Coordinate spatial/anatomical-coordinate@0.1	CHANGES_REQD	–	IIICCCICCCXX	W~PF-F	23/31/6/5
Cellular Coordinate spatial/cellular-coordinate@0.1	CHANGES_REQD	–	IIICCCICCCXX	W~PF-F	23/31/6/5
Concentration Field spatial/concentration-field@0.1	BLOCK	P-MEAS P-CONV	IIICICIICCIX	F~PFFF	21/31/6/5
Deformation Field spatial/deformation-field@0.1	CHANGES_REQD	–	IIICCCICCCIX	W~PF-F	21/31/6/5
Density Field spatial/density-field@0.1	CHANGES_REQD	–	IIICCCICCCIX	W~PF-F	21/31/6/5
Displacement Field spatial/displacement-field@0.1	CHANGES_REQD	–	IIICCCICCCIX	W~PF-F	21/31/6/5
Distance Matrix spatial/distance-matrix@0.1	BLOCK	P-MEAS P-CONV P-AXES	IIICICIICCIX	F~PFFF	21/31/6/5
Genomic Coordinate spatial/genomic-coordinate@0.1	CHANGES_REQD	–	IIICCCICCCXX	W~PF-F	23/31/6/5
Geographic Coordinate spatial/geographic-coordinate@0.1	CHANGES_REQD	–	IIIXXXICCIX	W~PF-F	23/31/6/5
Region Annotation spatial/region-annotation@0.1	CHANGES_REQD	–	IIIXXXICCIX	W~PF-F	23/31/6/5
Spatial Abundance Map spatial/spatial-abundance-map@0.1	BLOCK	P-MEAS P-CONV	IIICICIICCIX	F~PFFF	21/31/6/5
Spatial Cell Type Map spatial/spatial-cell-type-map@0.1	CHANGES_REQD	–	IIICCCICCCIX	W~PF-F	21/31/6/5
Spatial Colocalization spatial/spatial-colocalization@0.1	CHANGES_REQD	–	IIICCCICCCXX	W~PF-F	23/31/6/5
Spatial Coordinate spatial/spatial-coordinate@0.1	CHANGES_REQD	–	IIICCCICCCXX	W~PF-F	23/31/6/5
Spatial Expression Map spatial/spatial-expression-map@0.1	CHANGES_REQD	–	IIICCCICCCIX	W~PF-F	21/31/6/5
Spatial Gradient spatial/spatial-gradient@0.1	CHANGES_REQD	–	IIICCCICCCXX	W~PF-F	23/31/6/5

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Spatial Grid spatial/spatial-grid@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	23/31/6/5
Spatial Mesh spatial/spatial-mesh@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	23/31/6/5
Spatial Neighborhood spatial/spatial-neighborhood@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	23/31/6/5
Spatial Point Set spatial/spatial-point-set@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	23/31/6/5
Spatial Region spatial/spatial-region@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	23/31/6/5
Spatial Trajectory spatial/spatial-trajectory@0.1	CHANGES_REQD	–	IIICCCIICCIXX	W~PF-F	21/31/6/5
Tissue Coordinate spatial/tissue-coordinate@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	23/31/6/5
Velocity Field spatial/velocity-field@0.1	CHANGES_REQD	–	IIICCCICCCIXX	W~PF-F	21/31/6/5
transcriptome 25 profiles					
Alternative Splicing Event transcriptome/alternative-splicing-event@0.1	CHANGES_REQD	–	IIIXCXIICCXXC	W~PF-F	16/33/10/2
Differential Expression transcriptome/differential-expression@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/33/10/2
Expression Signature transcriptome/expression-signature@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/33/10/2
Expression Trajectory transcriptome/expression-trajectory@0.1	CHANGES_REQD	–	IIICCCIICCIXC	W~PF-F	18/38/11/2
Fold Change Vector transcriptome/fold-change-vector@0.1	CHANGES_REQD	–	IIICCCICCCIXC	W~PF-F	18/38/11/2
Gene Expression Counts transcriptome/gene-expression-counts@0.1	CHANGES_REQD	–	IIICICICCCIXC	W~PF-F	14/33/10/2
Gene Set Activity transcriptome/gene-set-activity@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	14/33/10/2
lncRNA Expression transcriptome/lncrna-expression@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/33/10/2
microRNA Expression transcriptome/microrna-expression@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/33/10/2
Nascent Transcription transcriptome/nascent-transcription@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/33/10/2
Normalized Gene Expression transcriptome/normalized-gene-expression@0.1	CHANGES_REQD	–	IIICICICCCXXC	W~PF-F	14/33/10/2
Pathway Activity transcriptome/pathway-activity@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/33/10/2
Pseudobulk Expression transcriptome/pseudobulk-expression@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/33/10/2
RNA Editing Event transcriptome/rna-editing-event@0.1	CHANGES_REQD	–	IIIXCXIICCXXC	W~PF-F	16/33/10/2
RNA Sequence transcriptome/rna-sequence@0.1	CHANGES_REQD	–	IIIXIXIICCIXC	W~PF-F	14/33/10/2
RNA Structure transcriptome/rna-structure@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/33/10/2

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
RNA Velocity transcriptome/rna-velocity@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/33/10/2
Single-cell Expression Matrix transcriptome/single-cell-expression-matrix@0.1	CHANGES_REQD	–	IIIIICCCICCCIXC	W~PF-F	18/38/11/2
Small RNA Abundance transcriptome/small-rna-abundance@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCIXC	F~PFFF	13/33/10/2
Spatial Transcriptomics Matrix transcriptome/spatial-transcriptomics-matrix@0.1	CHANGES_REQD	–	IIIIICCCICCCIXC	W~PF-F	18/38/11/2
Splice Junction Counts transcriptome/splice-junction-counts@0.1	CHANGES_REQD	–	IIICCCICCCIXC	W~PF-F	16/33/10/2
Transcript Expression transcriptome/transcript-expression@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/33/10/2
Transcript Isoform Usage transcriptome/transcript-isoform-usage@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/33/10/2
Transcription Factor Activity transcriptome/transcription-factor-activity@0.1	CHANGES_REQD	–	IIICCCICCCXXC	W~PF-F	16/33/10/2
Transcriptional State transcriptome/transcriptional-state@0.1	CHANGES_REQD	–	IIIXCXICCCIXC	W~PF-F	16/33/10/2
virology 25 profiles					
Antiviral Susceptibility virology/antiviral-susceptibility@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/38/12/5
Host Range virology/host-range@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/38/12/5
Immune Escape Score virology/immune-escape-score@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	16/38/12/5
Infection Rate virology/infection-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	16/38/12/5
Infection Stage virology/infection-stage@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	19/38/12/5
Infectious Titer virology/infectious-titer@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	16/38/12/5
Multiplicity of Infection virology/multiplicity-of-infection@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/38/12/5
Neutralization Assay virology/neutralization-assay@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/38/12/5
Plaque Forming Units virology/plaque-forming-units@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/38/12/5
Quasispecies Profile virology/quasispecies-profile@0.1	CHANGES_REQD	–	IIICCCICCCIXX	W~PF-F	19/38/12/5
Receptor Binding virology/receptor-binding@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/38/12/5
Replication Rate virology/replication-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	16/38/12/5
Shedding Time Series virology/shedding-time-series@0.1	BLOCK	P-MEAS P-CONV	IIIIICIIICCCXXX	F~PFFF	16/38/12/5
Tissue Tropism virology/tissue-tropism@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/38/12/5
Transmission Rate virology/transmission-rate@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCXXX	F~PFFF	16/38/12/5

PROFILE	VERDICT	OWN FINDINGS	CONTRACT AREAS	FIXTURE AUDIT	ITEMS R/C/Rec/O
Viral Entry virology/viral-entry@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/38/12/5
Viral Fitness virology/viral-fitness@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/38/12/5
Viral Genome virology/viral-genome@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/38/12/5
Viral Haplotype virology/viral-haplotype@0.1	CHANGES_REQD	–	IIIXXXICCCIXX	W~PF-F	19/38/12/5
Viral Load virology/viral-load@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/38/12/5
Viral Phylogeny virology/viral-phylogeny@0.1	CHANGES_REQD	–	IIICCCICCCXXX	W~PF-F	19/38/12/5
Viral Protein Abundance virology/viral-protein-abundance@0.1	BLOCK	P-MEAS P-CONV	IIICIIICCCIXX	F~PFFF	14/38/12/5
Viral Sequence Variant virology/viral-sequence-variant@0.1	CHANGES_REQD	–	IIIXIXICCCIXX	W~PF-F	17/38/12/5
Viral Transcript Abundance virology/viral-transcript-abundance@0.1	BLOCK	P-MEAS P-CONV	IIICCIICCCIXX	F~PFFF	16/38/12/5
Virus Identity virology/virus-identity@0.1	CHANGES_REQD	–	IIIXIXICCCIXX	W~PF-F	17/38/12/5

HOW EACH CONTRACT AREA WAS DECIDED, AND FOR HOW MANY PROFILES

Contract-area decisions

Every profile records one decision per contract area, with a reason. The decisions follow the rules below, so each rule is listed once with the number of profiles it applies to. An exclusion always carries its scientific reason.

Structure

DECISION	PROFILES	REASON
INCLUDED	650	signal_type, dtype and shape in model.yaml carry the transport shape; the contract adds meaning on top and must not restate them.

Scientific meaning

DECISION	PROFILES	REASON
INCLUDED	650	The scientific identity of the value. Included, but the concept must be re-based on an external vocabulary (S1) and subject must be term-bound (S2).

Representation

DECISION	PROFILES	REASON
INCLUDED	650	Which encodings are interchangeable, and when a visible adapter is needed (S11).

Dimensions and feature order

DECISION	PROFILES	REASON
CONDITIONAL	427	Required whenever this profile's value is exchanged as a vector, matrix, tensor or image; not applicable to the scalar case.
EXCLUDED	134	This profile's value is a single scalar, event or record with no indexed axes, so axis, coordinate and feature-order fields cannot change compatibility.
INCLUDED	89	The axes define what each position indexes; required for this profile's non-scalar values.

Identifiers and versions

DECISION	PROFILES	REASON
CONDITIONAL	486	Required once the value is indexed by entities (features, genes, taxa, codes) or mapped between namespaces; otherwise inapplicable.
INCLUDED	98	The identifier namespace and release define what the entity labels mean.
EXCLUDED	66	The value is not indexed by an external entity identifier, so namespace and release cannot change compatibility.

Quantity, unit, scale, normalisation, endpoint

DECISION	PROFILES	REASON
CONDITIONAL	394	Required whenever the port carries a numeric value: unit, scale and normalization must be compared when both sides declare them (S3).
EXCLUDED	132	The value is categorical, structural or an identity assertion with no measured magnitude, so unit and scale do not apply.
INCLUDED	124	Quantity, unit and scale are required, and must be corrected per the per-profile table.

Biological and experimental context

DECISION	PROFILES	REASON
INCLUDED	575	Species is required; it needs subsumption and role typing (S5), plus conditional tissue, cell type, assay and compartment.
EXCLUDED	50	Core, chemical and simulation values are defined independently of any organism; a species field would assert biological scope the value does not have.
CONDITIONAL	25	Chemical entities are organism-independent, but assay and system context become required once a measured value is attached.

Time and lifecycle meaning

DECISION	PROFILES	REASON
CONDITIONAL	601	Required where the value is observed at or aggregated over a time window; not a gate for a single static value.
INCLUDED	49	This profile carries time-indexed or event data: time unit, origin, sampling and temporal meaning must be required (S14).

Origin and provenance

DECISION	PROFILES	REASON
CONDITIONAL	650	origin.type and the transformation chain must be required wherever the intended use distinguishes measured from simulated or inferred values; recommended elsewhere.

Uncertainty and missingness

DECISION	PROFILES	REASON
CONDITIONAL	650	Required where the value carries censoring, detection limits or a missingness convention (S13); optional otherwise.

Artifact format

DECISION	PROFILES	REASON
EXCLUDED	417	The value is exchanged in-memory as a typed contract, not as a file, so format and checksum fields cannot change compatibility.
INCLUDED	233	This profile admits artifact or file exchange, so format, format version, schema and digest must be required.

Cross-field constraints

DECISION	PROFILES	REASON
EXCLUDED	625	No cross-field relation in this profile changes compatibility beyond the field-level rules already declared.
CONDITIONAL	25	Declarative cross-field constraints are needed where bounds or field dependencies are part of the science (bounded scores, dose-response pairs).

Security, consent, licensing, data use

DECISION	PROFILES	REASON
CONDITIONAL	350	Required whenever the port may carry human-subject data: de-identification level and workspace boundary gate the exchange.
EXCLUDED	250	The value cannot carry human-subject or licensed data in this profile's intended use, so consent and data-use fields do not apply.
INCLUDED	50	Consent scope and data use are required and must be DUO-bound with subsumption comparison (S12).

ONE DECISION PER CANDIDATE ITEM PATH, APPLIED IN EVERY PACKET THAT LISTS IT

Candidate item decisions

The packets list 240 distinct candidate items, 20 to 79 per profile. Decisions: 70 required, 135 conditional, 26 recommended, 9 optional, 0 excluded. Conditional names the condition under which an item becomes required. The Packets column counts how many of the 650 packets list the item.

artifact

ITEM	DECISION	PACKETS	REASON
artifact.byte_size	RECOMMENDED	75	Operational check.
artifact.compression	CONDITIONAL	75	Required where the consumer must decompress.
artifact.encryption	CONDITIONAL	75	Required where the artifact is encrypted at rest.
artifact.format	REQUIRED	75	For artifact and file ports the format is the contract; should be an EDAM format term or a media type.
artifact.format_version	REQUIRED	75	Format versions change what can be read.
artifact.media_type	RECOMMENDED	75	Transport-level type.
artifact.partitioning	CONDITIONAL	75	Required for partitioned datasets.
artifact.schema_ref	REQUIRED	75	A tabular or structured artifact needs its schema.
artifact.schema_sha256	REQUIRED	75	Pins the schema bytes.
artifact.sha256	REQUIRED	75	Pins the artifact bytes; the standard already requires digests for published references.
artifact.staging	OPTIONAL	75	Deployment concern.

biological_context

ITEM	DECISION	PACKETS	REASON
biological_context.age	RECOMMENDED	475	Context that supports interpretation without gating compatibility.
biological_context.assay	CONDITIONAL	475	Required where the assay defines the quantity (titres, scores, intensities).
biological_context.cell_line	CONDITIONAL	475	Required for cell-line experiments; should be a Cellosaurus identifier.
biological_context.cell_type	CONDITIONAL	475	Required for cell-resolved measurements.
biological_context.cohort	CONDITIONAL	525	Required for population-level measures.
biological_context.compartment	CONDITIONAL	475	Required where the same analyte differs by compartment (plasma versus tissue, intra- versus extracellular).
biological_context.confluence	RECOMMENDED	75	Affects growth and signalling assays.
biological_context.culture_system	CONDITIONAL	75	2D, 3D and organoid systems are not interchangeable.
biological_context.denominator	REQUIRED	100	For any rate or proportion, the denominator is part of the quantity.
biological_context.disease	CONDITIONAL	475	Required where disease state changes the value's meaning.
biological_context.geography	CONDITIONAL	100	Required for spatially indexed epidemiological data.
biological_context.intervention	CONDITIONAL	475	Required for perturbation data, with agent, dose, duration and route.

ITEM	DECISION	PACKETS	REASON
biological_context.intervention.agent	CONDITIONAL	50	Part of the perturbation identity.
biological_context.intervention.dose	CONDITIONAL	50	Part of the perturbation identity.
biological_context.intervention.duration	CONDITIONAL	50	Part of the perturbation identity.
biological_context.intervention.route	CONDITIONAL	50	Part of the perturbation identity.
biological_context.intervention.schedule	CONDITIONAL	25	Part of the perturbation identity for repeat dosing.
biological_context.lineage	CONDITIONAL	75	Required for lineage-resolved developmental data.
biological_context.medium	CONDITIONAL	75	Medium composition changes metabolic measurements.
biological_context.oxygen	CONDITIONAL	75	Oxygen tension changes metabolic and hypoxia-related values.
biological_context.passage	RECOMMENDED	75	Affects cell-line phenotype drift.
biological_context.population	CONDITIONAL	100	Required for population-level measures.
biological_context.sampling_frame	CONDITIONAL	100	Required for surveillance and survey data.
biological_context.sex	RECOMMENDED	475	As above; may become conditional for sex-specific references.
biological_context.strain	CONDITIONAL	475	Required for microbial, viral and inbred-animal work.
biological_context.stratum	CONDITIONAL	100	Required for stratified estimates.
biological_context.substrate	CONDITIONAL	75	Required for adhesion, migration and biofilm assays.
biological_context.tissue	CONDITIONAL	475	Required where the value is tissue-dependent.

constraints

ITEM	DECISION	PACKETS	REASON
constraints[].expected	CONDITIONAL	25	The expected relation.
constraints[].id	CONDITIONAL	25	Required when the profile declares cross-field constraints.
constraints[].inputs	CONDITIONAL	25	The fields the constraint reads.
constraints[].operator	CONDITIONAL	25	Must come from the published operator vocabulary.
constraints[].profile_ref	CONDITIONAL	25	The profile the constraint belongs to.
constraints[].reason	CONDITIONAL	25	Human-readable justification.
constraints[].remediation	CONDITIONAL	25	What the producer should change.
constraints[].severity	CONDITIONAL	25	Error versus warning.

dimensions

ITEM	DECISION	PACKETS	REASON
dimensions.axes	REQUIRED	36	Required for every non-scalar value.
dimensions.axes[].coordinates_ref	CONDITIONAL	118	Required for physical or genomic coordinate axes.
dimensions.axes[].dynamic	RECOMMENDED	118	Declares whether the size is fixed at bind time.

ITEM	DECISION	PACKETS	REASON
dimensions.axes[].labels_ref	CONDITIONAL	118	Required where the axis is labelled by entities rather than positions.
dimensions.axes[].labels_sha256	CONDITIONAL	118	Pins the label set so two ports can prove they share a feature universe.
dimensions.axes[].meaning	REQUIRED	118	What the axis indexes is the scientific content of the shape.
dimensions.axes[].name	REQUIRED	118	An axis without a name cannot be aligned.
dimensions.axes[].ordering	REQUIRED	118	Order determines whether position i is comparable across ports.
dimensions.axes[].size	CONDITIONAL	118	Required where a fixed size is part of the contract.
dimensions.axes[].unit	CONDITIONAL	118	Required for physical axes (time, length, concentration, m/z).
dimensions.dose_axis	REQUIRED	50	For dose- and concentration-response profiles, the dose axis and its unit.
dimensions.feature_axis	REQUIRED	125	Identifies which axis carries features.
dimensions.feature_labels_ref	REQUIRED	125	The feature universe must be resolvable, not implied by position.
dimensions.feature_labels_sha256	REQUIRED	125	Pins the feature universe by digest.
dimensions.orientation	REQUIRED	125	Patient or specimen orientation prevents left-right and anterior-posterior errors.
dimensions.origin	REQUIRED	125	The coordinate origin anchors the grid in space.
dimensions.pixel_spacing	REQUIRED	125	Without spacing, voxel indices carry no physical scale.
dimensions.sample_axis	REQUIRED	125	Identifies which axis carries samples.
dimensions.slice_thickness	REQUIRED	125	Part of the physical geometry of a volume.
dimensions.spatial_axes	REQUIRED	125	For imaging and spatial data, which axes are spatial and in what order.
dimensions.time_axis	REQUIRED	50	Identifies the time axis where one exists.

identifiers

ITEM	DECISION	PACKETS	REASON
identifiers.ambiguity_handling	CONDITIONAL	350	Required where one-to-many mappings are possible.
identifiers.canonicalization	CONDITIONAL	350	Required where an identifier has several textual forms.
identifiers.case_policy	RECOMMENDED	350	Case handling is a frequent source of silent mismatches.
identifiers.coding_system	REQUIRED	50	For clinical codes, the code system is the meaning (ICD-10 versus SNOMED CT).
identifiers.coding_version	REQUIRED	50	Clinical code semantics change between versions.
identifiers.deprecated_handling	CONDITIONAL	350	Required where retired identifiers may appear.
identifiers.edge_namespace	CONDITIONAL	154	Where edges carry their own identifiers.
identifiers.entity_type	REQUIRED	350	Gene, transcript, protein and variant identifiers are not interchangeable.
identifiers.mapping_refs	CONDITIONAL	350	Required whenever the two ports use different namespaces (S3).

ITEM	DECISION	PACKETS	REASON
identifiers.mapping_refs[].cardinality	REQUIRED	50	One-to-many mappings are lossy and must be declared as such.
identifiers.mapping_refs[].coverage	REQUIRED	50	Partial coverage makes a mapping lossy.
identifiers.mapping_refs[].license	RECOMMENDED	50	Redistribution terms for the mapping.
identifiers.mapping_refs[].ref	CONDITIONAL	50	The mapping resource must be resolvable.
identifiers.mapping_refs[].release	CONDITIONAL	50	Mappings change between releases.
identifiers.mapping_refs[].sha256	CONDITIONAL	50	Pins the mapping bytes.
identifiers.mapping_refs[].source_namespace	CONDITIONAL	50	Defines the mapping's direction.
identifiers.mapping_refs[].target_namespace	CONDITIONAL	50	Defines the mapping's direction.
identifiers.namespace	REQUIRED	259	Must be a Bioregistry-registered prefix (S12).
identifiers.namespace_version	REQUIRED	259	Identifier semantics change between releases.
identifiers.node_namespace	REQUIRED	154	For graphs, what the node identifiers are.
identifiers.structure_hash	CONDITIONAL	75	Pins a chemical or macromolecular structure.
identifiers.taxonomy_namespace	REQUIRED	100	For community profiles, the taxonomy and its release define the labels.
identifiers.taxonomy_version	REQUIRED	100	Taxonomies are renamed and re-parented between releases.
identifiers.transcript_reference	REQUIRED	25	HGVS-style notation is meaningless without the transcript and version.
identifiers.unmapped_handling	REQUIRED	350	What happens to unmapped features decides whether a mapping is lossy.
identifiers.variant_notation	REQUIRED	25	The notation system (HGVS, SPDI, VCF) defines how to read the variant.

lifecycle

ITEM	DECISION	PACKETS	REASON
lifecycle.causality	RECOMMENDED	350	Whether the value may depend on future samples (filtering versus smoothing).
lifecycle.consumption	OPTIONAL	350	Runtime scheduling concern.
lifecycle.evolutionary_time	CONDITIONAL	100	Required for phylogenetic time axes, with the calibration.
lifecycle.freshness	OPTIONAL	350	Operational rather than scientific.
lifecycle.interpolation	CONDITIONAL	350	Required for irregular series consumed as continuous signals.
lifecycle.kind	CONDITIONAL	350	State versus event distinction.
lifecycle.observation_period	CONDITIONAL	100	Required for incidence and surveillance measures.
lifecycle.production	OPTIONAL	350	Runtime scheduling concern.
lifecycle.sampling	REQUIRED	350	Regular versus irregular sampling and the interval.
lifecycle.temporal_meaning	REQUIRED	350	Whether the value is instantaneous, an interval average or cumulative.

ITEM	DECISION	PACKETS	REASON
lifecycle.time_origin	REQUIRED	350	The epoch that t=0 refers to.
lifecycle.time_unit	REQUIRED	350	Required for every time-indexed value (S14).
lifecycle.window	CONDITIONAL	350	Required where the value is computed over a window (firing rates, moving averages).

measurement

ITEM	DECISION	PACKETS	REASON
measurement.aggregation	CONDITIONAL	450	Required where the value summarises repeats or a population (mean, median, per-cell versus per-sample).
measurement.baseline	CONDITIONAL	450	Required for fold changes, deltas and response values, which are undefined without their reference.
measurement.batch_correction	CONDITIONAL	125	Required for matrices: batch-corrected values cannot be pooled with uncorrected ones.
measurement.calibration_ref	CONDITIONAL	475	Required where results are only comparable after calibration to a standard (immunoassays, PET SUV, WHO units).
measurement.control	CONDITIONAL	325	Required for assay readouts expressed relative to a control.
measurement.detection_limits	CONDITIONAL	450	Required for concentrations, titres and MIC values, where values below the limit are not zeros.
measurement.evolutionary_distance	CONDITIONAL	100	Required where the distance's substitution model must travel with the value.
measurement.normalization	CONDITIONAL	475	Required for every abundance, intensity or expression value: raw and normalised values are not interchangeable (S3).
measurement.quantity	REQUIRED	349	The quantity kind; must resolve to an external quantity term rather than a self-reference (S1).
measurement.rate_basis	REQUIRED	100	For every rate profile: per-capita, per person-time, per contact and per cell are different quantities.
measurement.reference_range	RECOMMENDED	450	Clinically useful; not a compatibility gate.
measurement.response_direction	CONDITIONAL	50	Required for response curves, where the sign convention changes interpretation.
measurement.scale	REQUIRED	349	Stevens level, separated from transform and value domain (S6).
measurement.standardization	CONDITIONAL	100	Required for age- or sex-standardised epidemiological measures.
measurement.tolerance	OPTIONAL	450	Numeric tolerance is a policy choice, not part of the scientific contract.
measurement.unit	REQUIRED	349	Required for every numeric quantity; must be a UCUM expression whose dimension matches the declared quantity (S4).

origin

ITEM	DECISION	PACKETS	REASON
origin.absolute_tolerance	CONDITIONAL	50	Solver tolerance affects reproducibility of trajectories.
origin.audit_ref	RECOMMENDED	50	Governance trail.
origin.edge_evidence	CONDITIONAL	154	Required for interaction graphs: predicted and experimental edges differ.
origin.estimated_method	CONDITIONAL	25	Required for estimated quantities.
origin.evidence_refs	RECOMMENDED	325	Supports audit.
origin.fit_method	CONDITIONAL	50	Required for fitted parameters.
origin.generated_at	RECOMMENDED	100	Build-time provenance.
origin.instrument	RECOMMENDED	325	Supports interpretation and batch reasoning.
origin.instrument_model	RECOMMENDED	325	As above.
origin.method	CONDITIONAL	100	Required where the method defines the quantity.
origin.method_version	CONDITIONAL	100	Required where method versions change values.
origin.model_digest	CONDITIONAL	50	Required where a model produced the value.
origin.observed_at	CONDITIONAL	100	Required where the observation time matters.
origin.operator	OPTIONAL	100	Personnel identity is not a compatibility property and carries privacy risk.
origin.parameter_digest	CONDITIONAL	50	Pins the parameters used.
origin.phylogenetic_method	CONDITIONAL	100	Required for tree-derived values.
origin.processing_pipeline_ref	CONDITIONAL	125	Required for imaging and omics pipelines.
origin.protocol_ref	CONDITIONAL	325	Required where a protocol defines the assay.
origin.protocol_version	CONDITIONAL	325	As above.
origin.random_seed	CONDITIONAL	50	Required for reproducible stochastic simulation.
origin.registration_ref	CONDITIONAL	125	Required where spatial registration underlies the value.
origin.relative_tolerance	CONDITIONAL	50	As above.
origin.segmentation_method	CONDITIONAL	125	Required where segmentation produces the value.
origin.software	RECOMMENDED	325	Provenance for computed values.
origin.software_version	CONDITIONAL	325	Required for computed values whose results change between versions.
origin.solver	CONDITIONAL	50	Required for simulation outputs.
origin.solver_version	CONDITIONAL	50	As above.
origin.source_ref	CONDITIONAL	100	Required for derived datasets.
origin.source_sha256	CONDITIONAL	100	Pins the source bytes for derived data.
origin.substitution_model	CONDITIONAL	100	Required for evolutionary distances.
origin.surveillance_system	CONDITIONAL	100	Required for public-health data.
origin.transformation_chain	REQUIRED	100	The applied transformations are what distinguish raw from processed values (S3).

ITEM	DECISION	PACKETS	REASON
origin.type	REQUIRED	100	Measured, simulated, inferred and transformed values must never silently substitute for one another.

representation

ITEM	DECISION	PACKETS	REASON
representation.alphabet	CONDITIONAL	100	Required for sequence ports (DNA, RNA, protein, extended alphabets).
representation.alternate_allele	REQUIRED	25	Part of variant identity.
representation.canonicalization	CONDITIONAL	125	Required where the same value has several textual forms (sequences, SMILES, variants).
representation.channel_order	REQUIRED	125	For multichannel images, channel order defines what each plane is.
representation.circularity	CONDITIONAL	100	Required for circular genomes and plasmids.
representation.conformer_method	RECOMMENDED	75	Provenance for generated 3D structures.
representation.coordinate_system	REQUIRED	225	For any positional data: assembly, origin and 0- versus 1-based conventions silently corrupt coordinates.
representation.directed	REQUIRED	154	Directedness changes the claim a graph makes.
representation.edge_schema	REQUIRED	154	For graphs, the edge schema defines what an edge asserts.
representation.encoding	CONDITIONAL	125	Required for sequence and artifact ports.
representation.endianness	OPTIONAL	125	Byte order is an artifact-level concern handled by the format.
representation.feature_space	REQUIRED	250	Required for every vector or matrix: the feature universe and its version define what position i means.
representation.formal_charge	CONDITIONAL	75	Required for structure-based modelling.
representation.hydrogen_policy	CONDITIONAL	75	Required for structure-based modelling.
representation.interval_convention	REQUIRED	100	Half-open versus closed intervals is the classic off-by-one in genomic ranges.
representation.isotope_state	CONDITIONAL	75	Required for tracer and exact-mass work.
representation.layout	CONDITIONAL	125	Required for matrices and tensors where row-major versus column-major changes interpretation of a flat buffer.
representation.left_normalized	REQUIRED	25	Variant normalisation determines whether two records are the same variant.
representation.masking	RECOMMENDED	100	Soft- versus hard-masking affects downstream use.
representation.multigraph	CONDITIONAL	154	Required where parallel edges are meaningful.
representation.node_schema	REQUIRED	154	For graphs, the node schema defines what a node is.
representation.normalization_tool	RECOMMENDED	25	Provenance of variant normalisation.
representation.normalization_version	RECOMMENDED	25	As above.
representation.ordering	REQUIRED	207	Required whenever the value is an ordered array; an unordered feature vector cannot be aligned.
representation.precision	RECOMMENDED	125	Precision loss matters for checkpoint and artifact exchange but rarely gates scientific compatibility.

ITEM	DECISION	PACKETS	REASON
representation.protonation_state	CONDITIONAL	75	Required for docking and physics-based modelling.
representation.quality_encoding	CONDITIONAL	100	Required for read data, where the quality offset differs by platform.
representation.reference_allele	REQUIRED	25	Part of variant identity.
representation.reference_assembly	REQUIRED	100	Coordinates are meaningless without the assembly and its patch level.
representation.reference_sequence	REQUIRED	100	As above, at sequence level.
representation.self_loops	CONDITIONAL	154	Required where self-interaction is meaningful.
representation.sparsity	CONDITIONAL	125	Required for sparse encodings, together with whether an absent entry is a zero or an unobserved value.
representation.state_order	CONDITIONAL	50	Required where categorical states have a defined order.
representation.stereochemistry	REQUIRED	75	For chemical structures, stereochemistry changes the molecule and its activity.
representation.strand	REQUIRED	100	Strand changes the biological claim for stranded data.
representation.structure_format	CONDITIONAL	75	Required for structure artifacts.
representation.tautomer_state	CONDITIONAL	75	Required for docking and property prediction.
representation.tree_format	CONDITIONAL	100	Required for phylogeny artifacts.
representation.weight_semantics	REQUIRED	154	An edge weight may be a probability, a correlation, a count or a distance; these are not interchangeable.

security

ITEM	DECISION	PACKETS	REASON
security.classification	CONDITIONAL	50	Required for controlled data.
security.deidentification	REQUIRED	50	For human-subject data, the de-identification level is a governance precondition, not a preference.
security.export_control	CONDITIONAL	50	Required for controlled biological or clinical data.
security.license	CONDITIONAL	50	Required where redistribution terms restrict use.
security.license_ref	CONDITIONAL	50	As above.
security.redaction	CONDITIONAL	50	Required where fields are redacted.
security.residency	CONDITIONAL	50	Required where data may not leave a jurisdiction.
security.retention	CONDITIONAL	50	Required where retention limits apply.
security.workspace_boundary	REQUIRED	50	The boundary that a compatibility decision must not cross silently.

semantic

ITEM	DECISION	PACKETS	REASON
semantic.ancestral_state	CONDITIONAL	100	Required for phylogenetic reconstructions, where the state's polarity is part of the claim.
semantic.cell_state	OPTIONAL	75	Useful context; not a compatibility gate on its own.

ITEM	DECISION	PACKETS	REASON
semantic.consequence	CONDITIONAL	25	Required where the consequence term is the value's meaning.
semantic.endpoint	CONDITIONAL	650	Required where the value is an assay or clinical endpoint whose definition changes the number (response, survival, toxicity); optional elsewhere.
semantic.entity	CONDITIONAL	75	Required where a specific molecular or cellular entity identifies the value.
semantic.intended_use	RECOMMENDED	50	Declared intended use supports the governance boundary but must not act as a compatibility gate.
semantic.lineage	CONDITIONAL	100	Required for lineage-resolved measurements.
semantic.model_formalism	CONDITIONAL	50	Required for simulation ports, where the formalism determines what the state means.
semantic.ontology_snapshot_ref	CONDITIONAL	50	Required whenever ontology_terms are compared, since term-equivalent and term-subsumes need a pinned snapshot.
semantic.ontology_snapshot_sha256	CONDITIONAL	50	As above: the snapshot must be pinned by digest for a reproducible comparison.
semantic.ontology_terms	REQUIRED	650	Carries the external scientific meaning that semantic.concept does not (S1); at least one CURIE from a registered prefix with ontology, version and relation.
semantic.ontology_terms[].label	RECOMMENDED	50	A human-readable label aids review but must never be the compared value.
semantic.ontology_terms[].ontology	REQUIRED	50	The source ontology determines which snapshot is used for subsumption.
semantic.ontology_terms[].relation	REQUIRED	50	Exact, broad and narrow matches support different compatibility conclusions.
semantic.ontology_terms[].uri	REQUIRED	50	The term IRI or CURIE is the compared value; without it the mapping cannot be resolved.
semantic.ontology_terms[].version	REQUIRED	50	Ontology releases obsolete and re-parent terms, so the release must be pinned.
semantic.pk_parameter	REQUIRED	25	For PK profiles the specific parameter (CL, Vd, AUC, Cmax) is the identity of the value.
semantic.process	CONDITIONAL	650	Required where the biological process defines the quantity (rates, fluxes, transitions).
semantic.qualifiers	OPTIONAL	650	Free qualifiers cannot be compared safely until they are term-bound; keep optional and non-comparing.
semantic.role	RECOMMENDED	650	Distinguishes the port's role (input, output, covariate) without changing the scientific identity of the value.
semantic.variant_class	CONDITIONAL	25	Required where variant class changes interpretation.

uncertainty

ITEM	DECISION	PACKETS	REASON
uncertainty.branch_support	CONDITIONAL	100	Required for phylogenetic trees.
uncertainty.censoring	REQUIRED	400	For time-to-event and survival values (S13).
uncertainty.confidence_level	CONDITIONAL	400	An interval without its level is uninterpretable.
uncertainty.distribution	CONDITIONAL	400	Required for distributional outputs.
uncertainty.edge_confidence	CONDITIONAL	154	Required for predicted interaction edges.
uncertainty.fit_quality	CONDITIONAL	50	Required for fitted parameters.
uncertainty.imputation	CONDITIONAL	400	Imputed values must not silently substitute for observed ones.
uncertainty.interval	CONDITIONAL	425	Required where an interval is provided.
uncertainty.kind	CONDITIONAL	400	Required wherever an uncertainty value is present, to say what it is.
uncertainty.missingness	REQUIRED	475	For matrices and vectors: whether an absent entry is a zero, a non-detect or unobserved (S13).
uncertainty.parameter_covariance	CONDITIONAL	25	Required where parameter correlations matter.
uncertainty.parameters	CONDITIONAL	400	Required with a declared distribution.
uncertainty.quality_flags	RECOMMENDED	525	Supports filtering without gating compatibility.
uncertainty.segmentation_confidence	CONDITIONAL	125	Required for derived imaging measurements.
uncertainty.variance	CONDITIONAL	400	Required where variance travels with the value.

EVERY QUESTION IN QUESTIONS_FOR_REVIEWERS IS ANSWERED FOR EVERY PACKET THAT ASKS IT

Reviewer questions

The packets use 13 question templates. Each is shown with the number of packets that ask it and the answer recorded for one representative profile. The answer text for every packet is in change-request.json, where it is specialised to the profile's own fields, units, axes and findings.

Does the proposed concept identify {profile} narrowly enough to prevent a different scientific quantity from matching?

Asked in 650 packets. Example answer for `neuroscience/firing-rate@0.1`:

No. The concept is the self-referential IRI `.../neuroscience/firing-rate/v0.1#concept` with no definition, label or axioms, and the equal rule on it is tautological for two valid contracts. It is narrow in the sense that nothing else matches it, which is also why it carries no scientific meaning: it restates profile identity. It is also version-bound, so a v0.2 profile would report every v0.1 port as INCOMPATIBLE. See S1.

Are the allowed representations complete, and can any two allowed representations connect without an explicit adapter?

Asked in 650 packets. Example answer for `neuroscience/firing-rate@0.1`:

The allowed set is scalar, dense_vector, sparse_vector, matrix, tensor, array, record, table, derived from the coarse applies_to family rather than from this profile's meaning. Two allowed representations cannot always connect: dense_vector and sparse_vector are lossless re-encodings but compare INCOMPATIBLE under the equal operator, while record and table require a declared schema mapping. See S11.

Which authoritative standards, ontologies, databases or primary publications support each required field and comparison rule?

Asked in 650 packets. Example answer for `neuroscience/firing-rate@0.1`:

Recorded per decision in this review: UCUM 2.2 and the SI Brochure for units, VIM and ISO 80000-1 for quantity and dimension, Stevens (1946) for measurement level, NCBI Taxonomy for species, UBERON/CL/OBI for specimen and assay semantics, Bioregistry for namespaces, GA4GH DUO for data use, plus the domain sources listed against each profile. No source can be pinned by SHA-256 until the documents are retrieved at sign-off.

Which missing value must return UNKNOWN, and which known contradiction must return INCOMPATIBLE?

Asked in 650 packets. Example answer for `neuroscience/firing-rate@0.1`:

UNKNOWN: any required field absent on either side (currently only the source side is fixture-tested, S10); a species of 'any' facing a specific target (currently INCOMPATIBLE, wrong, S5); an ontology or mapping comparison with no pinned snapshot. INCOMPATIBLE: a different concept or subject term with no subsumption path; units of a different dimension; a different measurement scale or normalisation; conflicting compartment or intervention context. Convertible units must never be INCOMPATIBLE, which they currently are for g vs kg, Cel vs K, 1/s vs /min and uM vs mol/L (S4, S8).

What uses and scientific claims must this profile explicitly exclude?

Asked in 650 packets. Example answer for `neuroscience/firing-rate@0.1`:

That a passing comparison implies Firing Rate values are biologically equivalent, measured under comparable conditions, or fit for clinical or regulatory use. The contract is an interface check only: it cannot detect batch effects, assay bias, population mismatch, or that an upstream model was wrong.

Which species, tissue, cell type, disease, intervention, assay, cohort or compartment fields are required or conditional?

Asked in 575 packets. Example answer for `neuroscience/firing-rate@0.1`:

Species: required, but with taxonomic subsumption and, where host and organism differ, role typing (S5).

Conditional-required: tissue and cell type where the value is tissue-dependent; assay where the assay defines the quantity; compartment where the same analyte differs by compartment; intervention (agent, dose, duration, route) for perturbation data; cohort and denominator for population measures. Age and sex: recommended context.

Does this profile need a versioned identifier namespace or an ordered feature universe?

Asked in 552 packets. Example answer for `neuroscience/firing-rate@0.1`:

Not for the scalar case. It becomes required as soon as the value is indexed by entities or exchanged as an array.

Should axes, labels, feature order, coordinates or artifact schema be required for non-scalar values?

Asked in 539 packets. Example answer for `neuroscience/firing-rate@0.1`:

Yes for every non-scalar exchange: axes, per-axis meaning and ordering, plus a pinned label set. This profile does not currently require them even though its allowed representations include arrays and matrices.

Should quantity, unit, scale, normalization, baseline or endpoint be required for this profile?

Asked in 526 packets. Example answer for `clinical/diagnosis-code@0.1`:

Yes. This profile permits numeric values but requires no unit or scale, so two ports differing in unit, normalisation or baseline compare `DIRECT_COMPATIBLE` (S3). Require quantity, unit (UCUM) and scale whenever a magnitude is carried, and normalization and baseline for any derived or relative value.

Are the quantity, unit, scale, normalization, baseline and endpoint definitions sufficient for this measurement?

Asked in 124 packets. Example answer for `neuroscience/firing-rate@0.1`:

No. Current unit '1/s' and scale 'ratio' are wrong or under-specified: proposed unit Hz (1/s), proposed scale ratio. Unit is correct. The blocking problem is the transformation fixture asserting a nanomolar-to-micromolar conversion for this profile (S7), and the absence of the counting window, which is part of the quantity. Normalization, baseline and endpoint are not required at all, and the quantity is a self-referential IRI rather than an external quantity term.

Which identifier namespaces, releases, canonicalization rules and mapping-loss rules are permitted?

Asked in 98 packets. Example answer for `clinical/diagnosis-code@0.1`:

Only Bioregistry-registered prefixes resolved against a pinned registry snapshot, each with the namespace's own release identifier. Canonicalization must be declared where an identifier has several textual forms.

Cross-namespace mapping must declare the mapping resource, release, digest, cardinality, coverage and the handling of unmapped features; one-to-many or partial mappings are lossy and must require approval rather than resolving as direct.

Are axis names, order, labels, coordinates and dynamic-size rules fully specified?

Asked in 89 packets. Example answer for `imaging/ct-volume@0.1`:

No. `[x, y]` should be `[x, y, z]` with pixel spacing, slice thickness and patient orientation: A CT volume is three-dimensional; two axes cannot index a voxel grid. Axis names are declared but per-axis meaning, unit, label reference, label digest and coordinate reference are only candidates, so two ports can agree on axis names while indexing different feature universes.

Are event/state meaning, observation time, sampling, window, freshness and interpolation rules complete?

Asked in 76 packets. Example answer for `neuroscience/firing-rate@0.1`:

No. `lifecycle.temporal_meaning`, `time_unit`, `time_origin` and `sampling` are candidates rather than requirements, so an instantaneous value, an interval average and a cumulative total are all exchangeable under this profile (S14).

FORMAL STANDARDS, MAINTAINED ONTOLOGIES, DATABASES, GUIDANCE AND PRIMARY PUBLICATIONS

Sources

Pinned by scripts/pin_sources.py and recorded in sources/sources.lock.json. Read the pinning column before relying on a digest: document pins the reviewed text, citation metadata pins only the Crossref record for the DOI, and licensed standards (ISO 80000-1, ISO 20776-1, SNOMED CT) cannot be pinned here at all. Where a decision rests on a paywalled or licensed text, the reviewer pins their own copy.

ID	TITLE AND LOCATION	VERSION	SHA-256
ucum-2.2	The Unified Code for Units of Measure (UCUM) STANDARD https://github.com/ucum-org/ucum/releases/tag/v2.2	Version 2.2, released 2024-06-28	sha256:dfcc ea1b5dc2842 4... document
si-brochure-9	The International System of Units (SI), 9th edition STANDARD https://doi.org/10.59161/AUEZ1291	9th edition (2019), current revision	sha256:1122 cf38e25b23d 7... document
vim-jcgm-200-2012	International vocabulary of metrology (VIM), JCGM 200:2012 STANDARD https://www.bipm.org/en/committees/jc/jcgm/publications	3rd edition, 2012 (2008 version with minor corrections)	sha256:8a60 3e99d236a65 d... document
iso-80000-1-2022	ISO 80000-1:2022 Quantities and units - Part 1: General STANDARD https://www.iso.org/standard/76921.html	2022 edition	not pinned
stevens-1946	On the Theory of Scales of Measurement PRIMARY PUBLICATION https://doi.org/10.1126/science.103.2684.677	Science 103(2684):677-680, 1946	sha256:ae8e 3afbb9bf55a 4... citation metadata
ncbitaxon-schoch-2020	NCBI Taxonomy: a comprehensive update on curation, resources and tools DATABASE https://doi.org/10.1093/database/baaa062	Database, 2020; NCBITaxon OBO release to be pinned	sha256:c911 7251b10f7bf a... citation metadata
obo-fp-003	OBO Foundry Principle 3: URI/Identifier Space STANDARD https://obofoundry.org/principles/fp-003-uris.html	Principles v1.4	sha256:75e5 a56f6ec2bf7 b... document
w3c-cooluris	Cool URIs for the Semantic Web STANDARD https://www.w3.org/TR/cooluris/	W3C Interest Group Note, 3 December 2008	sha256:71c2 5a8c9ab567a 2... document
bioregistry-2022	Unifying the identification of biomedical entities with the Bioregistry DATABASE https://doi.org/10.1038/s41597-022-01807-3	Scientific Data 9:714, 2022	sha256:26b2 f2deb98cdde 6... citation metadata
duo-2021	The Data Use Ontology to streamline responsible access to human biomedical datasets ONTOLOGY https://doi.org/10.1016/j.xgen.2021.100028	Cell Genomics 1(2):100028, 2021; DUO OWL release to be pinned	sha256:f191 554ab2cc6ee f... citation metadata
uberon-2012	Uberon, an integrative multi-species anatomy ontology ONTOLOGY https://doi.org/10.1186/gb-2012-13-1-r5	Genome Biology 13:R5, 2012; release to be pinned	sha256:dfca a2c743785a4 c... citation metadata
cl-2016	The Cell Ontology 2016: enhanced content, modularization, and ontology interoperability ONTOLOGY https://doi.org/10.1186/s13326-016-0088-7	J Biomed Semantics 7:44, 2016; release to be pinned	sha256:9fe8 06947749769 d... citation metadata

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obi-2016	The Ontology for Biomedical Investigations ONTOLOGY https://doi.org/10.1371/journal.pone.0154556	PLoS ONE 11(4):e0154556, 2016; release to be pinned	sha256:a72f918dbff24a3e... citation metadata
uo-2012	The Units Ontology: a tool for integrating units of measurement in science ONTOLOGY https://doi.org/10.1093/database/bas033	Database, 2012; release to be pinned	sha256:f5d8a15ae26404b5... citation metadata
edam-2013	EDAM: an ontology of bioinformatics operations, types of data and identifiers, topics and formats ONTOLOGY https://doi.org/10.1093/bioinformatics/btt113	Bioinformatics 29(10):1325-1332, 2013; release to be pinned	sha256:3d8dc67f32e418d2... citation metadata
iuphar-neubig-2003	IUPHAR Committee on Receptor Nomenclature and Drug Classification. XXXVIII. Update on terms and symbols in quantitative pharmacology STANDARD https://doi.org/10.1124/pr.55.4.4	Pharmacol Rev 55(4):597-606, 2003	sha256:3df1324ff70bf63b... citation metadata
iso-20776-1-2019	ISO 20776-1:2019 Broth micro-dilution reference method for testing in vitro activity of antimicrobial agents STANDARD https://www.iso.org/standard/70464.html	2019 edition	not pinned
kdigo-2024-ckd	KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease REGULATORY GUIDANCE https://doi.org/10.1016/j.kint.2023.10.018	Kidney Int 105(4S):S117-S314, 2024	sha256:0b0d05b3a208f2b5... citation metadata
ats-ers-spirometry-2019	Standardization of Spirometry 2019 Update REGULATORY GUIDANCE https://doi.org/10.1164/rccm.201908-1590ST	Am J Respir Crit Care Med 200(8):e70-e88, 2019	sha256:7f4346e7b7577a0b... citation metadata
fda-bmv-2018	Bioanalytical Method Validation: Guidance for Industry REGULATORY GUIDANCE https://www.fda.gov/regulatory-information/search-fda-guidance-documents/bioanalytical-method-validation-guidance-industry	May 2018	sha256:3083d8513b407d20... landing page
mzml-2011	mzML - a community standard for mass spectrometry data STANDARD https://doi.org/10.1074/mcp.R110.000133	Mol Cell Proteomics 10(1):R110.000133, 2011	sha256:9c5c8e9a4157b1cc... citation metadata
goldbook-mr	IUPAC Compendium of Chemical Terminology: relative molecular mass STANDARD https://doi.org/10.1351/goldbook.R05271	Gold Book, 2nd ed. (1997), online corrected version	sha256:ba9f1f8023adb06a... citation metadata
du-2010-mvalue	Comparison of Beta-value and M-value methods for quantifying methylation levels by microarray analysis PRIMARY PUBLICATION https://doi.org/10.1186/1471-2105-11-587	BMC Bioinformatics 11:587, 2010	sha256:f72d5e666c5262e4... citation metadata
cdc-epi-3ed	Principles of Epidemiology in Public Health Practice, 3rd Edition, Lesson 3: Measures of Risk REGULATORY GUIDANCE https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson3/section2.html	3rd edition, 2012 update	sha256:76755f77c88da7ae... document

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reed-muench-1938	A simple method of estimating fifty per cent endpoints PRIMARY PUBLICATION https://doi.org/10.1093/oxfordjournals.aje.a118408	Am J Epidemiol 27(3):493-497, 1938	sha256:c926 a92627057e6 9... citation metadata
who-is-antibody-2021	WHO International Standard for anti-SARS-CoV-2 immunoglobulin PRIMARY PUBLICATION https://doi.org/10.1016/S0140-6736(21)00527-4	Lancet 397(10282):1347-1348, 2021	sha256:02ed d34a693a41e b... citation metadata
pgs-catalog-2021	The Polygenic Score Catalog as an open database for reproducibility and systematic evaluation DATABASE https://doi.org/10.1038/s41588-021-00783-5	Nat Genet 53:420-425, 2021	sha256:165f 85b0bf5c0ac 4... citation metadata
orth-2010-fba	What is flux balance analysis? PRIMARY PUBLICATION https://doi.org/10.1038/nbt.1614	Nat Biotechnol 28:245-248, 2010	sha256:97ca 90d0dcf09b2 5... citation metadata
wagner-2012-tpm	Measurement of mRNA abundance using RNA-seq data: RPKM measure is inconsistent among samples PRIMARY PUBLICATION https://doi.org/10.1007/s12064-012-0162-3	Theory Biosci 131:281-285, 2012	sha256:4d5b fac12332d94 c... citation metadata
robinson-2010-tmm	A scaling normalization method for differential expression analysis of RNA-seq data PRIMARY PUBLICATION https://doi.org/10.1186/gb-2010-11-3-r25	Genome Biology 11:R25, 2010	sha256:580d 63d6fb98dbc 6... citation metadata
svensson-2020-zero s	Droplet scRNA-seq is not zero-inflated PRIMARY PUBLICATION https://doi.org/10.1038/s41587-019-0379-5	Nat Biotechnol 38:147-150, 2020	sha256:c49c 9d619a2ab6b e... citation metadata
lieberman-aiden-2009	Comprehensive mapping of long-range interactions reveals folding principles of the human genome PRIMARY PUBLICATION https://doi.org/10.1126/science.1181369	Science 326(5950):289-293, 2009	sha256:8a08 1e541a161a2 c... citation metadata
dicom-ps3.3	DICOM PS3.3 Information Object Definitions STANDARD https://dicom.nema.org/medical/dicom/current/output/html/part03.html	current edition; exact release to be pinned	sha256:6492 0e6e123ee39 c... document
boellaard-2015-suv	FDG PET/CT: EANM procedure guidelines for tumour imaging: version 2.0 REGULATORY GUIDANCE https://doi.org/10.1007/s00259-014-2961-x	Eur J Nucl Med Mol Imaging 42:328-354, 2015	sha256:2b9b a5f45949f2a 6... citation metadata
bids-ieeg-2019	iEEG-BIDS, extending the Brain Imaging Data Structure specification to human intracranial electrophysiology STANDARD https://doi.org/10.1038/s41597-019-0105-7	Scientific Data 6:102, 2019	sha256:c08f 154b9ce9f96 3... citation metadata
nwb-2022	The Neurodata Without Borders ecosystem for neurophysiological data science STANDARD https://doi.org/10.7554/eLife.78362	eLife 11:e78362, 2022	sha256:4306 d801c1af80a 4... citation metadata
clark-2003-survival	Survival Analysis Part I: Basic concepts and first analyses PRIMARY PUBLICATION https://doi.org/10.1038/sj.bjc.6601118	Br J Cancer 89:232-238, 2003	sha256:fe86 893b338c800 4... citation metadata

ID	TITLE AND LOCATION	VERSION	SHA-256
lynch-2010-mutrate	Evolution of the mutation rate PRIMARY PUBLICATION https://doi.org/10.1016/j.tig.2010.05.003	Trends Genet 26(8):345-352, 2010	sha256:a542 04f3ee0a0b1 4... citation metadata
ensembl-2024	Ensembl 2024 DATABASE https://doi.org/10.1093/nar/gkad1049	Nucleic Acids Res 52(D1):D891-D899, 2024; release to be pinned	sha256:7adc cc8ce058865 d... citation metadata

Approval of a profile would mean only that it gives a defensible and precise compatibility contract for its stated intended use. It would not mean that any model using it is scientifically valid, clinically safe or suitable for regulatory use.